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DTK

DTK 26.10
CUDA 程序适配使用手册

DTK-261000-017-0915-932

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版本更新

- DTK-26.10
  - 升级至 CUDA12.9，支持 fp8。
  - 新增 cufftw、nvencode、nvtx3、nvjpegk2000 等库。
  - MMA 支持 gfx938、gfx92a。
  - cub/thrust 支持高性能模式，通过 USE_GPU_FUSION_PLATFORM 宏切换。
- DTK-26.04
  - TensorCore 接口封装库 DUMMA，提供 DCU 专用矩阵计算单元能力。
  - 新增 nvvm、nvcomp、nvjpej、nvimgcodec、cuvid 等库，npp 库接口新增图像处理相关若干接口。
  - 新增 reduce、mma、fp8 头文件支持，CG 功能对齐 CUDA12.6。
  - thrust、cub、libcudacxx 头文件库对齐 NV CCCL 2.4 版本。
  - 提供 unset_cuda_env.sh 环境变量卸载脚本。
- DTK-25.04.2：
  - 只保留 CUDA12.6 支持，删除 CUDA11.8 支持。
- DTK-25.04.1：
  - 对标 CUDA12.6，支持若干基础库 API，需要通过加载环境变量使用，详情可见 2.1 描述，新增函数详情可见 5.15 描述。
  - 支持 nvrtc 原生编译选项。
  - Nvcc 修复若干兼容性 bug，包括-options-file、-arch-ls、-code-ls 等编译选项，提高对高版本 cmake 的兼容性。
  - 新增 warp-32 兼容功能，可通过编译选项“-D GPUFUSION_ENABLE_WARPSIZE32”开启，详情可见 4.2.9 描述。
- DTK-25.04：
  - 补充若干 CUDA11.8 接口的支持，runtime、driver、cublas、cufft、cusparse、curand、nvrtc、nccl 核心基础库功能覆盖率均达到 95%以上。
  - 新增或完善 cublasLt、cupti、libcudacxx、npp 等非基础库支持。
  - 优化部分 nvcc 选项支持问题。

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- 修复 nccl 构建问题。
- 同一环境下不支持 nvcc 和 hipcc 一起使用，需要设置环境变量 DCC_PLATFORM=hipcc 或 nvcc，来切换编程环境。

■ DTK-24.04：

- 补充若干 CUDA11.8 接口的支持，不支持接口给出明确的警告。
- 支持分离编译、默认流、pipeline 等基础功能。
- 支持内嵌 PTX。
- 修复若干 CUDA11.8 接口正确性问题。
- 修复 bfloat16 等精度相关的头文件。
- 优化构建编译生态，包括支持高版本 cmake、修复若干 nvcc 选项、cmake 不再需要强制指定默认编译器 nvcc 等。
- 调整 nvcc 默认的 C++标准为 11，thrust、cub 等库使用时需用户手动追加 C++14 以上的编译选项。
- 支持部分标准库头文件的自动导入。
- 删除静态库。
- 使用手册中增加一些与原生 CUDA 用法差异的说明，合并 4、5 两节。

■ DTK-23.10：

- 对标 CUDA 11.8，全量覆盖各基础库的 API。
- 增加了对 nvml 库的支持。
- 修复若干 API 功能实现、计算正确性等问题。
- 补充 4.2“常见问题”和 5“限制”。

■ DTK-23.04：

- 支持 cub 功能。
- 对标 CUDA 10.2，全量覆盖各库的 API。
- 修复若干 API 功能实现、计算正确性等问题。
- 支持主要 CUDA 原生宏。
- 修正 6.2.3 标题为未支持的函数。

■ DTK-22.10.1：

- 提供完整的 CUDA 头文件。
- 提供 cudnn 和 cusolver 数学库接口，以及 nccl 通讯库接口。

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- 支持 cooperative groups 和 thrust 功能。

- DTK-22.10：
  - 提供 CUDA Runtime API 接口，提供 cublas、cufft、cusparse 和 curand 数学库接口。
  - 提供 nvcc 编译器和主要 cuda 头文件。
  - 提供 CUDA CMake 和 FindCUDA 编译构建系统支持。

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1 GPUfusion 简介

DTK（DCU Toolkit）是 DCU 硬件平台开发工具包。借助 DTK，开发人员可以在 DCU 上开发、优化和部署应用。目前，最新 DTK 已经支持多种计算框架，包括基础数学算子、科学计算、深度学习等众多领域。DTK 原生提供 HIP 编程模型的支持。

为了能够支持 CUDA 编程环境，设计了 CUDA 兼容框架“GPUfusion”，用户可以无感知地在 DTK 平台编写 CUDA 程序，并快速地将已有原生 CUDA 应用在 DTK 上进行部署，简化原生 CUDA 代码的移植流程。



图 1.1 融合 GPUfusion 的 DTK 软件栈

2 CUDA 兼容性

兼容的 CUDA 版本：12.9。

2.1 使用方法

基于 CUDA 兼容框架 GPUfusion，可以在不修改源码以及 CMakeLists.txt 的情况下，直接将 CUDA 源码编译为 DTK 平台的可执行程序。提高了系统兼容性，为程序移植工作提供极大的便利。

GPUfusion 已内置在 DTK 工具包内，使用前需要设置相应的环境变量。在 DTK 的 cuda 目录下面提供了设置相应环境变量的脚本 env.sh，设置方法如下：

1) 首先加载 DTK 目录下的 env.sh，用于设置 DTK 环境变量：

```
source dtk/env.sh
```

2) 然后加载 DTK 的 cuda 目录下的 env.sh，用于设置 CUDA 环境变量：

```
source dtk/cuda/env.sh
```

环境变量加载成功后，则可以按照 CUDA 的用法，进行程序构建、编译、运行和测试。目前主要覆盖了接口、编译参数等方面的兼容，来支持在 DTK 平台编译运行 CUDA 程序。可以使用 unset_cuda_env.sh 脚本卸载 CUDA 环境变量，进行 HIP 编程。

2.2 接口兼容

GPUfusion 提供的 CUDA API 可以分为三类：

1) 支持的 API

接口，行为和输出与原版的 CUDA 完全一致。

2) 未严格转换的 API

未严格转换的 API 是指行为和 CUDA 原版不完全一致，但不影响系统的正常执行。一般情况下不会影响程序的正确性。

在默认情况下，在编译期间会提示：

```
warning：This API is not support currently. This may result in inaccurate results.
```

用户可以通过环境变量 CUDA_STRICT 来禁用未严格转换的 API，这种情况下，未严格转换的 API 会进行报错处理，然后终止编译。

```
export CUDA_STRICT=1
```

3) 不支持的 API

这部分 API 目前 GPUfusion 还不支持，建议使用相关支持的 API 进行替换。详细 API 的支持情况，可以参考附录。

4) 表中未标注的则表示支持。

2.3 编译参数兼容

GPUfusion 提供对 CUDA nvcc 编译参数的支持，GPUfusion nvcc 会自动解析编译参数并根据内置规则进行转换，无法转换的参数将根据其功能进行分析，并根据其对编译结果的影响进行忽略或告警信息。GPUfusion nvcc 对参数的处理规则为：

1) “I”，表示不支持，且不影响编译正确性的，使用该参数将会直接忽略参数。
2) “W”，表示不支持，且可能会影响正确性的，使用该参数会给出警告，并丢弃该参数。
3) “E”，表示不支持，且直接影响编译行为的，直接进行报错处理。

表 2.1 参数转换表

CUDA 长选项参数	CUDA 短选项参数	参数处理情况
--allow-unsupported-compiler	-allow-unsupported-compiler	I
--archive-options	-Xarchive	W
--archiver-binary	-arbin	W
--clean-targets	-clean	W
--compile	-c	
--compile-as-tools-patch	-astoolspatch	W
--compiler-bindir	-ccbin	W
--compiler-options	-Xcompiler	
--cubin	-cubin	
--cuda	-cuda	
--cudadevrt	-cudadevrt	I
--cudart	-cudart	I
--debug	-g	
--default-stream	-default-stream	
--define-macro	-D	
--dependency-drive-prefix	-ddp	I
--dependency-output	-MF	
--dependency-target-name	-MT	
--device-c	-dc	
--device-debug	-G	
--device-link	-dlink	
--device-w	-dw	

--diag-error	-diag-error	I
--diag-suppress	-diag-suppress	I
--diag-warn	-diag-warn	I
--disable-warnings	-w	
--display-error-number	-err-no	I
--dlink-time-opt	-dlto	I
--dont-use-profile	-noprof	I
--drive-prefix	-dp	I
--dryrun	-dryrun	
--entries	-e	I
--expt-extended-lambda	-expt-extended-lambda	I
--expt-relaxed-constexpr	-expt-relaxed-constexpr	W
--extended-lambda	-extended-lambda	I
--extensible-whole-program	-ewp	I
--extra-device-vectorization	-extra-device-vectorization	I
--fatbin	-fatbin	
--fmad	-fmad	I
--forward-unknown-opts	-forward-unknown-opts	W
--forward-unknown-to-host-compiler	-forward-unknown-to-host-compiler	W
--forward-unknown-to-host-linker	-forward-unknown-to-host-linker	W
--ftemplate-backtrace-limit	-ftemplate-backtrace-limit	I
--ftemplate-depth	-ftemplate-depth	I
--ftz	-ftz	W
--generate-code	-gencode	
--generate-dependencies	-M	
--generate-dependencies-with-compile	-MD	
--generate-dependency-targets	-MP	
--generate-line-info	-lineinfo	I
--gpu-architecture	-arch	
--gpu-code	-code	
--help	-h	
--include-path	-I	
--input-drive-prefix	-idp	I
--keep	-keep	
--keep-device-functions	-keep-device-functions	W
--keep-dir	-keep-dir	W
--lib	-lib	
--libdevice-directory	-ldir	W

--library	-l	
--library-path	-L	
--link	-link	I
--linker-options	-Xlinker	
--list-gpu-arch	-arch-ls	W
--list-gpu-code	-code-ls	W
--lto	-lto	I
--m32	-m32	I
--m64	-m64	I
--machine		I
--maxrregcount	-maxrregcount	I
--no-align-double	--no-align-double	I
--no-compress	-no-compress	I
--no-device-link	-nodlink	
--no-display-error-number	-no-err-no	I
--no-exceptions	-noeh	W
--no-host-device-initializer-list	-nohdinitlist	I
--no-host-device-move-forward	-nohdmoveforward	I
--nvlink-options	-Xnvlink	W
--objdir-as-tempdir	-objtemp	W
--optimization-info	-opt-info	W
--optimize	-O	
--options-file	-optf	
--output-directory	-odir	W
--output-file	-o	
--pre-include	-include	
--prec-div	-prec-div	W
--prec-sqrt	-prec-sqrt	W
--preprocess	-E	
--profile	-pg	W
--ptx	-ptx	
--ptxas-options	-Xptxas	
--relocatable-device-code	-rdc	
--resource-usage	-res-usage	I
--restrict	-restrict	W
--run	-run	W
--run-args	-run-args	W
--save-temps	-save-temps	
--shared	-shared	
--source-in-ptx	-src-in-ptx	W
--std	-std	
--system-include	-isystem	

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--target-directory	-target-dir	W
--threads	-t	W
--time	-time	W
--undefine-macro	-U	
--use-local-env	-use-local-env	I
--use_fast_math	-use_fast_math	
--verbose	-v	
--version	-V	
--version-ident	-dQ	I
--Wdefault-stream-launch	-Wdefault-stream-launch	W
--Werror	-Werror	I
--Wext-lambda-captures-this	-Wext-lambda-captures-this	I
--Wmissing-launch-bounds	-Wmissing-launch-bounds	I
--Wno-deprecated-declarations	-Wno-deprecated-declarations	I
--Wno-deprecated-gpu-targets	-Wno-deprecated-gpu-targets	I
--Wreorder	-Wreorder	
--x	-x	I
--cuda-gpu-arch		

2.4 构建系统兼容

GPUfusion 目前已支持 CMake 的 FindCUDA，enable_language(CUDA)语言特性，可以在不修改构建系统的情况下，直接对项目进行构建。相关测试用例可以参考第三节的示例。

find_package(CUDA)特性对 CMake 3.16 至 CMake 3.24 之间的版本均支持。

enable_language(CUDA)特性在 CMake 3.16.2 到 CMake 3.24.2 范围的版本做了测试，其中通过测试的版本为 3.16.2，3.18.2，3.19.2，3.20.2，3.22.2，3.23.2，3.24.2。其他更多版本未做全面的测试，可尝试基本的用法。后续将会适配更多的 CMake 版本。

注：建议使用 CMake 3.16.2 版本。该版本经过较全面的测试。

2.5 核函数兼容

GPUfusion 支持 CUDA 的核函数调用语法。

```
kernel<<<gridSize, blockSize, shardMemorySize, Stream>>>(Args)
```

2.6 程序编译

编译期间，对于不支持的参数，会通过 warning 显示出来。比如常见的 warning 为：

```
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
```

目前所有的 CUDA GPU arch 均会转换为默认的架构名 gfx906，如果用户未设置 arch，或者 arch 中含有非 gfx906 的情型，程序将会给出 warning 以进行警示。对于未严格支持的 API，也会通过 warning 展示出来。

用户也可以通过环境变量 CUDA_STRICT 设置为严格模式，这样未严格支持的 API 会在编译期间直接报错。终止编译行为。

2.7 宏兼容

目前支持直接使用的 CUDA 原生宏如下表所示。

表 2.2 CUDA 宏兼容表

CUDA 宏	默认值	作用
__NVCC__	1	表示当前代码正在被 NVCC 编译器编译
__CUDA__	1	表示当前代码正在被 CUDA 编译器编译
__CUDACC__	1	表示当前处于主机端代码编译环境
__CUDACC_EXTENDED_LAMBDA__	未定义	当使用选项--extended-lambda 时，设置该宏值为 1，支持扩展 Lambda 表达式
__CUDA_ARCH__	520	表示当前处于设备端代码编译环境
__CUDACC_VER_MAJOR__	11	CUDA 主版本号
__CUDACC_VER_MINOR__	8	CUDA 次版本号
__CUDACC_VER_BUILD__	89	CUDA 构建号

其它 CUDA 宏相关功能不支持，暂不提供兼容。其中，__CUDA_ARCH__支持作为区分主机端和设备端代码的开关功能，且提供了默认值 750。但目前不支持根据架构型号进行功能区分，所以尽量不以__CUDA_ARCH__的值作为代码切换的条件。

3 示例

这里使用一个简单的 axpy.cu 程序作为示例，CMake 使用 3.16.2 版本：

```
#include <iostream>
__global__ void axpy(float a, float* x, float* y) {
    y[threadIdx.x] = a * x[threadIdx.x];
}
int main(int argc, char* argv[]) {
    const int kDataLen = 4;
    float a = 2.0f;
    float host_x[kDataLen] = {1.0f, 2.0f, 3.0f, 4.0f};
    float host_y[kDataLen];
    float host_y_res[kDataLen];
    // Copy input data to device.
    float* device_x;
    float* device_y;
    cudaMalloc((void**)&device_x, kDataLen * sizeof(float));
    cudaMalloc((void**)&device_y, kDataLen * sizeof(float));
    cudaMemcpy(device_x, host_x, kDataLen * sizeof(float),
                        cudaMemcpyHostToDevice);
    // Launch the kernel.
    axpy<<<1, kDataLen>>>(a, device_x, device_y);
    // Copy output data to host.
    cudaDeviceSynchronize();
    cudaMemcpy(host_y, device_y, kDataLen * sizeof(float),
                        cudaMemcpyDeviceToHost);
    for(int i=0; i< kDataLen; ++i) {
        host_y_res[i] = a * host_x[i];
    }
    int errors = 0;
```

```
    double eps = 1.0e-6;
    for(int i=0; i< kDataLen; ++i) {
        if (std::abs(host_y_res[i] - host_y[i]) > eps)
            errors++;
    }
    if (errors != 0) {
        std::cout << "FAILED: " << errors << " errors" << std::endl;
    }
    else {
        std::cout << "PASSED!"<< std::endl;
    }
    cudaDeviceReset();
    return 0;
}
```

3.1 直接编译 CUDA 程序

```
$ nvcc axpy.cu -o axpy
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
$ ./axpy
PASSED!
```

3.2 使用 FindCUDA 的 cmake 程序

3.2.1 CMakeLists.txt 内容

```
cmake_minimum_required(VERSION 3.0)
project(MyApp)
find_package(CUDA)
cuda_add_executable(axpy axpy.cu)
```

3.2.2 cmake 输出

```
-- The C compiler identification is GNU 7.3.1
```

```
-- The CXX compiler identification is GNU 7.3.1
-- Check for working C compiler: /opt/rh/devtoolset-7/root/usr/bin/cc
-- Check for working C compiler: /opt/rh/devtoolset-7/root/usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /opt/rh/devtoolset-7/root/usr/bin/c++
-- Check for working CXX compiler: /opt/rh/devtoolset-7/root/usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Found CUDA: cuda (found version "10.2")
-- Configuring done
-- Generating done
-- Build files have been written to: /root/document/build
```

3.2.3 make 及其执行结果

```
$ make
[ 50%] Building NVCC (Device) object CMakeFiles/axpy.dir/axpy_generated_axpy.cu.o
Warning: -ccbin Not Support
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
Warning: -ccbin Not Support
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
Scanning dependencies of target axpy
[100%] Linking CXX executable axpy
[100%] Built target axpy
```

```
[root@liuhw_22.10_test build]$ ./axpy
PASSED!
```

3.3 使用 enable_language 的 cmake 程序

3.3.1 CMakeLists.txt 内容

```
cmake_minimum_required(VERSION 3.0)
project(MyApp)
enable_language(CUDA)
add_executable(axpy axpy.cu)
```

3.3.2 cmake 输出

```
-- The C compiler identification is GNU 7.3.1
-- The CXX compiler identification is GNU 7.3.1
-- Check for working C compiler: /opt/rh/devtoolset-7/root/usr/bin/cc
-- Check for working C compiler: /opt/rh/devtoolset-7/root/usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /opt/rh/devtoolset-7/root/usr/bin/c++
-- Check for working CXX compiler: /opt/rh/devtoolset-7/root/usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- The CUDA compiler identification is NVIDIA 10.2.89
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc
-- Check for working CUDA compiler: /usr/local//cuda/bin/nvcc -- works
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
```

```
-- Configuring done
-- Generating done
-- Build files have been written to: /root/document/build
```

3.3.3 make 及其执行结果

```
$ make
Scanning dependencies of target axpy
[ 50%] Building CUDA object CMakeFiles/axpy.dir/axpy.cu.o
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
[100%] Linking CUDA executable axpy
[100%] Built target axpy
$ ./axpy
PASSED!
```

3.4 使用 project 设置语言的 cmake 程序

3.4.1 CMakeLists.txt 内容

```
cmake_minimum_required(VERSION 3.0)
project(MyApp CUDA)
add_executable(axpy axpy.cu)
```

3.4.2 cmake 输出

```
-- The CUDA compiler identification is NVIDIA 10.2.89
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc -- works
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
-- Configuring done
-- Generating done
-- Build files have been written to: /root/document/build
```

3.4.3 make 及其执行结果

```
$ make
Scanning dependencies of target axpy
[ 50%] Building CUDA object CMakeFiles/axpy.dir/axpy.cu.o
Warning: nvcc current only support gfx906 and gfx926 arch. All architecture parameters will be replaced by gfx906 arch.
[100%] Linking CUDA executable axpy
[100%] Built target axpy
$ ./axpy
PASSED!
```

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4 CUDA 程序适配

从 2 和 3 两节可以看出，在配置好环境变量之后，GPUfusion 提供了和原生 CUDA 基本一致的用法，能够满足 CUDA 用户的基本需求。但仍有部分用法存在差异，这些地方暂时不能很好兼容，本节将针对这些进行说明并给出修改建议。

4.1 适配流程

4.1.1 程序构建

- 基于 Makefile

对于以 Makefile 方式构建的应用来说，相关的 cuda 路径已经固化在 Makefile 中，因此需要用户手动进行替换。用户需要手动替换 cuda path，以及以 cuda path 为基础的 nvcc 路径和 libs 路径。

- 基于 CMake

GPUfusion 支持 find_package(CUDA)和 enable_language(CUDA)的 CMake 特性。可以在不修改任何构建系统的基础上，引导 cmake 将相关目录指向 GPUfusion 的文件夹。使用方法与原生 CUDA 一致。

由于 CMake 版本众多，且在不断迭代更新中，不能保证每一个版本都能完全适用。像最新版本的 CMake，虽然也支持简单的用法，但部分复杂特性的使用尚未充分验证。因此建议参考表 4.1 中已验证过的版本中进行选择，优先建议使用 3.16.2 版本。

表 4.1 GPUfusion 支持的 CMake 版本

VERSION	3.16.2	3.17.2	3.18.2	3.19.2	3.20.2	3.21.2	3.22.2	3.23.2	3.24.2
find_package(CUDA)	支持	支持	支持	支持	支持	不支持	支持	支持	支持
enable_language(CUDA)	支持	不支持	支持	支持	支持	不支持	支持	支持	支持

对于 CMake3.18 之后的版本，不建议用户自己用 CMAKE_CUDA_ARCHITECTURES 设置架构，GPUfusion 已提供默认架构值 75。

另外，CMake3.21 部分子版本对异构编译器的识别存在问题，CMake 官方后续版本也对该版本的问题进行了说明。GPUfusion 依赖于 DTK 环境，所以也受到该问题的影响。请用户尽量避开使用该版本的 CMake。

```
The AMD ROCm Platform ``hipcc`` compiler was identifed by CMake 3.21.0 through 3.21.2 as a distinct compiler with id ``ROCMClang``. This has been removed because it caused regressions.
```

4.1.2 程序编译

在程序编译过程中如果出现错误，建议先使用单线程输出详细信息，以输出真正错误处的构建指令，方便问题分析。指令如下：

```
make VERBOSE=1
```

注：GPUfusion 目前支持部分内嵌 PTX。若报错 PTX 指令不支持，需要用户手动将其修改为等价的 C++语言。具体 PTX 语法请参考 CUDA 官方相关文档。

4.1.3 程序运行

GPUfusion 为了程序最高的效率，默认线程的最大值为 256，在程序运行中如果出现以下提示，则需要调整最大值的大小。

```
Launch params (1024, 1, 1) are larger than launch bounds (256) for kernel _ZL14timedReductionPKfPfPl please add __launch_bounds__ to kernel define or use --gpu-max-threads-per-block recompile program !
```

修改的方法有两种：

1. 在编译命令中添加如下参数，然后重新编译：

```
--gpu-max-threads-per-block=1024
```

2. 在核函数头部添加：__launch_bounds__(1024)。示例如下：

```
__launch_bounds__(1024)
__global__ static void timedReduction(const float *input, float *output) {
```

4.1.4 运行结果检查

如果程序运行结果不正确，此时需要检查程序中 warp size 的设置。默认原生 CUDA 的 warp size 为 32，GPUfusion 适用的 warp size 为 64。用户需要确保项目中所有的 warp size 修改为 64，以保证程序的正确性。

以上是通用的 CUDA 程序移植检查流程，下面 4.2 节，将更详细的介绍与原生 CUDA 用法的一些差异，并给出修改建议。

4.2 原生 CUDA 用法差异与修改建议

4.2.1 接口功能差异

1. runtime/drive

（1） cudaGetDeviceProperties 接口获取设备属性，进而打印 prop.name（设备名称和设备架构信息），可以同时获取到设备名称和设备架构信息。

（2） cuCtxCreate_v3、cuCtxGetExecAffinity、cuDeviceGetExecAffinitySupport 接口受限于硬件，目前不支持执行亲和性特性。

（3） cudaDeviceGetAttribute 接口部分属性不支持，包括 cudaDevAttrCanUseHostPointerForRegisteredMem、cudaDevAttrComputePreemptionSupported、cudaDevAttrGlobalL1CacheSupported 等。

2. nvrtc

（1） 使用 nvrtcGetPTXSize 和 nvrtcGetPTX 接口，需要在调用 nvrtcCompileProgram 时编译选项添加"-fgpu-rdc"选项。

（2） 使用 nvrtcAddNameExpression 和 nvrtcGetLoweredName 接口，不能在调用 nvrtcCompileProgram 时添加"-fgpu-rdc"选项。

（3） 调用 nvrtcCreateProgram 时，第二个参数指向的源码字符串不能为空字符串，否则会报错"NVRTC_ERROR_INVALID_INPUT"。

（4） cuModuleLoadDataEx 目前仅支持加载 CUBIN 类型的二进制数据，即 nvrtcGetCUBIN 得到的二进制数据可以通过 cuModuleLoadDataEx 加载到 CUmodule 中，不支持从 PTX 中加载。且在调用 nvrtcCompileProgram 时不能添加"-fgpu-rdc"选项。

3. 数学库

（1） cublasLtOrder_t 属性只支持 CUBLASLT_ORDER_COL，其它属性暂不支持（矩阵 CUBLASLT_ORDER_ROW 属性目前是转成矩阵 CUBLASLT_ORDER_COL 属性处理的）

（2） cublasXt 目前是单卡实现，cublasXt<T>spmm 只支持 csr 格式

（3） cublasLtMatmulDescCreate 功能实现与 cuda 有差异

（4） Cufft 不支持获取 getworksize

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（5） cufftXt 不支持空流操作，cufftXtMalloc 实数正反变化不支持，目前只支持 CUFFT_XT_FOTMAT_INPLACE ， CUFFT_XT_FOTMAT_INPUT ， CUFFT_XT_FOTMAT_OUTPUT，三种类型。

4. device 函数

GPUfusion 算力当前大致对标 sm_75，不支持或者有差异的 device 函数主要包括如下几个。

（1） atomicDec_system、atomicInc_system 由 atomicCAS_system 软实现，性能较差；

（2） mma、awbarrier 等函数不支持，详细看附录。

4.2.2 分离编译

如果项目中存在 cuda 源文件之间 device 函数符号相互调用的情况，编译时需要主动开启分离编译特性。否则，在编译过程中会出现 device 函数符号找不到的情况。

分离编译特性一般依赖于“-dc”和“-dlink”选项，基本用法如下。详情可参考 CUDA 官网的描述。

```
$ nvcc --gpu-architecture=sm_50 --device-c a.cu b.cu
$ nvcc --gpu-architecture=sm_50 --device-link a.o b.o --output-file link.o
$ g++ a.o b.o link.o --library-path=/usr/local/cuda/lib64 --library=cudart
```

如果想让 CMake 项目开启分离编译特性，可参考如下两种方式。

```
# enable_language(CUDA)的形式
set_target_properties(tgt PROPERTIES CUDA_SEPARABLE_COMPILATION ON)
set_target_properties(tgt PROPERTIES CUDA_RESOLVE_DEVICE_SYMBOLS ON)
```

或者

```
# find_package(CUDA)的形式
SET(CUDA_SEPARABLE_COMPILATION ON)
SET(CUDA_RESOLVE_DEVICE_SYMBOLS ON)
```

构建流程中会生成 cmake_device_link.o 等分离编译的中间产物，最终被链接成可执行文件或者动态库的形式。

GPUfusion 支持 CUDA 原生分离编译的基本用法。但相比于原生用法，GPUfusion 需要用户额外指定链接器为 nvcc。手动编译和 makefile 指定较为简单，这里主要对 CMake 的用法进行说明。

若没有按照 GPUfusion 的要求进行分离编译配置，往往会有以下类似的报错。

1）程序链接时

```
undefined reference to `__hip_fatbin'
```

2）链接时没报错，但运行时明显报错核函数找不到

```
Cannot find Symbol with name: vectorAddKernel
```

若指定默认编译器为 nvcc，则能够自动支持分离编译功能。

```
cmake ../ -DCMAKE_CXX_COMPILER=nvcc
```

若不指定默认编译器为 nvcc，那么需要指定链接器为 nvcc。

1）当 target 是可执行程序时

```
cmake -DCMAKE_LINKER=nvcc -DCMAKE_CXX_LINK_EXECUTABLE="<CMAKE_LINKER> <FLAGS> <CMAKE_CXX_LINK_FLAGS> <LINK_FLAGS> <OBJECTS> -o <TARGET> <LINK_LIBRARIES>" ..
```

也可以直接在 cmake 文件中添加如下代码，来指定链接器为 nvcc。

```
set(CMAKE_LINKER $ENV{CUDA_PATH}/bin/nvcc)
set(CMAKE_CXX_LINK_EXECUTABLE "<CMAKE_LINKER> <FLAGS> <CMAKE_CXX_LINK_FLAGS> <LINK_FLAGS> <OBJECTS> -o <TARGET> <LINK_LIBRARIES>")
```

2）当 target 是动态库时

```
cmake -DCMAKE_LINKER=nvcc -DCMAKE_CXX_CREATE_SHARED_LIBRARY="<CMAKE_LINKER> <CMAKE_SHARED_LIBRARY_CXX_FLAGS> <LANGUAGE_COMPILE_FLAGS> <LINK_FLAGS> <CMAKE_SHARED_LIBRARY_CREATE_CXX_FLAGS> <SONAME_FLAG><TARGET_SONAME> -o <TARGET> <OBJECTS> <LINK_LIBRARIES>" ..
```

或者

```
set(CMAKE_LINKER $ENV{CUDA_PATH}/bin/nvcc)
set(CMAKE_CXX_CREATE_SHARED_LIBRARY
```

```
"<CMAKE_LINKER> <CMAKE_SHARED_LIBRARY_CXX_FLAGS>
<LANGUAGE_COMPILE_FLAGS> <LINK_FLAGS>
<CMAKE_SHARED_LIBRARY_CREATE_CXX_FLAGS>
<SONAME_FLAG><TARGET_SONAME> -o <TARGET> <OBJECTS>
<LINK_LIBRARIES>")
```

4.2.3 host 和 device 关键字

GPUfusion nvcc 底层依赖于 clang 进行编译，而原生 nvcc 和 clang 在编译模型上有差别，这使得部分编译场景的处理也会存在差异。具体可参考 clang cuda 官网对差异的详细描述。

目前影响较大的是对__host__和__device__关键字处理的差异，GPUfusion 的语法检查更加严格，以下几种场景会有编译报错，暂时希望用户能够对代码进行调整。

情况 1：编译报错“call to __device__ function from __host__ function”

代码示例

```
void hostFunc() {
    float b = cos(3);
}
```

编译器认为存在 host 函数调用 device 函数。当缺少 C++标准库头文件引入时，编译器判断用户想在 host 函数中调用 device 的标准函数，从而报错。这里应该主动引入“math.h”头文件。

修改建议：正确的引入对应的标准库头文件。目前部分标准库头文件已经支持自动引入。

情况 2：编译报错“call to __host__ function from __global__/__device__ function”

代码示例

```
__host__ void testFunc() {}

#ifdef __CUDA_ARCH__
__device__ void testFunc() {}
#endif

__global__ void kernelFunc() {
```

```
    testFunc();
}
```

编译器认为存在 global 函数或者 device 函数调用 host 函数，需要检查对应的 device 函数定义是否存在。在 host 编译模式时，没有__CUDA_ARCH__宏定义，所以缺少了 testFunc 函数的 device 端定义。

修改建议：在这种情况下，可以使用__CUDACC__作为宏判断条件，或者不需要宏判断条件。

情况 3：编译报错“reference to __host__ function in __host__ __device__ function ”

代码示例

```
__host__ __device__void Func() {
    // host  数组
    vector<int> host_array;
    // device  数组
    int *device_array[100];
}
```

编译器认为函数被定义成__host__ __deivce__的形式后，应该能够同时支持 host 端和 device 端的编译以及语法检查，否则编译报错。

修改建议：在这种情况下，可以使用__CUDA_ARCH__对代码进行隔离。不要存在 host 调用 device 或者 device 调用 host 的情况。如果是类或者结构体中的多个成员函数涉及这种情况，建议用户彻底分离 host 和 device 的成分，减少共用结构体或者函数的情况。

```
__host__ __device__void Func() {
#ifdef __CUDA_ARCH__
    int *device_array[100];
#else
    vector<int> host_array;
#endif
}
```

4.2.4 C++标准库以及工具链

GPUfusion nvcc 默认的 C++标准是 11。在使用 GPUfusion 的 thrust、cub 等库时，请确保默认的 C++标准在 14 以上，否则部分高级的 C++语法特性不能被正常识别。

另外，若编译中存在和 C++标准库成员缺失相关的报错，请检查是否为项目指定了正确的 C++标准。

```
error: ‘forward’ is not a member of ‘std’
```

基础环境也尽量在 g++7 以上，以免部分高级的 C++语法特性不支持。

若环境中的 gcc 安装路径不是系统路径，那么最好使用“--gcc-toolchain”选项，显示指明 gcc 安装路径，以便于 nvcc 能够正确的找到 gnu 工具链。

4.2.5 文件编译模式

在选择编译模式的时候，GPUfusion nvcc 已默认和原生 nvcc 一致，根据文件后缀或者“-x”选项进行判断，而不是默认用异构模式进行编译。

目前也支持引入 CUDA 头文件的 cpp 文件被 g++或者其它编译器编译。因此除了分离编译的情况外，尽量不要用 GPUfusion nvcc 去编译非 cu 后缀的文件。从而减少编译器差异导致的报错。

如果用户想强制用 CUDA 模式编译文件，可以直接加上“-x cu”的选项或者 cmake 文件中添加“add_compile_options(-xcu)”。

4.2.6 静态库

GPUfusion 目前不支持提供静态库，例如 libcudart_static.a 等。如果项目中有用到，那么可以替换成等效的动态库链接形式。

4.2.7 VLA 类型初始化

GPUfusion nvcc 目前不支持 VLA 类型初始化，可用 auto 进行规避。

代码示例

```
static void resid(void* pointer_u, void* pointer_v, void* pointer_r, int n1, int n2, int n3, double a[4], int k){
    // compile fail
    double (*u)[n2][n1] = (double (*)[n2][n1])pointer_u;
    double (*v)[n2][n1] = (double (*)[n2][n1])pointer_v;
```

```
        double (*r)[n2][n1] = (double (*)[n2][n1])pointer_r;
        // compile success
        auto u = (double (*)[n2][n1])pointer_u;
        auto v = (double (*)[n2][n1])pointer_v;
        auto r= (double (*)[n2][n1])pointer_r;
}
```

4.2.8 Share 类型变量初始化

GPUfusion nvcc 仅支持基础类型的 share 变量初始化。对于声明为 share 的类或者结构体，仅支持构造函数为空的情况，不支持非空构造函数。

而原生 nvcc 也仅支持上述情况编译通过，实际并未调用非空构造函数，正确性无法保证。因此不建议使用这种方式。

4.2.9 Warpsize 差异

CUDA 程序隐式地利用 warp 执行来获得加速，但同样可以显式地使用 warp 级编程，来获得更高的性能。但 CUDA 线程束大小为 32，GPUfusion 线程束大小为 64。当进行 CUDA 程序移植时，若不修改用户的代码逻辑，直接把 warp-32 逻辑的 CUDA 程序运行在 GPUfusion（warp-64）上，会导致计算结果错误，甚至崩溃。

当前版本兼容了用户以 CUDA warpsize 取 32 编写的代码，需要在编译时加上“-D GPUFUSION_ENABLE_WARPSIZE32”。该方法仅作为实验性质的功能，软兼容会有较大性能损耗。GPUfusion 默认 warpsize 为 64，建议用户按照 64 的逻辑来修改代码，以达到最优性能。

目前已知该方案无法适用于“warp 间存在数据依赖”的场景。

4.3 已知限制

1) 不支持 Dynamic Parallel 功能。
2) GPUfusion 默认的 Warp Size 为 64，移植时需要注意调整。
3) 构建编译环境的 GCC 版本应该不低于 7.0，且保证编译时高版本的 C、C++标准头文件和库被优先选择；cmake 最好参考表 4.1 中的推荐版本。
4) 尽管当前对标 CUDA 版本为 12.6，但像 libcublasLt.so、libcufftw.so 等非基础库，目前还未支持或支持效果不好。若涉及相关接口的调用，请先尝试用其它基础接口进

行替换。另外，目前已删除对静态库的支持。

5) 不支持 CUDA 和 HIP 混合使用。

6) 内嵌 PTX 功能开启需要主动加“-finline-asm-ptx”选项。

7) 使用 cupti profiler 接口时，需要在编译好的程序前设置环境变量 HSA_TOOLS_LIB=librocprofiler64v2.so,才能获取性能信息。

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5 兼容性支持列表

支持的函数：提供完整的函数声明与定义，并支持函数正常调用。

未支持的函数：提供函数声明与定义，但函数调用时提示 unsupported。

5.1 Runtime 函数

5.1.1 支持的函数

cudaGraphKernelNodeGetParams	cudaGraphKernelNodeSetAttribute	cudaGraphKernelNodeSetParams
cudaGraphLaunch	cudaGraphLaunch_ptsz	cudaGraphMemAllocNodeGetParams
cudaGraphMemFreeNodeGetParams	cudaGraphMemcpyNodeGetParams	cudaGraphMemcpyNodeSetParams
cudaGraphMemcpyNodeSetParams1D	cudaGraphMemcpyNodeSetParamsFromSymbol	cudaGraphMemcpyNodeSetParamsToSymbol
cudaGraphMemsetNodeGetParams	cudaGraphMemsetNodeSetParams	cudaGraphNodeFindInClone
cudaGraphNodeGetDependencies	cudaGraphNodeGetDependencies_v2	cudaGraphNodeGetDependentNodes
cudaGraphNodeGetDependentNodes_v2	cudaGraphNodeGetEnabled	cudaGraphNodeGetType
cudaGraphNodeSetEnabled	cudaGraphNodeSetParams	cudaGraphReleaseUserObject
cudaGraphRemoveDependencies	cudaGraphRemoveDependencies_v2	cudaGraphRetainUserObject
cudaGraphUpload	cudaGraphUpload_ptsz	cudaHostAlloc
cudaHostGetDevicePointer	cudaHostGetFlags	cudaHostRegister
cudaHostUnregister	cudaImportExternalMemory	cudaImportExternalSemaphore
cudaInitDevice	cudaIpcCloseMemHandle	cudaIpcGetEventHandle

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cudaIpcGetMemHandle	cudaIpcOpenEventHandle	cudaIpcOpenMemHandle
cudaLaunchCooperativeKernel	cudaLaunchCooperativeKernelMultiDevice	cudaLaunchCooperativeKernel_ptsz
cudaLaunchHostFunc	cudaLaunchHostFunc_ptsz	cudaLaunchKernel
cudaLaunchKernelExC	cudaLaunchKernelExC_ptsz	cudaLaunchKernel_ptsz
cudaMalloc	cudaMalloc3D	cudaMalloc3DArray
cudaMallocArray	cudaMallocAsync	cudaMallocAsync_ptsz
cudaMallocFromPoolAsync	cudaMallocFromPoolAsync_ptsz	cudaMallocHost
cudaMallocManaged	cudaMallocMipmappedArray	cudaMallocPitch
cudaMemAdvise	cudaMemAdvise_v2	cudaMemGetInfo
cudaMemPoolDestroy	cudaMemPoolExportPointer	cudaMemPoolExportToShareableHandle
cudaMemPoolGetAccess	cudaMemPoolGetAttribute	cudaMemPoolImportFromShareableHandle
cudaMemPoolImportPointer	cudaMemPoolSetAccess	cudaMemPoolSetAttribute
cudaMemPoolTrimTo	cudaMemPrefetchAsync	cudaMemPrefetchAsync_ptsz
cudaMemPrefetchAsync_v2	cudaMemPrefetchAsync_v2_ptsz	cudaMemRangeGetAttribute
cudaMemRangeGetAttributes	cudaMemcpy	cudaMemcpy2D
cudaMemcpy2DArrayToArray	cudaMemcpy2DArrayToArray_ptds	cudaMemcpy2DAsync
cudaMemcpy2DAsync_ptsz	cudaMemcpy2DFromArray	cudaMemcpy2DFromArrayAsync
cudaMemcpy2DFromArrayAsync_ptsz	cudaMemcpy2DFromArray_ptds	cudaMemcpy2DToArray
cudaMemcpy2DToArrayAsync	cudaMemcpy2DToArrayAsync_ptsz	cudaMemcpy2DToArray_ptds
cudaMemcpy2D_ptds	cudaMemcpy3D	cudaMemcpy3DAsync

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cudaMemcpy3DAsync_ptsz	cudaMemcpy3DPeer	cudaMemcpy3DPeerAsync
cudaMemcpy3DPeerAsync_ptsz	cudaMemcpy3DPeer_ptds	cudaMemcpy3D_ptds
cudaMemcpyArrayToArray	cudaMemcpyArrayToArray_ptds	cudaMemcpyAsync
cudaMemcpyAsync_ptsz	cudaMemcpyFromArray	cudaMemcpyFromArrayAsync
cudaMemcpyFromArrayAsync_ptsz	cudaMemcpyFromArray_ptds	cudaMemcpyFromSymbol
cudaMemcpyFromSymbolAsync	cudaMemcpyFromSymbolAsync_ptsz	cudaMemcpyFromSymbol_ptds
cudaMemcpyPeer	cudaMemcpyPeerAsync	cudaMemcpyToArray
cudaMemcpyToArrayAsync	cudaMemcpyToArrayAsync_ptsz	cudaMemcpyToArray_ptds
cudaMemcpyToSymbol	cudaMemcpyToSymbolAsync	cudaMemcpyToSymbolAsync_ptsz
cudaMemcpyToSymbol_ptds	cudaMemcpy_ptds	cudaMemset
cudaMemset2D	cudaMemset2DAsync	cudaMemset2DAsync_ptsz
cudaMemset2D_ptds	cudaMemset3D	cudaMemset3DAsync
cudaMemset3DAsync_ptsz	cudaMemset3D_ptds	cudaMemsetAsync
cudaMemsetAsync_ptsz	cudaMemset_ptds	cudaMipmappedArrayGetMemoryRequirements
cudaOccupancyMaxActiveBlocksPerMultiprocessor	cudaOccupancyMaxActiveBlocksPerMultiprocessorWithFlags	cudaPeekAtLastError
cudaPointerGetAttributes	cudaProfilerStart	cudaProfilerStop
cudaRuntimeGetVersion	cudaSetDevice	cudaSetDeviceFlags
cudaSetValidDevices	cudaSignalExternalSemaphoresAsync	cudaSignalExternalSemaphoresAsync_v2
cudaSignalExternalSemaphoresAsync_v2_ptsz	cudaStreamAddCallback	cudaStreamAddCallback_ptsz

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cudaStreamAttachMemAsync	cudaStreamAttachMemAsync_ptsz	cudaStreamBeginCapture
cudaStreamBeginCaptureToGraph	cudaStreamBeginCaptureToGraph_ptsz	cudaStreamBeginCapture_ptsz
cudaStreamCopyAttributes	cudaStreamCopyAttributes_ptsz	cudaStreamCreate
cudaStreamCreateWithFlags	cudaStreamCreateWithPriority	cudaStreamDestroy
cudaStreamEndCapture	cudaStreamEndCapture_ptsz	cudaStreamGetAttribute
cudaStreamGetAttribute_ptsz	cudaStreamGetCaptureInfo	cudaStreamGetCaptureInfo_v2
cudaStreamGetCaptureInfo_v2_ptsz	cudaStreamGetCaptureInfo_v3	cudaStreamGetCaptureInfo_v3_ptsz
cudaStreamGetFlags	cudaStreamGetFlags_ptsz	cudaStreamGetId
cudaStreamGetId_ptsz	cudaStreamGetPriority	cudaStreamGetPriority_ptsz
cudaStreamIsCapturing	cudaStreamIsCapturing_ptsz	cudaStreamQuery
cudaStreamQuery_ptsz	cudaStreamSetAttribute	cudaStreamSetAttribute_ptsz
cudaStreamSynchronize	cudaStreamSynchronize_ptsz	cudaStreamUpdateCaptureDependencies
cudaStreamUpdateCaptureDependencies_v2	cudaStreamUpdateCaptureDependencies_v2_ptsz	cudaStreamWaitEvent
cudaStreamWaitEvent_ptsz	cudaThreadExchangeStreamCaptureMode	cudaThreadExit
cudaThreadGetCacheConfig	cudaThreadGetLimit	cudaThreadSetCacheConfig
cudaThreadSetLimit	cudaThreadSynchronize	cudaUserObjectCreate
cudaUserObjectRelease	cudaUserObjectRetain	cudaWaitExternalSemaphoresAsync
cudaWaitExternalSemaphoresAsync_v2	cudaWaitExternalSemaphoresAsync_v2_ptsz	

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5.1.2 未支持的函数

cudaArrayGetPlane	cudaArrayGetSparseProperties	cudaCtxResetPersistingL2Cache
cudaDeviceFlushGPUDirectRDMAWrites	cudaDeviceGetNvSciSyncAttributes	cudaDeviceGetTexture1DLinearMaxWidth
cudaGLGetDevices	cudaGetExportTable	cudaGraphicsGLRegisterBuffer
cudaGraphicsGLRegisterImage	cudaGraphicsMapResources	cudaGraphicsResourceGetMappedMipmappedArray
cudaGraphicsResourceGetMappedPointer	cudaGraphicsResourceSetMapFlags	cudaGraphicsSubResourceGetMappedArray
cudaGraphicsUnmapResources	cudaGraphicsUnregisterResource	cudaMipmappedArrayGetSparseProperties
cudaOccupancyAvailableDynamicSMemPerBlock	cudaOccupancyMaxActiveClusters	cudaOccupancyMaxPotentialClusterSize
cudaSetDoubleForDevice	cudaSetDoubleForHost	

5.2 Device 函数

5.2.1 支持的函数

__all	__any	__assert_fail	__assertfail	__ballot
__bfloat1622float2	__bfloat162bfloat162	__bfloat162float	__bfloat162int_rd	__bfloat162int_rn
__bfloat162int_ru	__bfloat162int_rz	__bfloat162ll_rd	__bfloat162ll_rn	__bfloat162ll_ru
__bfloat162ll_rz	__bfloat162short_rd	__bfloat162short_rn	__bfloat162short_ru	__bfloat162short_rz
__bfloat162uint_rd	__bfloat162uint_rn	__bfloat162uint_ru	__bfloat162uint_rz	__bfloat162ull_rd
__bfloat162ull_rn	__bfloat162ull_ru	__bfloat162ull_rz	__bfloat162ushort_rd	__bfloat162ushort_rn
__bfloat162ushort	__bfloat162ushort	__bfloat16_as_sh	__bfloat16_as_us	__brev

_ru	_rz	ort	hort	
__brevll	__brkpt	__byte_perm	__clz	__clzll
__cosf	__cvta_constant_to_generic	__cvta_generic_to_constant	__cvta_generic_to_global	__cvta_generic_to_local
__cvta_generic_to_shared	__cvta_global_to_generic	__cvta_local_to_generic	__cvta_shared_to_generic	__dadd_rd
__dadd_rn	__dadd_ru	__dadd_rz	__ddiv_rd	__ddiv_rn
__ddiv_ru	__ddiv_rz	__dmul_rd	__dmul_rn	__dmul_ru
__dmul_rz	__double2bfloat16	__double2float_rd	__double2float_rn	__double2float_ru
__double2float_rz	__double2half	__double2hiint	__double2int_rd	__double2int_rn
__double2int_ru	__double2int_rz	__double2ll_rd	__double2ll_rn	__double2ll_ru
__double2ll_rz	__double2loint	__double2uint_rd	__double2uint_rn	__double2uint_ru
__double2uint_rz	__double2ull_rd	__double2ull_rn	__double2ull_ru	__double2ull_rz
__double_as_longlong	__dp2a_hi	__dp2a_lo	__dp4a	__drcp_rd
__drcp_rn	__drcp_ru	__drcp_rz	__dsqrt_rd	__dsqrt_rn
__dsqrt_ru	__dsqrt_rz	__dsub_rd	__dsub_rn	__dsub_ru
__dsub_rz	__exp10f	__expf	__fadd_rd	__fadd_rn
__fadd_ru	__fadd_rz	__fdiv_rd	__fdiv_rn	__fdiv_ru
__fdiv_rz	__fdividef	__ffs	__ffsll	__finite
__finitef	__finitel	__float22bfloat162_rn	__float22half2_rn	__float2bfloat16
__float2bfloat162_rn	__float2bfloat16_rd	__float2bfloat16_rn	__float2bfloat16_ru	__float2bfloat16_rz
__float2half	__float2half2_rn	__float2half_rd	__float2half_rn	__float2half_ru
__float2half_rz	__float2int_rd	__float2int_rn	__float2int_ru	__float2int_rz
__float2ll_rd	__float2ll_rn	__float2ll_ru	__float2ll_rz	__float2uint_rd

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__float2uint_rn	__float2uint_ru	__float2uint_rz	__float2ull_rd	__float2ull_rn
__float2ull_ru	__float2ull_rz	__float_as_int	__float_as_uint	__floats2bfloat162_rn
__floats2half2_rn	__fma_rd	__fma_rn	__fma_ru	__fma_rz
__fmaf_ieee_rd	__fmaf_ieee_rn	__fmaf_ieee_ru	__fmaf_ieee_rz	__fmaf_rd
__fmaf_rn	__fmaf_ru	__fmaf_rz	__fmul_rd	__fmul_rn
__fmul_ru	__fmul_rz	__fns	__frcp_rd	__frcp_rn
__frcp_ru	__frcp_rz	__frsqrt_rn	__fsqrt_rd	__fsqrt_rn
__fsqrt_ru	__fsqrt_rz	__fsub_rd	__fsub_rn	__fsub_ru
__fsub_rz	__funnelshift_l	__funnelshift_lc	__funnelshift_r	__funnelshift_rc
__h2div	__habs	__habs2	__hadd	__hadd2
__hadd2_rn	__hadd2_sat	__hadd_rn	__hadd_sat	__half22float2
__half2float	__half2half2	__half2int_rd	__half2int_rn	__half2int_ru
__half2int_rz	__half2ll_rd	__half2ll_rn	__half2ll_ru	__half2ll_rz
__half2short_rd	__half2short_rn	__half2short_ru	__half2short_rz	__half2uint_rd
__half2uint_rn	__half2uint_ru	__half2uint_rz	__half2ull_rd	__half2ull_rn
__half2ull_ru	__half2ull_rz	__half2ushort_rd	__half2ushort_rn	__half2ushort_ru
__half2ushort_rz	__half_as_short	__half_as_ushort	__halves2bfloat162	__halves2half2
__hbeq2	__hbequ2	__hbge2	__hbgeu2	__hbgt2
__hbgtu2	__hble2	__hbleu2	__hblt2	__hbltu2
__hbne2	__hbneu2	__hcmadd	__hdiv	__heq
__heq2	__hequ	__hequ2	__hfma	__hfma2
__hfma2_relu	__hfma2_sat	__hfma_relu	__hfma_sat	__hge
__hge2	__hgeu	__hgeu2	__hgt	__hgt2
__hgtu	__hgtu2	__high2bfloat16	__high2bfloat162	__high2float
__high2half	__high2half2	__highs2bfloat162	__highs2half2	__hiloint2double

__hisinf	__hisnan	__hisnan2	__hle	__hle2
__hleu	__hleu2	__hlt	__hlt2	__hltu
__hltu2	__hmax	__hmax2	__hmax2_nan	__hmax_nan
__hmin	__hmin2	__hmin2_nan	__hmin_nan	__hmul
__hmul2	__hmul2_rn	__hmul2_sat	__hmul_rn	__hmul_sat
__hne	__hne2	__hneg	__hneg2	__hneu
__hneu2	__hsub	__hsub2	__hsub2_rn	__hsub2_sat
__hsub_rn	__hsub_sat	__int2bfloat16_rd	__int2bfloat16_rn	__int2bfloat16_ru
__int2bfloat16_rz	__int2double_rn	__int2float_rd	__int2float_rn	__int2float_ru
__int2float_rz	__int2half_rd	__int2half_rn	__int2half_ru	__int2half_rz
__int_as_float	__isGlobal	__isLocal	__isShared	__isinf
__isinff	__isinfl	__isnan	__isnanf	__isnanl
__ldca	__ldcg	__ldcs	__ldcv	__ldg
__ldlu	__ll2bfloat16_rd	__ll2bfloat16_rn	__ll2bfloat16_ru	__ll2bfloat16_rz
__ll2double_rd	__ll2double_rn	__ll2double_ru	__ll2double_rz	__ll2float_rd
__ll2float_rn	__ll2float_ru	__ll2float_rz	__ll2half_rd	__ll2half_rn
__ll2half_ru	__ll2half_rz	__log10f	__log2f	__logf
__longlong_as_double	__low2bfloat16	__low2bfloat162	__low2float	__low2half
__low2half2	__lowhigh2highlow	__lows2bfloat162	__lows2half2	__mul24
__mul64hi	__mulhi	__popc	__popcll	__powf
__rhadd	__sad	__saturatef	__shfl	__shfl_down
__shfl_up	__shfl_xor	__short2bfloat16_rd	__short2bfloat16_rn	__short2bfloat16_ru
__short2bfloat16_rz	__short2half_rd	__short2half_rn	__short2half_ru	__short2half_rz
__short_as_bfloat	__short_as_half	__signbit	__signbitf	__signbitl

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__sincosf	__sinf	__stcg	__stcs	__stwb
__stwt	__syncthreads	__syncthreads_and	__syncthreads_count	__syncthreads_or
__tanf	__threadfence	__threadfence_block	__threadfence_system	__trap
__uhadd	__uint2bfloat16_rd	__uint2bfloat16_rn	__uint2bfloat16_ru	__uint2bfloat16_rz
__uint2double_rn	__uint2float_rd	__uint2float_rn	__uint2float_ru	__uint2float_rz
__uint2half_rd	__uint2half_rn	__uint2half_ru	__uint2half_rz	__uint_as_float
__ull2bfloat16_rd	__ull2bfloat16_rn	__ull2bfloat16_ru	__ull2bfloat16_rz	__ull2double_rd
__ull2double_rn	__ull2double_ru	__ull2double_rz	__ull2float_rd	__ull2float_rn
__ull2float_ru	__ull2float_rz	__ull2half_rd	__ull2half_rn	__ull2half_ru
__ull2half_rz	__umul24	__umul64hi	__umulhi	__urhadd
__usad	__ushort2bfloat16_rd	__ushort2bfloat16_rn	__ushort2bfloat16_ru	__ushort2bfloat16_rz
__ushort2half_rd	__ushort2half_rn	__ushort2half_ru	__ushort2half_rz	__ushort_as_bfloat16
__ushort_as_half	__vabs2	__vabs4	__vabsdiffs2	__vabsdiffs4
__vabsdiffu2	__vabsdiffu4	__vabsss2	__vabsss4	__vadd2
__vadd4	__vaddss2	__vaddss4	__vaddus2	__vaddus4
__vavgs2	__vavgs4	__vavgu2	__vavgu4	__vcmpeq2
__vcmpeq4	__vcmpges2	__vcmpges4	__vcmpgeu2	__vcmpgeu4
__vcmpgts2	__vcmpgts4	__vcmpgtu2	__vcmpgtu4	__vcmples2
__vcmples4	__vcmpleu2	__vcmpleu4	__vcmplts2	__vcmplts4
__vcmpltu2	__vcmpltu4	__vcmpne2	__vcmpne4	__vhaddu2
__vhaddu4	__vmaxs2	__vmaxs4	__vmaxu2	__vmaxu4
__vmins2	__vmins4	__vminu2	__vminu4	__vneg2

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__vneg4	__vnegss2	__vnegss4	__vsads2	__vsads4
__vsadu2	__vsadu4	__vseteq2	__vseteq4	__vsetges2
__vsetges4	__vsetgeu2	__vsetgeu4	__vsetgts2	__vsetgts4
__vsetgtu2	__vsetgtu4	__vsetles2	__vsetles4	__vsetleu2
__vsetleu4	__vsetlts2	__vsetlts4	__vsetltu2	__vsetltu4
__vsetne2	__vsetne4	__vsub2	__vsub4	__vsubss2
__vsubss4	__vsubus2	__vsubus4	abs	acos
acosf	acosh	acoshf	asin	asinf
asinh	asinhf	atan	atan2	atan2f
atanf	atanh	atanhf	atomicAdd	atomicAdd_block
atomicAdd_system	atomicAnd	atomicAnd_block	atomicAnd_system	atomicCAS
atomicCAS_block	atomicCAS_system	atomicDec	atomicDec_block	atomicDec_system
atomicExch	atomicExch_block	atomicExch_system	atomicInc	atomicInc_block
atomicInc_system	atomicMax	atomicMax_block	atomicMax_system	atomicMin
atomicMin_block	atomicMin_system	atomicOr	atomicOr_block	atomicOr_system
atomicSub	atomicSub_block	atomicSub_system	atomicXor	atomicXor_block
atomicXor_system	ballot	cbrt	cbrtf	ceil
ceilf	clock	clock64	copysign	copysignf
cos	cosf	cosh	coshf	cospi
cospif	cyl_bessel_i0	cyl_bessel_i0f	cyl_bessel_i1	cyl_bessel_i1f
dadd	delete	delete[]	dmul	double2int

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double2ll	double2uint	double2ull	dsub	erf
erfc	erfcf	erfcinv	erfcinvf	erfcx
erfcxf	erff	erfinv	erfinvf	exp
exp10	exp10f	exp2	exp2f	expf
expm1	expm1f	fabs	fabsf	fdim
fdimf	fdivide	fdividef	float2double	float2int
float2uint	float_as_int	float_as_uint	floor	floorf
fma	fmaf	fmax	fmaxf	fmin
fminf	fmod	fmodf	free	frexp
frexpf	h2ceil	h2cos	h2exp	h2exp10
h2exp2	h2floor	h2log	h2log10	h2log2
h2rcp	h2rint	h2rsqrt	h2sin	h2sqrt
h2trunc	hceil	hcos	hexp	hexp10
hexp2	hfloor	hlog	hlog10	hlog2
hrcp	hrint	hrsqrt	hsin	hsqrt
htrunc	hypot	hypotf	ilogb	ilogbf
int2double	int2float	int_as_float	isfinite	isinf
isnan	j0	j0f	j1	j1f
jn	jnf	labs	ldexp	ldexpf
lgamma	lgammaf	ll2double	llabs	llmax
llmin	llrint	llrintf	llround	llroundf
log	log10	log10f	log1p	log1pf
log2	log2f	logb	logbf	logf
lrint	lrintf	lround	lroundf	make_bfloat162
make_half2	malloc	max	memcpy	memset
min	modf	modff	mul24	mul64hi
mulhi	nan	nanf	nearbyint	nearbyintf
new	new[]	nextafter	nextafterf	norm

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norm3d	norm3df	norm4d	norm4df	normcdf
normcdff	normcdfinv	normcdfinvf	normf	operator
pow	powf	printf	rcbrt	rcbrtf
remainder	remainderf	remquo	remquof	rhypot
rhypotf	rint	rintf	rnorm	rnorm3d
rnorm3df	rnorm4d	rnorm4df	rnormf	round
roundf	rsqrt	rsqrtf	saturate	scalbln
scalblnf	scalbn	scalbnf	signbit	sin
sincos	sincosf	sincospi	sincospif	sinf
sinh	sinhf	sinpi	sinpif	sqrt
sqrtf	syncthreads_and	syncthreads_count	syncthreads_or	tan
tanf	tanh	tanhf	tgamma	tgammaf
trunc	truncf	uint2double	uint2float	uint_as_float
ull2double	ullmax	ullmin	umax	umin
umul24	y0	y0f	y1	y1f
yn	ynf	atomicAdd	atomicAdd_block	atomicAdd_system
__vimax_s32_relu	__vimax_s16x2_relu	__vimin_s32_relu	__vimin_s16x2_relu	__vimax3_s32
__vimax3_s16x2	__vimax3_u32	__vimax3_u16x2	__vimin3_s32	__vimin3_s16x2
__vimin3_u32	__vimin3_u16x2	__vimax3_s32_relu	__vimax3_s16x2_relu	__vimin3_s32_relu
__vimin3_s16x2_relu	__viaddmax_s32	__viaddmax_s16x2	__viaddmax_u32	__viaddmax_u16x2
__viaddmin_s32	__viaddmin_s16x2	__viaddmin_u32	__viaddmin_u16x2	__viaddmax_s32_relu
__viaddmax_s16x2_relu	__viaddmin_s32_relu	__viaddmin_s16x2_relu	__vibmax_s32	__vibmax_u32

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__vibmin_s32	__vibmin_u32	__vibmax_s16x2	__vibmax_u16x2	__vibmin_s16x2
__vibmin_u16x2	__half2char_rz	__half2uchar_rz	__heq2_mask	__hne2_mask
__hle2_mask	__hge2_mask	__hlt2_mask	__hgt2_mask	__hequ2_mask
__hneu2_mask	__hleu2_mask	__hgeu2_mask	__hltu2_mask	__hgtu2_mask
__bfloat162char_rz	__bfloat162uchar_rz	__heq2_mask	__hne2_mask	__hle2_mask
__hge2_mask	__hlt2_mask	__hgt2_mask	__hequ2_mask	__hneu2_mask
__hleu2_mask	__hgeu2_mask	__hltu2_mask	__hgtu2_mask	atomicAdd
atomicAdd_block	atomicAdd_system			

5.2.2 未严格支持的函数

下面函数仅支持参数 mask 全 1 的情况。

__activemask	__all_sync	__any_sync	__ballot_sync	__barrier_sync
__match_all_sync	__match_any_sync	__shfl_down_sync	__shfl_sync	__shfl_up_sync
__shfl_xor_sync	__syncwarp			

5.2.3 未支持的函数

__barrier_sync_count	__clusterDim	__clusterDimIsSpecified	__clusterGridDimInClusters	__clusterIdx
__clusterRelativeBlockIdx	__clusterRelativeBlockRank	__clusterSizeInBlocks	__cluster_barrier_arrive	__cluster_barrier_wait
__cluster_map_shared_multicast	__cluster_map_shared_rank	__cluster_query_shared_rank	__cvta_generic_to_grid_constant	__cvta_grid_constant_to_generic
__isClusterShared	__isConstant	__isCtaShared	__isGridConstant	__nanosleep
__nv_aligned_device_malloc	__nv_cvt_bfloat16raw2_to_fp8x2	__nv_cvt_bfloat16raw_to_fp8	__nv_cvt_double2_to_fp8x2	__nv_cvt_double_to_fp8
__nv_cvt_float2_to_fp8x2	__nv_cvt_float_to_fp8	__nv_cvt_fp8_to_halfraw	__nv_cvt_fp8x2_to_halfraw2	__nv_cvt_halfraw2_to_fp8x2

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__nv_cvt_halfraw_to_fp8	__pm0	__pm1	__pm2	__pm3
__prof_trigger	__reduce_add_sync	__reduce_and_sync	__reduce_max_sync	__reduce_min_sync
__reduce_or_sync	__reduce_xor_sync	__threadfence_cluster	__uni_sync	

5.3 Driver 函数

5.3.1 支持的函数

cuArray3DCreate	cuArray3DCreate_v2	cuArray3DGetDescriptor
cuArray3DGetDescriptor_v2	cuArrayCreate	cuArrayCreate_v2
cuArrayDestroy	cuArrayGetDescriptor	cuArrayGetDescriptor_v2
cuArrayGetMemoryRequirements	cuCtxAttach	cuCtxCreate
cuCtxCreate_v2	cuCtxCreate_v3	cuCtxCreate_v4
cuCtxDestroy	cuCtxDestroy_v2	cuCtxDetach
cuCtxDisablePeerAccess	cuCtxEnablePeerAccess	cuCtxFromGreenCtx
cuCtxGetApiVersion	cuCtxGetCacheConfig	cuCtxGetCurrent
cuCtxGetDevResource	cuCtxGetDevice	cuCtxGetExecAffinity
cuCtxGetFlags	cuCtxGetId	cuCtxGetLimit
cuCtxGetSharedMemConfig	cuCtxGetStreamPriorityRange	cuCtxPopCurrent
cuCtxPopCurrent_v2	cuCtxPushCurrent	cuCtxPushCurrent_v2
cuCtxRecordEvent	cuCtxSetCacheConfig	cuCtxSetCurrent
cuCtxSetFlags	cuCtxSetLimit	cuCtxSetSharedMemConfig
cuCtxSynchronize	cuCtxWaitEvent	cuDestroyExternalMemory
cuDestroyExternalSemaphore	cuDevResourceGenerateDesc	cuDevSmResourceSplitByCount
cuDeviceCanAccessPeer	cuDeviceComputeCapability	cuDeviceGet
cuDeviceGetAttribute	cuDeviceGetByPCIBusId	cuDeviceGetCount
cuDeviceGetDefaultMemPool	cuDeviceGetDevResource	cuDeviceGetExecAffinitySuppo

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cuDeviceGetGraphMemAttribute	cuDeviceGetMemPool	cuDeviceGetName
cuDeviceGetP2PAttribute	cuDeviceGetPCIBusId	cuDeviceGetProperties
cuDeviceGetTexture1DLinearMaxWidth	cuDeviceGetUuid	cuDeviceGetUuid_v2
cuDeviceGraphMemTrim	cuDevicePrimaryCtxGetState	cuDevicePrimaryCtxRelease
cuDevicePrimaryCtxRelease_v2	cuDevicePrimaryCtxReset	cuDevicePrimaryCtxReset_v2
cuDevicePrimaryCtxRetain	cuDevicePrimaryCtxSetFlags	cuDevicePrimaryCtxSetFlags_v2
cuDeviceRegisterAsyncNotification	cuDeviceSetGraphMemAttribute	cuDeviceSetMemPool
cuDeviceTotalMem	cuDeviceTotalMem_v2	cuDeviceUnregisterAsyncNotification
cuDriverGetVersion	cuEventCreate	cuEventDestroy
cuEventDestroy_v2	cuEventElapsedTime	cuEventQuery
cuEventRecord	cuEventRecordWithFlags	cuEventRecordWithFlags_ptsz
cuEventRecord_ptsz	cuEventSynchronize	cuExternalMemoryGetMappedBuffer
cuExternalMemoryGetMappedMipmappedArray	cuFuncGetAttribute	cuFuncGetModule
cuFuncGetName	cuFuncGetParamInfo	cuFuncIsLoaded
cuFuncLoad	cuFuncSetBlockShape	cuFuncSetSharedSize
cuGetErrorName	cuGetErrorString	cuGetProcAddress
cuGetProcAddress_v2	cuGraphAddBatchMemOpNode	cuGraphAddChildGraphNode
cuGraphAddDependencies	cuGraphAddDependencies_v2	cuGraphAddEmptyNode
cuGraphAddEventRecordNode	cuGraphAddEventWaitNode	cuGraphAddExternalSemaphore

		sSignalNode
cuGraphAddExternalSemaphoresWaitNode	cuGraphAddHostNode	cuGraphAddKernelNode
cuGraphAddKernelNode_v2	cuGraphAddMemAllocNode	cuGraphAddMemFreeNode
cuGraphAddMemcpyNode	cuGraphAddMemsetNode	cuGraphAddNode
cuGraphAddNode_v2	cuGraphBatchMemOpNodeGetParams	cuGraphBatchMemOpNodeSetParams
cuGraphChildGraphNodeGetGraph	cuGraphClone	cuGraphConditionalHandleCreate
cuGraphCreate	cuGraphDebugDotPrint	cuGraphDestroy
cuGraphDestroyNode	cuGraphEventRecordNodeGetEvent	cuGraphEventRecordNodeSetEvent
cuGraphEventWaitNodeGetEvent	cuGraphEventWaitNodeSetEvent	cuGraphExecBatchMemOpNodeSetParams
cuGraphExecChildGraphNodeSetParams	cuGraphExecDestroy	cuGraphExecEventRecordNodeSetEvent
cuGraphExecEventWaitNodeSetEvent	cuGraphExecExternalSemaphoresSignalNodeSetParams	cuGraphExecExternalSemaphoresWaitNodeSetParams
cuGraphExecGetFlags	cuGraphExecHostNodeSetParams	cuGraphExecKernelNodeSetParams
cuGraphExecKernelNodeSetParams_v2	cuGraphExecMemcpyNodeSetParams	cuGraphExecMemsetNodeSetParams
cuGraphExecNodeSetParams	cuGraphExecUpdate	cuGraphExecUpdate_v2
cuGraphExternalSemaphoresSignalNodeGetParams	cuGraphExternalSemaphoresSignalNodeSetParams	cuGraphExternalSemaphoresWaitNodeGetParams
cuGraphExternalSemaphoresWaitNodeSetParams	cuGraphGetEdges	cuGraphGetEdges_v2
cuGraphGetNodes	cuGraphGetRootNodes	cuGraphHostNodeGetParams

cuGraphHostNodeSetParams	cuGraphInstantiate	cuGraphInstantiateWithFlags
cuGraphInstantiateWithParams	cuGraphInstantiateWithParams_ptsz	cuGraphInstantiate_v2
cuGraphKernelNodeCopyAttributes	cuGraphKernelNodeGetAttribute	cuGraphKernelNodeGetParams
cuGraphKernelNodeGetParams_v2	cuGraphKernelNodeSetAttribute	cuGraphKernelNodeSetParams
cuGraphKernelNodeSetParams_v2	cuGraphLaunch	cuGraphLaunch_ptsz
cuGraphMemAllocNodeGetParams	cuGraphMemFreeNodeGetParams	cuGraphMemsetNodeGetParams
cuGraphMemsetNodeSetParams	cuGraphNodeFindInClone	cuGraphNodeGetDependencies
cuGraphNodeGetDependencies_v2	cuGraphNodeGetDependentNodes	cuGraphNodeGetDependentNodes_v2
cuGraphNodeGetEnabled	cuGraphNodeGetType	cuGraphNodeSetEnabled
cuGraphNodeSetParams	cuGraphReleaseUserObject	cuGraphRemoveDependencies
cuGraphRemoveDependencies_v2	cuGraphRetainUserObject	cuGraphUpload
cuGraphUpload_ptsz	cuGreenCtxCreate	cuGreenCtxDestroy
cuGreenCtxGetDevResource	cuGreenCtxRecordEvent	cuGreenCtxStreamCreate
cuGreenCtxWaitEvent	cuImportExternalMemory	cuImportExternalSemaphore
cuInit	cuIpcCloseMemHandle	cuIpcGetEventHandle
cuIpcGetMemHandle	cuIpcOpenEventHandle	cuIpcOpenMemHandle
cuIpcOpenMemHandle_v2	cuKernelGetAttribute	cuKernelGetFunction
cuKernelGetLibrary	cuKernelGetName	cuKernelGetParamInfo
cuKernelSetAttribute	cuLaunch	cuLaunchCooperativeKernel
cuLaunchCooperativeKernelM	cuLaunchCooperativeKernel_pts	cuLaunchGrid

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cuLaunchGridAsync	cuLaunchHostFunc	cuLaunchHostFunc_ptsz
cuLaunchKernel	cuLaunchKernelEx	cuLaunchKernelEx_ptsz
cuLaunchKernel_ptsz	cuLibraryEnumerateKernels	cuLibraryGetGlobal
cuLibraryGetKernel	cuLibraryGetKernelCount	cuLibraryGetManaged
cuLibraryGetModule	cuLibraryLoadData	cuLibraryLoadFromFile
cuLibraryUnload	cuLinkAddData	cuLinkAddData_v2
cuLinkAddFile	cuLinkAddFile_v2	cuLinkComplete
cuLinkCreate	cuLinkCreate_v2	cuLinkDestroy
cuMemAddressFree	cuMemAddressReserve	cuMemAdvise
cuMemAdvise_v2	cuMemAlloc	cuMemAllocAsync
cuMemAllocAsync_ptsz	cuMemAllocFromPoolAsync	cuMemAllocFromPoolAsync_ptsz
cuMemAllocHost	cuMemAllocHost_v2	cuMemAllocManaged
cuMemAllocPitch	cuMemAllocPitch_v2	cuMemAlloc_v2
cuMemCreate	cuMemFree	cuMemFreeAsync
cuMemFreeAsync_ptsz	cuMemFreeHost	cuMemFree_v2
cuMemGetAccess	cuMemGetAddressRange	cuMemGetAddressRange_v2
cuMemGetAllocationGranularity	cuMemGetAllocationPropertiesFromHandle	cuMemGetHandleForAddressRange
cuMemGetInfo	cuMemGetInfo_v2	cuMemHostAlloc
cuMemHostGetDevicePointer	cuMemHostGetDevicePointer_v2	cuMemHostGetFlags
cuMemHostRegister	cuMemHostRegister_v2	cuMemHostUnregister
cuMemMap	cuMemMapArrayAsync	cuMemMapArrayAsync_ptsz
cuMemPoolCreate	cuMemPoolExportPointer	cuMemPoolExportToShareableHandle
cuMemPoolGetAccess	cuMemPoolGetAttribute	cuMemPoolImportFromShareab

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		leHandle
cuMemPoolImportPointer	cuMemPoolSetAccess	cuMemPoolSetAttribute
cuMemPoolTrimTo	cuMemPrefetchAsync	cuMemPrefetchAsync_ptsz
cuMemPrefetchAsync_v2	cuMemPrefetchAsync_v2_ptsz	cuMemRangeGetAttribute
cuMemRangeGetAttributes	cuMemRelease	cuMemRetainAllocationHandle
cuMemSetAccess	cuMemUnmap	cuMemcpy
cuMemcpy2D	cuMemcpy2DAsync	cuMemcpy2DAsync_v2
cuMemcpy2DAsync_v2_ptsz	cuMemcpy2DUnaligned	cuMemcpy2DUnaligned_v2
cuMemcpy2DUnaligned_v2_ptds	cuMemcpy2D_v2	cuMemcpy2D_v2_ptds
cuMemcpy3D	cuMemcpy3DAsync	cuMemcpy3DAsync_v2
cuMemcpy3DAsync_v2_ptsz	cuMemcpy3DPeer	cuMemcpy3DPeerAsync
cuMemcpy3DPeerAsync_ptsz	cuMemcpy3DPeer_ptds	cuMemcpy3D_v2
cuMemcpy3D_v2_ptds	cuMemcpyAsync	cuMemcpyAsync_ptsz
cuMemcpyAtoA	cuMemcpyAtoA_v2	cuMemcpyAtoA_v2_ptds
cuMemcpyAtoD	cuMemcpyAtoD_v2	cuMemcpyAtoD_v2_ptds
cuMemcpyAtoH	cuMemcpyAtoHAsync	cuMemcpyAtoHAsync_v2
cuMemcpyAtoHAsync_v2_ptsz	cuMemcpyAtoH_v2	cuMemcpyAtoH_v2_ptds
cuMemcpyDtoA	cuMemcpyDtoA_v2	cuMemcpyDtoA_v2_ptds
cuMemcpyDtoD	cuMemcpyDtoDAsync	cuMemcpyDtoDAsync_v2
cuMemcpyDtoDAsync_v2_ptsz	cuMemcpyDtoD_v2	cuMemcpyDtoD_v2_ptds
cuMemcpyDtoH	cuMemcpyDtoHAsync	cuMemcpyDtoHAsync_v2
cuMemcpyDtoHAsync_v2_ptsz	cuMemcpyDtoH_v2	cuMemcpyDtoH_v2_ptds
cuMemcpyHtoA	cuMemcpyHtoAAsync	cuMemcpyHtoAAsync_v2
cuMemcpyHtoAAsync_v2_pts	cuMemcpyHtoA_v2	cuMemcpyHtoA_v2_ptds

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cuMemcpyHtoD	cuMemcpyHtoDAsync	cuMemcpyHtoDAsync_v2
cuMemcpyHtoDAsync_v2_ptsz	cuMemcpyHtoD_v2	cuMemcpyHtoD_v2_ptds
cuMemcpyPeer	cuMemcpyPeerAsync	cuMemcpyPeerAsync_ptsz
cuMemcpyPeer_ptds	cuMemcpy_ptds	cuMemsetD16
cuMemsetD16Async	cuMemsetD16Async_ptsz	cuMemsetD16_v2
cuMemsetD16_v2_ptds	cuMemsetD2D16	cuMemsetD2D16Async
cuMemsetD2D16Async_ptsz	cuMemsetD2D16_v2	cuMemsetD2D16_v2_ptds
cuMemsetD2D32	cuMemsetD2D32Async	cuMemsetD2D32Async_ptsz
cuMemsetD2D32_v2	cuMemsetD2D32_v2_ptds	cuMemsetD2D8
cuMemsetD2D8Async	cuMemsetD2D8Async_ptsz	cuMemsetD2D8_v2
cuMemsetD2D8_v2_ptds	cuMemsetD32	cuMemsetD32Async
cuMemsetD32Async_ptsz	cuMemsetD32_v2	cuMemsetD32_v2_ptds
cuMemsetD8	cuMemsetD8Async	cuMemsetD8Async_ptsz
cuMemsetD8_v2	cuMemsetD8_v2_ptds	cuMipmappedArrayCreate
cuMipmappedArrayDestroy	cuMipmappedArrayGetLevel	cuMipmappedArrayGetMemoryRequirements
cuModuleEnumerateFunctions	cuModuleGetFunction	cuModuleGetFunctionCount
cuModuleGetGlobal	cuModuleGetGlobal_v2	cuModuleGetLoadingMode
cuModuleGetSurfRef	cuModuleGetTexRef	cuModuleLoad
cuModuleLoadData	cuModuleLoadDataEx	cuModuleLoadFatBinary
cuModuleUnload	cuOccupancyMaxActiveBlocksPerMultiprocessor	cuOccupancyMaxActiveBlocksPerMultiprocessorWithFlags
cuOccupancyMaxPotentialBlockSize	cuOccupancyMaxPotentialBlockSizeWithFlags	cuParamSetSize
cuParamSetTexRef	cuParamSetf	cuParamSeti
cuParamSetv	cuPointerGetAttribute	cuPointerGetAttributes

cuPointerSetAttribute	cuProfilerStart	cuProfilerStop
cuSignalExternalSemaphoresAsync	cuSignalExternalSemaphoresAsync_ptsz	cuStreamAddCallback
cuStreamAddCallback_ptsz	cuStreamAttachMemAsync	cuStreamAttachMemAsync_ptsz
cuStreamBatchMemOp	cuStreamBatchMemOp_v2	cuStreamBatchMemOp_v2_ptsz
cuStreamBeginCapture	cuStreamBeginCaptureToGraph	cuStreamBeginCaptureToGraph_ptsz
cuStreamBeginCapture_ptsz	cuStreamBeginCapture_v2	cuStreamBeginCapture_v2_ptsz
cuStreamCopyAttributes	cuStreamCopyAttributes_ptsz	cuStreamCreate
cuStreamCreateWithPriority	cuStreamDestroy	cuStreamDestroy_v2
cuStreamEndCapture	cuStreamEndCapture_ptsz	cuStreamGetAttribute
cuStreamGetAttribute_ptsz	cuStreamGetCaptureInfo	cuStreamGetCaptureInfo_v2
cuStreamGetCaptureInfo_v2_ptsz	cuStreamGetCaptureInfo_v3	cuStreamGetCaptureInfo_v3_ptsz
cuStreamGetCtx	cuStreamGetCtx_ptsz	cuStreamGetCtx_v2
cuStreamGetCtx_v2_ptsz	cuStreamGetFlags	cuStreamGetFlags_ptsz
cuStreamGetGreenCtx	cuStreamGetId	cuStreamGetId_ptsz
cuStreamGetPriority	cuStreamGetPriority_ptsz	cuStreamIsCapturing
cuStreamIsCapturing_ptsz	cuStreamQuery	cuStreamQuery_ptsz
cuStreamSetAttribute	cuStreamSetAttribute_ptsz	cuStreamSynchronize
cuStreamSynchronize_ptsz	cuStreamUpdateCaptureDependencies	cuStreamUpdateCaptureDependencies_ptsz
cuStreamUpdateCaptureDependencies_v2	cuStreamUpdateCaptureDependencies_v2_ptsz	cuStreamWaitEvent
cuStreamWaitEvent_ptsz	cuStreamWaitValue32	cuStreamWaitValue32_v2
cuStreamWaitValue32_v2_ptsz	cuStreamWaitValue64	cuStreamWaitValue64_v2
cuStreamWaitValue64_v2_ptsz	cuStreamWriteValue32	cuStreamWriteValue32_v2
cuStreamWriteValue32_v2_pts	cuStreamWriteValue64	cuStreamWriteValue64_v2

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cuStreamWriteValue64_v2_ptsz	cuSurfObjectCreate	cuSurfObjectDestroy
cuSurfObjectGetResourceDesc	cuSurfRefGetArray	cuSurfRefSetArray
cuTexObjectCreate	cuTexObjectDestroy	cuTexObjectGetResourceDesc
cuTexObjectGetResourceViewDesc	cuTexObjectGetTextureDesc	cuTexRefCreate
cuTexRefDestroy	cuTexRefGetAddress	cuTexRefGetAddressMode
cuTexRefGetAddress_v2	cuTexRefGetArray	cuTexRefGetFilterMode
cuTexRefGetFlags	cuTexRefGetFormat	cuTexRefGetMaxAnisotropy
cuTexRefGetMipmapFilterMode	cuTexRefGetMipmapLevelBias	cuTexRefGetMipmapLevelClamp
cuTexRefGetMipmappedArray	cuTexRefSetAddress	cuTexRefSetAddress2D_v3
cuTexRefSetAddressMode	cuTexRefSetAddress_v2	cuTexRefSetArray
cuTexRefSetBorderColor	cuTexRefSetFilterMode	cuTexRefSetFlags
cuTexRefSetFormat	cuTexRefSetMaxAnisotropy	cuTexRefSetMipmapFilterMode
cuTexRefSetMipmapLevelBias	cuTexRefSetMipmapLevelClamp	cuTexRefSetMipmappedArray
cuThreadExchangeStreamCaptureMode	cuUserObjectCreate	cuUserObjectRelease
cuUserObjectRetain	cuWaitExternalSemaphoresAsync	cuWaitExternalSemaphoresAsync_ptsz

5.3.2 未支持的函数

cuArrayGetPlane	cuArrayGetSparseProperties	cuCoredumpGetAttribute
cuCoredumpGetAttributeGlobal	cuCoredumpSetAttribute	cuCoredumpSetAttributeGlobal
cuCtxResetPersistingL2Cache	cuDeviceGetLuid	cuDeviceGetNvSciSyncAttributes
cuFlushGPUDirectRDMAWrites	cuFuncSetAttribute	cuFuncSetCacheConfig

cuFuncSetSharedMemConfig	cuGLGetDevices	cuGLGetDevices_v2
cuGetExportTable	cuGraphicsGLRegisterBuffer	cuGraphicsGLRegisterImage
cuGraphicsMapResources	cuGraphicsMapResources_ptsz	cuGraphicsResourceGetMappedMipmappedArray
cuGraphicsResourceGetMappedPointer	cuGraphicsResourceGetMappedPointer_v2	cuGraphicsResourceSetMapFlags
cuGraphicsResourceSetMapFlags_v2	cuGraphicsSubResourceGetMappedArray	cuGraphicsUnmapResources
cuGraphicsUnmapResources_ptsz	cuGraphicsUnregisterResource	cuKernelSetCacheConfig
cuLibraryGetUnifiedFunction	cuMemExportToShareableHandle	cuMemImportFromShareableHandle
cuMipmappedArrayGetSparseProperties	cuMulticastAddDevice	cuMulticastBindAddr
cuMulticastBindMem	cuMulticastCreate	cuMulticastGetGranularity
cuMulticastUnbind	cuOccupancyAvailableDynamicSMemPerBlock	cuOccupancyMaxActiveClusters
cuOccupancyMaxPotentialClusterSize	cuProfilerInitialize	cuTensorMapEncodeIm2col
cuTensorMapEncodeTiled	cuTensorMapReplaceAddress	cuTexRefGetBorderColor

5.4 cuBLAS 函数

5.4.1 支持的函数

cublasAxpyEx	cublasAxpyEx_64	cublasCaxpy_v2
cublasCaxpy_v2_64	cublasCcopy_v2	cublasCcopy_v2_64
cublasCdgmm	cublasCdgmm_64	cublasCdotc_v2
cublasCdotc_v2_64	cublasCdotu_v2	cublasCdotu_v2_64
cublasCgbmv_v2	cublasCgbmv_v2_64	cublasCgeam

cublasCgeam_64	cublasCgelsBatched	cublasCgemm_v2
cublasCgemm_v2_64	cublasCgemm3m	cublasCgemm3m_64
cublasCgemm3mStridedBatched	cublasCgemm3mStridedBatched_64	cublasCgemmBatched
cublasCgemmBatched_64	cublasCgemmEx	cublasCgemmEx_64
cublasCgemmStridedBatched	cublasCgemmStridedBatched_64	cublasCgemv_v2
cublasCgemv_v2_64	cublasCgemvBatched	cublasCgemvBatched_64
cublasCgemvStridedBatched	cublasCgemvStridedBatched_64	cublasCgeqrfBatched
cublasCgerc_v2	cublasCgerc_v2_64	cublasCgeru_v2
cublasCgeru_v2_64	cublasCgetrfBatched	cublasCgetriBatched
cublasCgetrsBatched	cublasChbmv_v2	cublasChbmv_v2_64
cublasChemm_v2	cublasChemm_v2_64	cublasChemv_v2
cublasChemv_v2_64	cublasCher_v2	cublasCher_v2_64
cublasCher2_v2	cublasCher2_v2_64	cublasCher2k_v2
cublasCher2k_v2_64	cublasCherk_v2	cublasCherk_v2_64
cublasCherk3mEx	cublasCherk3mEx_64	cublasCherkEx
cublasCherkEx_64	cublasCherkx	cublasCherkx_64
cublasChpmv_v2	cublasChpmv_v2_64	cublasChpr_v2
cublasChpr_v2_64	cublasChpr2_v2	cublasChpr2_v2_64
cublasCmatinvBatched	cublasCreate_v2	cublasCrot_v2
cublasCrot_v2_64	cublasCrotg_v2	cublasCscal_v2
cublasCscal_v2_64	cublasCsrot_v2	cublasCsrot_v2_64
cublasCsscal_v2	cublasCsscal_v2_64	cublasCswap_v2
cublasCswap_v2_64	cublasCsymm_v2	cublasCsymm_v2_64
cublasCsymv_v2	cublasCsymv_v2_64	cublasCsyr_v2
cublasCsyr_v2_64	cublasCsyr2_v2	cublasCsyr2_v2_64

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cublasCsyr2k_v2	cublasCsyr2k_v2_64	cublasCsyrk_v2
cublasCsyrk_v2_64	cublasCsyrk3mEx	cublasCsyrk3mEx_64
cublasCsyrkEx	cublasCsyrkEx_64	cublasCsyrkx
cublasCsyrkx_64	cublasCtbmv_v2	cublasCtbmv_v2_64
cublasCtbsv_v2	cublasCtbsv_v2_64	cublasCtpmv_v2
cublasCtpmv_v2_64	cublasCtpsv_v2	cublasCtpsv_v2_64
cublasCtpttr	cublasCtrmm_v2	cublasCtrmm_v2_64
cublasCtrmv_v2	cublasCtrmv_v2_64	cublasCtrsm_v2
cublasCtrsm_v2_64	cublasCtrsmBatched	cublasCtrsmBatched_64
cublasCtrsv_v2	cublasCtrsv_v2_64	cublasCtrttp
cublasDasum_v2	cublasDasum_v2_64	cublasDaxpy_v2
cublasDaxpy_v2_64	cublasDcopy_v2	cublasDcopy_v2_64
cublasDdgmm	cublasDdgmm_64	cublasDdot_v2
cublasDdot_v2_64	cublasDestroy_v2	cublasDgbmv_v2
cublasDgbmv_v2_64	cublasDgeam	cublasDgeam_64
cublasDgelsBatched	cublasDgemm_v2	cublasDgemm_v2_64
cublasDgemmBatched	cublasDgemmBatched_64	cublasDgemmGroupedBatched
cublasDgemmGroupedBatched_64	cublasDgemmStridedBatched	cublasDgemmStridedBatched_64
cublasDgemv_v2	cublasDgemv_v2_64	cublasDgemvBatched
cublasDgemvBatched_64	cublasDgemvStridedBatched	cublasDgemvStridedBatched_64
cublasDgeqrfBatched	cublasDger_v2	cublasDger_v2_64
cublasDgetrfBatched	cublasDgetriBatched	cublasDgetrsBatched
cublasDmatinvBatched	cublasDnrm2_v2	cublasDnrm2_v2_64
cublasDotcEx	cublasDotcEx_64	cublasDotEx
cublasDotEx_64	cublasDrot_v2	cublasDrot_v2_64
cublasDrotg_v2	cublasDrotm_v2	cublasDrotm_v2_64
cublasDrotmg_v2	cublasDsbmv_v2	cublasDsbmv_v2_64

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cublasDscal_v2	cublasDscal_v2_64	cublasDspmv_v2
cublasDspmv_v2_64	cublasDspr_v2	cublasDspr_v2_64
cublasDspr2_v2	cublasDspr2_v2_64	cublasDswap_v2
cublasDswap_v2_64	cublasDsymm_v2	cublasDsymm_v2_64
cublasDsymv_v2	cublasDsymv_v2_64	cublasDsyr_v2
cublasDsyr_v2_64	cublasDsyr2_v2	cublasDsyr2_v2_64
cublasDsyr2k_v2	cublasDsyr2k_v2_64	cublasDsyrk_v2
cublasDsyrk_v2_64	cublasDsyrkx	cublasDsyrkx_64
cublasDtbmv_v2	cublasDtbmv_v2_64	cublasDtbsv_v2
cublasDtbsv_v2_64	cublasDtpmv_v2	cublasDtpmv_v2_64
cublasDtpsv_v2	cublasDtpsv_v2_64	cublasDtpttr
cublasDtrmm_v2	cublasDtrmm_v2_64	cublasDtrmv_v2
cublasDtrmv_v2_64	cublasDtrsm_v2	cublasDtrsm_v2_64
cublasDtrsmBatched	cublasDtrsmBatched_64	cublasDtrsv_v2
cublasDtrsv_v2_64	cublasDtrttp	cublasDzasum_v2
cublasDzasum_v2_64	cublasDznrm2_v2	cublasDznrm2_v2_64
cublasGemmBatchedEx	cublasGemmBatchedEx_64	cublasGemmEx
cublasGemmEx_64	cublasGemmGroupedBatchedEx	cublasGemmGroupedBatchedEx_64
cublasGemmStridedBatchedEx	cublasGemmStridedBatchedEx_64	cublasGetAtomicsMode
cublasGetCudartVersion	cublasGetMathMode	cublasGetMatrix
cublasGetMatrix_64	cublasGetMatrixAsync	cublasGetMatrixAsync_64
cublasGetPointerMode_v2	cublasGetProperty	cublasGetStatusName
cublasGetStatusString	cublasGetStream_v2	cublasGetVector
cublasGetVector_64	cublasGetVectorAsync	cublasGetVectorAsync_64
cublasGetVersion_v2	cublasHgemm	cublasHgemm_64
cublasHgemmBatched	cublasHgemmBatched_64	cublasHgemmStridedBatched

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cublasHgemmStridedBatched_64	cublasHSHgemvBatched	cublasHSHgemvBatched_64
cublasHSHgemvStridedBatched	cublasHSHgemvStridedBatched_64	cublasHSSgemvBatched
cublasHSSgemvBatched_64	cublasHSSgemvStridedBatched	cublasHSSgemvStridedBatched_64
cublasIcamax_v2	cublasIcamax_v2_64	cublasIcamin_v2
cublasIcamin_v2_64	cublasIdamax_v2	cublasIdamax_v2_64
cublasIdamin_v2	cublasIdamin_v2_64	cublasIsamax_v2
cublasIsamax_v2_64	cublasIsamin_v2	cublasIsamin_v2_64
cublasIzamax_v2	cublasIzamax_v2_64	cublasIzamin_v2
cublasIzamin_v2_64	cublasNrm2Ex	cublasNrm2Ex_64
cublasRotEx	cublasRotEx_64	cublasSasum_v2
cublasSasum_v2_64	cublasSaxpy_v2	cublasSaxpy_v2_64
cublasScalEx	cublasScalEx_64	cublasScasum_v2
cublasScasum_v2_64	cublasScnrm2_v2	cublasScnrm2_v2_64
cublasScopy_v2	cublasScopy_v2_64	cublasSdgmm
cublasSdgmm_64	cublasSdot_v2	cublasSdot_v2_64
cublasSetAtomicsMode	cublasSetMathMode	cublasSetMatrix
cublasSetMatrix_64	cublasSetMatrixAsync	cublasSetMatrixAsync_64
cublasSetPointerMode_v2	cublasSetStream_v2	cublasSetVector
cublasSetVector_64	cublasSetVectorAsync	cublasSetVectorAsync_64
cublasSetWorkspace_v2	cublasSgbmv_v2	cublasSgbmv_v2_64
cublasSgeam	cublasSgeam_64	cublasSgelsBatched
cublasSgemm_v2	cublasSgemm_v2_64	cublasSgemmBatched
cublasSgemmBatched_64	cublasSgemmEx	cublasSgemmEx_64
cublasSgemmGroupedBatched	cublasSgemmGroupedBatched_64	cublasSgemmStridedBatched

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cublasSgemmStridedBatched_64	cublasSgemv_v2	cublasSgemv_v2_64
cublasSgemvBatched	cublasSgemvBatched_64	cublasSgemvStridedBatched
cublasSgemvStridedBatched_64	cublasSgeqrfBatched	cublasSger_v2
cublasSger_v2_64	cublasSgetrfBatched	cublasSgetriBatched
cublasSgetrsBatched	cublasSmatinvBatched	cublasSnrm2_v2
cublasSnrm2_v2_64	cublasSrot_v2	cublasSrot_v2_64
cublasSrotg_v2	cublasSrotm_v2	cublasSrotm_v2_64
cublasSrotmg_v2	cublasSsbmv_v2	cublasSsbmv_v2_64
cublasSscal_v2	cublasSscal_v2_64	cublasSspmv_v2
cublasSspmv_v2_64	cublasSspr_v2	cublasSspr_v2_64
cublasSspr2_v2	cublasSspr2_v2_64	cublasSswap_v2
cublasSswap_v2_64	cublasSsymm_v2	cublasSsymm_v2_64
cublasSsymv_v2	cublasSsymv_v2_64	cublasSsyr_v2
cublasSsyr_v2_64	cublasSsyr2_v2	cublasSsyr2_v2_64
cublasSsyr2k_v2	cublasSsyr2k_v2_64	cublasSsyrk_v2
cublasSsyrk_v2_64	cublasSsyrkx	cublasSsyrkx_64
cublasStbmv_v2	cublasStbmv_v2_64	cublasStbsv_v2
cublasStbsv_v2_64	cublasStpmv_v2	cublasStpmv_v2_64
cublasStpsv_v2	cublasStpsv_v2_64	cublasStpttr
cublasStrmm_v2	cublasStrmm_v2_64	cublasStrmv_v2
cublasStrmv_v2_64	cublasStrsm_v2	cublasStrsm_v2_64
cublasStrsmBatched	cublasStrsmBatched_64	cublasStrsv_v2
cublasStrsv_v2_64	cublasStrttp	cublasTSSgemvBatched
cublasTSSgemvBatched_64	cublasTSSgemvStridedBatched	cublasTSSgemvStridedBatched_64
cublasTSTgemvBatched	cublasTSTgemvBatched_64	cublasTSTgemvStridedBatched
cublasTSTgemvStridedBatched_64	cublasZaxpy_v2	cublasZaxpy_v2_64

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cublasZcopy_v2	cublasZcopy_v2_64	cublasZdgmm
cublasZdgmm_64	cublasZdotc_v2	cublasZdotc_v2_64
cublasZdotu_v2	cublasZdotu_v2_64	cublasZdrot_v2
cublasZdrot_v2_64	cublasZdscal_v2	cublasZdscal_v2_64
cublasZgbmv_v2	cublasZgbmv_v2_64	cublasZgeam
cublasZgeam_64	cublasZgelsBatched	cublasZgemm_v2
cublasZgemm_v2_64	cublasZgemm3m	cublasZgemm3m_64
cublasZgemmBatched	cublasZgemmBatched_64	cublasZgemmStridedBatched
cublasZgemmStridedBatched_64	cublasZgemv_v2	cublasZgemv_v2_64
cublasZgemvBatched	cublasZgemvBatched_64	cublasZgemvStridedBatched
cublasZgemvStridedBatched_64	cublasZgeqrfBatched	cublasZgerc_v2
cublasZgerc_v2_64	cublasZgeru_v2	cublasZgeru_v2_64
cublasZgetrfBatched	cublasZgetriBatched	cublasZgetrsBatched
cublasZhbmv_v2	cublasZhbmv_v2_64	cublasZhemm_v2
cublasZhemm_v2_64	cublasZhemv_v2	cublasZhemv_v2_64
cublasZher_v2	cublasZher_v2_64	cublasZher2_v2
cublasZher2_v2_64	cublasZher2k_v2	cublasZher2k_v2_64
cublasZherk_v2	cublasZherk_v2_64	cublasZherkx
cublasZherkx_64	cublasZhpmv_v2	cublasZhpmv_v2_64
cublasZhpr_v2	cublasZhpr_v2_64	cublasZhpr2_v2
cublasZhpr2_v2_64	cublasZmatinvBatched	cublasZrot_v2
cublasZrot_v2_64	cublasZrotg_v2	cublasZscal_v2
cublasZscal_v2_64	cublasZswap_v2	cublasZswap_v2_64
cublasZsymm_v2	cublasZsymm_v2_64	cublasZsymv_v2
cublasZsymv_v2_64	cublasZsyr_v2	cublasZsyr_v2_64
cublasZsyr2_v2	cublasZsyr2_v2_64	cublasZsyr2k_v2
cublasZsyr2k_v2_64	cublasZsyrk_v2	cublasZsyrk_v2_64

cublasZsyrkx	cublasZsyrkx_64	cublasZtbmv_v2
cublasZtbmv_v2_64	cublasZtbsv_v2	cublasZtbsv_v2_64
cublasZtpmv_v2	cublasZtpmv_v2_64	cublasZtpsv_v2
cublasZtpsv_v2_64	cublasZtpttr	cublasZtrmm_v2
cublasZtrmm_v2_64	cublasZtrmv_v2	cublasZtrmv_v2_64
cublasZtrsm_v2	cublasZtrsm_v2_64	cublasZtrsmBatched
cublasZtrsmBatched_64	cublasZtrsv_v2	cublasZtrsv_v2_64
cublasZtrttp	cublasLtCreate	cublasLtDestroy
cublasLtGetCudartVersion	cublasLtGetProperty	cublasLtGetVersion
cublasLtMatmul	cublasLtMatmulAlgoCapGetAttribute	cublasLtMatmulAlgoCheck
cublasLtMatmulAlgoConfigGetAttribute	cublasLtMatmulAlgoConfigSetAttribute	cublasLtMatmulAlgoGetHeuristic
cublasLtMatmulAlgoGetIds	cublasLtMatmulAlgoInit	cublasLtMatmulDescCreate
cublasLtMatmulDescDestroy	cublasLtMatmulDescGetAttribute	cublasLtMatmulDescSetAttribute
cublasLtMatmulPreferenceCreate	cublasLtMatmulPreferenceDestroy	cublasLtMatmulPreferenceGetAttribute
cublasLtMatmulPreferenceSetAttribute	cublasLtMatrixLayoutCreate	cublasLtMatrixLayoutDestroy
cublasLtMatrixLayoutGetAttribute	cublasLtMatrixLayoutSetAttribute	cublasLtMatrixTransform
cublasLtMatrixTransformDescCreate	cublasLtMatrixTransformDescDestroy	cublasLtMatrixTransformDescGetAttribute
cublasLtMatrixTransformDescSetAttribute	cublasLtGetStatusName	cublasLtGetStatusString
cublasLtMatrixLayoutInit_internal	cublasLtMatmulDescInit_internal	cublasLtMatrixTransformDescInit_internal

cublasLtMatmulPreferenceInit_internal	cublasXtCreate	cublasXtDestroy
cublasXtGetNumBoards	cublasXtMaxBoards	cublasXtDeviceSelect
cublasXtSetBlockDim	cublasXtGetBlockDim	cublasXtGetPinningMemMode
cublasXtSetPinningMemMode	cublasXtSetCpuRoutine	cublasXtSetCpuRatio
cublasXtSgemm	cublasXtDgemm	cublasXtCgemm
cublasXtZgemm	cublasXtSsyrk	cublasXtDsyrk
cublasXtCsyrk	cublasXtZsyrk	cublasXtCherk
cublasXtZherk	cublasXtSsyr2k	cublasXtDsyr2k
cublasXtCsyr2k	cublasXtZsyr2k	cublasXtCherkx
cublasXtZherkx	cublasXtStrsm	cublasXtDtrsm
cublasXtCtrsm	cublasXtZtrsm	cublasXtSsymm
cublasXtDsymm	cublasXtCsymm	cublasXtZsymm
cublasXtChemm	cublasXtZhemm	cublasXtSsyrkx
cublasXtDsyrkx	cublasXtCsyrkx	cublasXtZsyrkx
cublasXtCher2k	cublasXtZher2k	cublasXtSspmm
cublasXtDspmm	cublasXtCspmm	cublasXtZspmm
cublasXtStrmm	cublasXtDtrmm	cublasXtCtrmm
cublasXtZtrmm		

5.4.2 未支持的函数

cublasInit	cublasShutdown	cublasGetError
cublasGetVersion	cublasAlloc	cublasFree
cublasSetKernelStream	cublasSnrm2	cublasDnrm2
cublasScnrm2	cublasDznrm2	cublasSdot
cublasDdot	cublasCdotu	cublasCdotc
cublasZdotu	cublasZdotc	cublasSscal
cublasDscal	cublasCscal	cublasZscal
cublasCsscal	cublasZdscal	cublasSaxpy

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cublasDaxpy	cublasCaxpy	cublasZaxpy
cublasScopy	cublasDcopy	cublasCcopy
cublasZcopy	cublasSswap	cublasDswap
cublasCswap	cublasZswap	cublasIsamax
cublasIdamax	cublasIcamax	cublasIzamax
cublasIsamin	cublasIdamin	cublasIcamin
cublasIzamin	cublasSasum	cublasDasum
cublasScasum	cublasDzasum	cublasSrot
cublasDrot	cublasCrot	cublasZrot
cublasCsrot	cublasZdrot	cublasSrotg
cublasDrotg	cublasCrotg	cublasZrotg
cublasSrotm	cublasDrotm	cublasSrotmg
cublasDrotmg	cublasSgemv	cublasDgemv
cublasCgemv	cublasZgemv	cublasSgbmv
cublasDgbmv	cublasCgbmv	cublasZgbmv
cublasStrmv	cublasDtrmv	cublasCtrmv
cublasZtrmv	cublasStbmv	cublasDtbmv
cublasCtbmv	cublasZtbmv	cublasStpmv
cublasDtpmv	cublasCtpmv	cublasZtpmv
cublasStrsv	cublasDtrsv	cublasCtrsv
cublasZtrsv	cublasStpsv	cublasDtpsv
cublasCtpsv	cublasZtpsv	cublasStbsv
cublasDtbsv	cublasCtbsv	cublasZtbsv
cublasSsymv	cublasDsymv	cublasChemv
cublasZhemv	cublasSsbmv	cublasDsbmv
cublasChbmv	cublasZhbmv	cublasSspmv
cublasDspmv	cublasChpmv	cublasZhpmv
cublasSger	cublasDger	cublasCgeru

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cublasCgerc	cublasZgeru	cublasZgerc
cublasSsyr	cublasDsyr	cublasCher
cublasZher	cublasSspr	cublasDspr
cublasChpr	cublasZhpr	cublasSsyr2
cublasDsyr2	cublasCher2	cublasZher2
cublasSspr2	cublasDspr2	cublasChpr2
cublasZhpr2	cublasSgemm	cublasDgemm
cublasCgemm	cublasZgemm	cublasSsyrk
cublasDsyrk	cublasCsyrk	cublasZsyrk
cublasCherk	cublasZherk	cublasSsyr2k
cublasDsyr2k	cublasCsyr2k	cublasZsyr2k
cublasCher2k	cublasZher2k	cublasSsymm
cublasDsymm	cublasCsymm	cublasZsymm
cublasChemm	cublasZhemm	cublasStrsm
cublasDtrsm	cublasCtrsm	cublasZtrsm
cublasStrmm	cublasDtrmm	cublasCtrmm
cublasZtrmm	cublasAsumEx	cublasAsumEx_64
cublasCgemm3mBatched	cublasCgemm3mBatched_64	cublasCgemm3mEx
cublasCgemm3mEx_64	cublasCopyEx	cublasCopyEx_64
cublasGetLoggerCallback	cublasGetSmCountTarget	cublasIamaxEx
cublasIamaxEx_64	cublasIaminEx	cublasIaminEx_64
cublasLoggerConfigure	cublasRotgEx	cublasRotmEx
cublasRotmEx_64	cublasRotmgEx	cublasSetLoggerCallback
cublasSetSmCountTarget	cublasSwapEx	cublasSwapEx_64
cublasUint8gemmBias	cublasXerbla	cublasLtHeuristicsCacheGetCapacity
cublasLtHeuristicsCacheSetCapacity	cublasLtLoggerSetCallback	cublasLtLoggerSetFile

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cublasLtLoggerOpenFile	cublasLtLoggerSetLevel	cublasLtLoggerSetMask
cublasLtLoggerForceDisable	cublasLtDisableCpuInstructionsSetMask	

5.5 cuFFT 函数

5.5.1 支持的函数

cufftCreate	cufftDestroy	cufftEstimate1d
cufftEstimate2d	cufftEstimate3d	cufftEstimateMany
cufftExecC2C	cufftExecC2R	cufftExecD2Z
cufftExecR2C	cufftExecZ2D	cufftExecZ2Z
cufftGetProperty	cufftGetSize	cufftGetSize1d
cufftGetSize2d	cufftGetSize3d	cufftGetSizeMany
cufftGetSizeMany64	cufftGetVersion	cufftMakePlan1d
cufftMakePlan2d	cufftMakePlan3d	cufftMakePlanMany
cufftMakePlanMany64	cufftPlan1d	cufftPlan2d
cufftPlan3d	cufftPlanMany	cufftSetAutoAllocation
cufftSetStream	cufftSetWorkArea	cufftXtClearCallback
cufftXtExec	cufftXtExecDescriptor	cufftXtExecDescriptorC2C
cufftXtExecDescriptorC2R	cufftXtExecDescriptorD2Z	cufftXtExecDescriptorR2C
cufftXtExecDescriptorZ2D	cufftXtExecDescriptorZ2Z	cufftXtFree
cufftXtGetSizeMany	cufftXtMakePlanMany	cufftXtMalloc
cufftXtMemcpy	cufftXtQueryPlan	cufftXtSetCallback
cufftXtSetCallbackSharedSize	cufftXtSetGPUs	cufftXtSetWorkArea
cufftXtSetWorkAreaPolicy	cufftGetPlanPropertyInt64	cufftSetPlanPropertyInt64
cufftResetPlanProperty	cufftXtSetJITCallback_12_7	cufftMpAttachComm
cufftXtSetDistribution	cufftXtSetSubformatDefault	

5.5.2 未支持函数

cufftMpCreateReshape	cufftMpAttachReshapeComm	cufftMpGetReshapeSize

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cufftMpMakeReshape	cufftMpExecReshapeAsync	cufftMpDestroyReshape

5.6 cuRAND 函数

5.6.1 支持的函数

curandCreateGenerator	curandCreateGeneratorHost	curandCreatePoissonDistribution
curandDestroyDistribution	curandDestroyGenerator	curandGenerate
curandGenerateLogNormal	curandGenerateLogNormalDouble	curandGenerateLongLong
curandGenerateNormal	curandGenerateNormalDouble	curandGeneratePoisson
curandGeneratePoissonMethod	curandGenerateSeeds	curandGenerateUniform
curandGenerateUniformDouble	curandGetDirectionVectors32	curandGetDirectionVectors64
curandGetProperty	curandGetScrambleConstants32	curandGetScrambleConstants64
curandGetVersion	curandSetGeneratorOffset	curandSetGeneratorOrdering
curandSetPseudoRandomGeneratorSeed	curandSetQuasiRandomGeneratorDimensions	curandSetStream

5.6.2 未支持的函数

curandGenerateBinomial	curandGenerateBinomialMethod	

5.7 cuSPARSE 函数

5.7.1 支持的函数

cusparseAxpby	cusparseBlockedEllGet	cusparseBsrSetStridedBatch
cusparseCbsr2csr	cusparseCbsric02	cusparseCbsric02_analysis
cusparseCbsric02_bufferSize	cusparseCbsrilu02	cusparseCbsrilu02_analysis
cusparseCbsrilu02_bufferSize	cusparseCbsrilu02_numericBoost	cusparseCbsrmm
cusparseCbsrmv	cusparseCbsrsm2_analysis	cusparseCbsrsm2_bufferSize

cusparseCbsrsm2_solve	cusparseCbsrsv2_analysis	cusparseCbsrsv2_bufferSize
cusparseCbsrsv2_bufferSizeExt	cusparseCbsrsv2_solve	cusparseCbsrxmv
cusparseCcsr2bsr	cusparseCcsr2csr_compress	cusparseCcsr2csru
cusparseCcsr2gebsr	cusparseCcsr2gebsr_bufferSize	cusparseCcsrcolor
cusparseCcsrgeam2	cusparseCcsrgeam2_bufferSizeExt	cusparseCcsric02
cusparseCcsric02_analysis	cusparseCcsric02_bufferSize	cusparseCcsric02_bufferSizeExt
cusparseCcsrilu02	cusparseCcsrilu02_analysis	cusparseCcsrilu02_bufferSize
cusparseCcsrilu02_bufferSizeExt	cusparseCcsrilu02_numericBoost	cusparseCcsru2csr
cusparseCcsru2csr_bufferSizeExt	cusparseCgebsr2csr	cusparseCgebsr2gebsc
cusparseCgebsr2gebsc_bufferSize	cusparseCgebsr2gebsr	cusparseCgebsr2gebsr_bufferSize
cusparseCgemvi	cusparseCgemvi_bufferSize	cusparseCgpsvInterleavedBatch
cusparseCgpsvInterleavedBatch_bufferSizeExt	cusparseCgtsv2	cusparseCgtsv2_bufferSizeExt
cusparseCgtsv2_nopivot	cusparseCgtsv2_nopivot_bufferSizeExt	cusparseCgtsv2StridedBatch
cusparseCgtsv2StridedBatch_bufferSizeExt	cusparseCgtsvInterleavedBatch	cusparseCgtsvInterleavedBatch_bufferSizeExt
cusparseCnnz	cusparseCnnz_compress	cusparseConstBlockedEllGet
cusparseConstCooGet	cusparseConstCscGet	cusparseConstCsrGet
cusparseConstDnMatGet	cusparseConstDnMatGetValues	cusparseConstDnVecGet
cusparseConstDnVecGetValues	cusparseConstSpMatGetValues	cusparseConstSpVecGet
cusparseConstSpVecGetValues	cusparseCooGet	cusparseCooSetPointers
cusparseCooSetStridedBatch	cusparseCreate	cusparseCreateBlockedEll

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cusparseCreateBsr	cusparseCreateBsric02Info	cusparseCreateBsrilu02Info
cusparseCreateBsrsm2Info	cusparseCreateBsrsv2Info	cusparseCreateColorInfo
cusparseCreateConstBlockedEll	cusparseCreateConstBsr	cusparseCreateConstCoo
cusparseCreateConstCsc	cusparseCreateConstCsr	cusparseCreateConstDnMat
cusparseCreateConstDnVec	cusparseCreateConstSlicedEll	cusparseCreateConstSpVec
cusparseCreateCoo	cusparseCreateCsc	cusparseCreateCsr
cusparseCreateCsric02Info	cusparseCreateCsrilu02Info	cusparseCreateCsru2csrInfo
cusparseCreateDnMat	cusparseCreateDnVec	cusparseCreateIdentityPermutation
cusparseCreateMatDescr	cusparseCreatePruneInfo	cusparseCreateSlicedEll
cusparseCreateSpVec	cusparseCscGet	cusparseCscSetPointers
cusparseCsr2cscEx2	cusparseCsr2cscEx2_bufferSize	cusparseCsrGet
cusparseCsrSetPointers	cusparseCsrSetStridedBatch	cusparseDbsr2csr
cusparseDbsric02	cusparseDbsric02_analysis	cusparseDbsric02_bufferSize
cusparseDbsrilu02	cusparseDbsrilu02_analysis	cusparseDbsrilu02_bufferSize
cusparseDbsrilu02_numericBoost	cusparseDbsrmm	cusparseDbsrmv
cusparseDbsrsm2_analysis	cusparseDbsrsm2_bufferSize	cusparseDbsrsm2_solve
cusparseDbsrsv2_analysis	cusparseDbsrsv2_bufferSize	cusparseDbsrsv2_bufferSizeExt
cusparseDbsrsv2_solve	cusparseDbsrxmv	cusparseDcsr2bsr
cusparseDcsr2csr_compress	cusparseDcsr2csru	cusparseDcsr2gebsr
cusparseDcsr2gebsr_bufferSize	cusparseDcsrcolor	cusparseDcsrgeam2
cusparseDcsrgeam2_bufferSizeExt	cusparseDcsric02	cusparseDcsric02_analysis
cusparseDcsric02_bufferSize	cusparseDcsric02_bufferSizeExt	cusparseDcsrilu02
cusparseDcsrilu02_analysis	cusparseDcsrilu02_bufferSize	cusparseDcsrilu02_bufferSizeEx

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cusparseDcsrilu02_numericBoost	cusparseDcsru2csr	cusparseDcsru2csr_bufferSizeExt
cusparseDenseToSparse_analysis	cusparseDenseToSparse_bufferSize	cusparseDenseToSparse_convert
cusparseDestroy	cusparseDestroyBsric02Info	cusparseDestroyBsrilu02Info
cusparseDestroyBsrsm2Info	cusparseDestroyBsrsv2Info	cusparseDestroyColorInfo
cusparseDestroyCsric02Info	cusparseDestroyCsrilu02Info	cusparseDestroyCsru2csrInfo
cusparseDestroyDnMat	cusparseDestroyDnVec	cusparseDestroyMatDescr
cusparseDestroyPruneInfo	cusparseDestroySpMat	cusparseDestroySpVec
cusparseDgebsr2csr	cusparseDgebsr2gebsc	cusparseDgebsr2gebsc_bufferSize
cusparseDgebsr2gebsr	cusparseDgebsr2gebsr_bufferSize	cusparseDgemvi
cusparseDgemvi_bufferSize	cusparseDgpsvInterleavedBatch	cusparseDgpsvInterleavedBatch_bufferSizeExt
cusparseDgtsv2	cusparseDgtsv2_bufferSizeExt	cusparseDgtsv2_nopivot
cusparseDgtsv2_nopivot_bufferSizeExt	cusparseDgtsv2StridedBatch	cusparseDgtsv2StridedBatch_bufferSizeExt
cusparseDgtsvInterleavedBatch	cusparseDgtsvInterleavedBatch_bufferSizeExt	cusparseDnMatGet
cusparseDnMatGetStridedBatch	cusparseDnMatGetValues	cusparseDnMatSetStridedBatch
cusparseDnMatSetValues	cusparseDnnz	cusparseDnnz_compress
cusparseDnVecGet	cusparseDnVecGetValues	cusparseDnVecSetValues
cusparseDpruneCsr2csr	cusparseDpruneCsr2csr_bufferSizeExt	cusparseDpruneCsr2csrByPercentage
cusparseDpruneCsr2csrByPercentage_bufferSizeExt	cusparseDpruneCsr2csrNnz	cusparseDpruneCsr2csrNnzByPercentage

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cusparseDpruneDense2csr	cusparseDpruneDense2csr_bufferSizeExt	cusparseDpruneDense2csrByPercentage
cusparseDpruneDense2csrByPercentage_bufferSizeExt	cusparseDpruneDense2csrNnz	cusparseDpruneDense2csrNnzByPercentage
cusparseGather	cusparseGetErrorName	cusparseGetErrorString
cusparseGetMatDiagType	cusparseGetMatFillMode	cusparseGetMatIndexBase
cusparseGetMatType	cusparseGetPointerMode	cusparseGetProperty
cusparseGetStream	cusparseGetVersion	cusparseRot
cusparseSbsr2csr	cusparseSbsric02	cusparseSbsric02_analysis
cusparseSbsric02_bufferSize	cusparseSbsrilu02	cusparseSbsrilu02_analysis
cusparseSbsrilu02_bufferSize	cusparseSbsrilu02_numericBoost	cusparseSbsrmm
cusparseSbsrmv	cusparseSbsrsm2_analysis	cusparseSbsrsm2_bufferSize
cusparseSbsrsm2_solve	cusparseSbsrsv2_analysis	cusparseSbsrsv2_bufferSize
cusparseSbsrsv2_bufferSizeExt	cusparseSbsrsv2_solve	cusparseSbsrxmv
cusparseScatter	cusparseScsr2bsr	cusparseScsr2csr_compress
cusparseScsr2csru	cusparseScsr2gebsr	cusparseScsr2gebsr_bufferSize
cusparseScsrcolor	cusparseScsrgeam2	cusparseScsrgeam2_bufferSizeExt
cusparseScsric02	cusparseScsric02_analysis	cusparseScsric02_bufferSize
cusparseScsric02_bufferSizeExt	cusparseScsrilu02	cusparseScsrilu02_analysis
cusparseScsrilu02_bufferSize	cusparseScsrilu02_bufferSizeExt	cusparseScsrilu02_numericBoost
cusparseScsru2csr	cusparseScsru2csr_bufferSizeExt	cusparseSDDMM
cusparseSDDMM_bufferSize	cusparseSDDMM_preprocess	cusparseSetMatDiagType
cusparseSetMatFillMode	cusparseSetMatIndexBase	cusparseSetMatType
cusparseSetPointerMode	cusparseSetStream	cusparseSgebsr2csr

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cusparseSgebsr2gebsc	cusparseSgebsr2gebsc_bufferSize	cusparseSgebsr2gebsr
cusparseSgebsr2gebsr_bufferSize	cusparseSgemvi	cusparseSgemvi_bufferSize
cusparseSgpsvInterleavedBatch	cusparseSgpsvInterleavedBatch_bufferSizeExt	cusparseSgtsv2
cusparseSgtsv2_bufferSizeExt	cusparseSgtsv2_nopivot	cusparseSgtsv2_nopivot_bufferSizeExt
cusparseSgtsv2StridedBatch	cusparseSgtsv2StridedBatch_bufferSizeExt	cusparseSgtsvInterleavedBatch
cusparseSgtsvInterleavedBatch_bufferSizeExt	cusparseSnnz	cusparseSnnz_compress
cusparseSparseToDense	cusparseSparseToDense_bufferSize	cusparseSpGEMM_compute
cusparseSpGEMM_copy	cusparseSpGEMM_createDescr	cusparseSpGEMM_destroyDescr
cusparseSpGEMM_estimateMemory	cusparseSpGEMM_getNumProducts	cusparseSpGEMMreuse_compute
cusparseSpGEMMreuse_copy	cusparseSpGEMMreuse_nnz	cusparseSpGEMMreuse_workEstimation
cusparseSpGEMM_workEstimation	cusparseSpMatGetAttribute	cusparseSpMatGetFormat
cusparseSpMatGetIndexBase	cusparseSpMatGetSize	cusparseSpMatGetStridedBatch
cusparseSpMatGetValues	cusparseSpMatSetAttribute	cusparseSpMatSetValues
cusparseSpMM	cusparseSpMM_bufferSize	cusparseSpMMOp
cusparseSpMMOp_createPlan	cusparseSpMMOp_destroyPlan	cusparseSpMM_preprocess
cusparseSpMV	cusparseSpMV_bufferSize	cusparseSpMV_preprocess

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cusparseSpruneCsr2csr	cusparseSpruneCsr2csr_bufferSizeExt	cusparseSpruneCsr2csrByPercentage
cusparseSpruneCsr2csrByPercentage_bufferSizeExt	cusparseSpruneCsr2csrNnz	cusparseSpruneCsr2csrNnzByPercentage
cusparseSpruneDense2csr	cusparseSpruneDense2csr_bufferSizeExt	cusparseSpruneDense2csrByPercentage
cusparseSpruneDense2csrByPercentage_bufferSizeExt	cusparseSpruneDense2csrNnz	cusparseSpruneDense2csrNnzByPercentage
cusparseSpSM_analysis	cusparseSpSM_bufferSize	cusparseSpSM_createDescr
cusparseSpSM_destroyDescr	cusparseSpSM_solve	cusparseSpSM_updateMatrix
cusparseSpSV_analysis	cusparseSpSV_bufferSize	cusparseSpSV_createDescr
cusparseSpSV_destroyDescr	cusparseSpSV_solve	cusparseSpSV_updateMatrix
cusparseSpVecGet	cusparseSpVecGetIndexBase	cusparseSpVecGetValues
cusparseSpVecSetValues	cusparseSpVV	cusparseSpVV_bufferSize
cusparseXbsric02_zeroPivot	cusparseXbsrilu02_zeroPivot	cusparseXbsrsm2_zeroPivot
cusparseXbsrsv2_zeroPivot	cusparseXcoo2csr	cusparseXcoosort_bufferSizeExt
cusparseXcoosortByColumn	cusparseXcoosortByRow	cusparseXcscsort
cusparseXcscsort_bufferSizeExt	cusparseXcsr2bsrNnz	cusparseXcsr2coo
cusparseXcsr2gebsrNnz	cusparseXcsrgeam2Nnz	cusparseXcsric02_zeroPivot
cusparseXcsrilu02_zeroPivot	cusparseXcsrsort	cusparseXcsrsort_bufferSizeExt
cusparseXgebsr2gebsrNnz	cusparseZbsr2csr	cusparseZbsric02
cusparseZbsric02_analysis	cusparseZbsric02_bufferSize	cusparseZbsrilu02
cusparseZbsrilu02_analysis	cusparseZbsrilu02_bufferSize	cusparseZbsrilu02_numericBoost
cusparseZbsrmm	cusparseZbsrmv	cusparseZbsrsm2_analysis
cusparseZbsrsm2_bufferSize	cusparseZbsrsm2_solve	cusparseZbsrsv2_analysis
cusparseZbsrsv2_bufferSize	cusparseZbsrsv2_bufferSizeExt	cusparseZbsrsv2_solve

cusparseZbsrxmv	cusparseZcsr2bsr	cusparseZcsr2csr_compress
cusparseZcsr2csru	cusparseZcsr2gebsr	cusparseZcsr2gebsr_bufferSize
cusparseZcsrcolor	cusparseZcsrgeam2	cusparseZcsrgeam2_bufferSizeExt
cusparseZcsric02	cusparseZcsric02_analysis	cusparseZcsric02_bufferSize
cusparseZcsric02_bufferSizeExt	cusparseZcsrilu02	cusparseZcsrilu02_analysis
cusparseZcsrilu02_bufferSize	cusparseZcsrilu02_bufferSizeExt	cusparseZcsrilu02_numericBoost
cusparseZcsru2csr	cusparseZcsru2csr_bufferSizeExt	cusparseZgebsr2csr
cusparseZgebsr2gebsc	cusparseZgebsr2gebsc_bufferSize	cusparseZgebsr2gebsr
cusparseZgebsr2gebsr_bufferSize	cusparseZgemvi	cusparseZgemvi_bufferSize
cusparseZgpsvInterleavedBatch	cusparseZgpsvInterleavedBatch_bufferSizeExt	cusparseZgtsv2
cusparseZgtsv2_bufferSizeExt	cusparseZgtsv2_nopivot	cusparseZgtsv2_nopivot_bufferSizeExt
cusparseZgtsv2StridedBatch	cusparseZgtsv2StridedBatch_bufferSizeExt	cusparseZgtsvInterleavedBatch
cusparseZgtsvInterleavedBatch_bufferSizeExt	cusparseZnnz	cusparseZnnz_compress

5.7.2 未支持的函数

cusparseCbsric02_bufferSizeExt	cusparseCbsrilu02_bufferSizeExt	cusparseCbsrsm2_bufferSizeExt
cusparseCcsr2gebsr_bufferSizeExt	cusparseCgebsr2gebsc_bufferSizeExt	cusparseCgebsr2gebsr_bufferSizeExt
cusparseDbsric02_bufferSizeExt	cusparseDbsrilu02_bufferSize	cusparseDbsrsm2_bufferSizeExt

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	Ext	
cusparseDcsr2gebsr_bufferSizeExt	cusparseDgebsr2gebsc_bufferSizeExt	cusparseDgebsr2gebsr_bufferSizeExt
cusparseHpruneCsr2csr	cusparseHpruneCsr2csr_bufferSizeExt	cusparseHpruneCsr2csrByPercentage
cusparseHpruneCsr2csrByPercentage_bufferSizeExt	cusparseHpruneCsr2csrNnz	cusparseHpruneCsr2csrNnzByPercentage
cusparseHpruneDense2csr	cusparseHpruneDense2csr_bufferSizeExt	cusparseHpruneDense2csrByPercentage
cusparseHpruneDense2csrByPercentage_bufferSizeExt	cusparseHpruneDense2csrNnz	cusparseHpruneDense2csrNnzByPercentage
cusparseLoggerForceDisable	cusparseLoggerOpenFile	cusparseLoggerSetCallback
cusparseLoggerSetFile	cusparseLoggerSetLevel	cusparseLoggerSetMask
cusparseSbsric02_bufferSizeExt	cusparseSbsrilu02_bufferSizeExt	cusparseSbsrsm2_bufferSizeExt
cusparseScsr2gebsr_bufferSizeExt	cusparseSgebsr2gebsc_bufferSizeExt	cusparseSgebsr2gebsr_bufferSizeExt
cusparseXgebsr2csr	cusparseZbsric02_bufferSizeExt	cusparseZbsrilu02_bufferSizeExt
cusparseZbsrsm2_bufferSizeExt	cusparseZcsr2gebsr_bufferSizeExt	cusparseZgebsr2gebsc_bufferSizeExt
cusparseZgebsr2gebsr_bufferSizeExt		

5.8 Complex 函数

5.8.1 支持的函数

cuCabs	cuCabsf	cuCadd	cuCaddf	cuCdiv
cuCdivf	cuCfma	cuCfmaf	cuCimag	cuCimagf

cuCmul	cuCmulf	cuComplexDoubleToFloat	cuCrealf	cuConj
cuConjf	cuCreal	cuComplexFloatToDouble	cuCsub	cuCsubf
make_cuComplex	make_cuDoubleComplex	make_cuFloatComplex		

5.9 CUPTI 函数

5.9.1 支持的函数

cuptiActivityEnable	cuptiActivityDisable	cuptiActivityRegisterCallbacks
cuptiActivityGetAttribute	cuptiActivitySetAttribute	cuptiRangeProfilerPopRange
cuptiActivityGetNextRecord	cuptiGetResultString	cuptiActivityGetNumDroppedRecords
cuptiActivityFlushAll	cuptiSubscribe	cuptiUnsubscribe
cuptiEnableCallback	cuptiEnableDomain	cuptiRangeProfilerCounterDataGetRangeInfo
cuptiActivityEnableDriverApi	cuptiActivityEnableRuntimeApi	cuptiRangeProfilerGetCounterDataInfo
cuptiGetErrorMessage	cuptiRangeProfilerCounterDataImageInitialize	cuptiRangeProfilerGetCounterDataSize
cuptiProfilerHostInitialize	cuptiProfilerHostDeinitialize	cuptiProfilerHostEvaluateToGpuValues
cuptiProfilerHostConfigAddMetrics	cuptiProfilerHostGetNumOfPasses	cuptiRangeProfilerSetConfig
cuptiRangeProfilerEnable	cuptiRangeProfilerDisable	cuptiRangeProfilerStart
cuptiRangeProfilerStop	cuptiRangeProfilerPushRange	

5.9.2 未支持的函数

cuptiActivityConfigurePCSamp	cuptiActivityConfigureUnified	cuptiActivityDisableContext

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ling	MemoryCounter	
cuptiActivityEnableContext	cuptiActivityEnableLatencyTimestamps	cuptiActivityFlush
cuptiActivityPopExternalCorrelationId	cuptiActivityPushExternalCorrelationId	cuptiEnableKernelReplayMode
cuptiComputeCapabilitySupported	cuptiDeviceEnumEventDomains	cuptiDeviceEnumMetrics
cuptiDeviceGetAttribute	cuptiDeviceGetEventDomainAttribute	cuptiDeviceGetNumEventDomains
cuptiDeviceGetNumMetrics	cuptiDeviceGetTimestamp	cuptiDeviceSupported
cuptiDeviceVirtualizationMode	cuptiDisableKernelReplayMode	cuptiEnableAllDomains
cuptiEnumEventDomains	cuptiEnumMetrics	cuptiEventDomainEnumEvents
cuptiEventDomainGetAttribute	cuptiEventDomainGetNumEvents	cuptiEventGetAttribute
cuptiEventGetIdFromName	cuptiEventGroupAddEvent	cuptiEventGroupCreate
cuptiEventGroupDestroy	cuptiEventGroupDisable	cuptiEventGroupEnable
cuptiEventGroupGetAttribute	cuptiEventGroupReadAllEvents	cuptiEventGroupReadEvent
cuptiEventGroupRemoveAllEvents	cuptiEventGroupRemoveEvent	cuptiEventGroupResetAllEvents
cuptiEventGroupSetAttribute	cuptiEventGroupSetDisable	cuptiEventGroupSetEnable
cuptiEventGroupSetsCreate	cuptiEventGroupSetsDestroy	cuptiFinalize
cuptiGetAutoBoostState	cuptiGetCallbackName	cuptiGetCallbackState
cuptiGetContextId	cuptiGetDeviceId	cuptiGetLastError
cuptiGetNumEventDomains	cuptiGetNumMetrics	cuptiOpenACCInitialize
cuptiGetStreamId	cuptiGetStreamIdEx	cuptiGetThreadIdType
cuptiGetVersion	cuptiKernelReplaySubscribeUpdate	cuptiMetricCreateEventGroupSets
cuptiMetricEnumEvents	cuptiMetricEnumProperties	cuptiMetricGetAttribute

cuptiMetricGetIdFromName	cuptiMetricGetNumEvents	cuptiMetricGetNumProperties
cuptiMetricGetRequiredEventGroupSets	cuptiMetricGetValue	cuptiMetricGetValue2
cuptiSetEventCollectionMode	cuptiSetThreadIdType	cuptiSupportedDomains
cuptiGetGraphExecId	cuptiActivityEnableDeviceGraph	cuptiActivityEnableAllocationSource
cuptiProfilerHostGetSupportedChips	cuptiProfilerHostGetBaseMetrics	cuptiProfilerHostGetSubMetrics
cuptiProfilerHostGetMetricProperties	cuptiProfilerHostGetRangeName	cuptiProfilerHostGetConfigImageSize
cuptiProfilerHostGetConfigImage	cuptiProfilerHostGetMaxNumHardwareMetricsPerPass	cuptiRangeProfilerDecodeData
NVPW_CounterData_ExtractCounterDataPrefix	NVPW_CounterDataCombiner_CopyIntoRange	NVPW_PeriodicSampler_Config_GetSocEstimatedSampleSize
NVPW_PeriodicSampler_Config_GetGpuEstimatedSampleSize	NVPW_Config_GetRawCounterInfo	NVPW_Config_GetRawCounters
NVPW_PeriodicSampler_CounterData_IsDataComplete	cuptiSassMetricsSetConfig	cuptiSassMetricsUnsetConfig
cuptiPmSamplingSetConfig	cuptiPmSamplingEnable	cuptiPmSamplingDisable
cuptiPmSamplingStart	cuptiPmSamplingStop	cuptiPmSamplingDecodeData
cuptiPmSamplingGetCounterAvailability	cuptiPmSamplingGetCounterDataSize	cuptiPmSamplingCounterDataImageInitialize
cuptiPmSamplingGetCounterDataInfo	cuptiPmSamplingCounterDataGetSampleInfo	
cuptiSassMetricsDisable	cuptiSassMetricsEnable	cuptiSassMetricsFlushData
cuptiSassMetricsGetDataProperties	cuptiSassMetricsGetMetrics	cuptiSassMetricsGetNumOfMetrics
cuptiSassMetricsGetProperties		

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5.10 cuSOLVER 函数

5.10.1 支持的函数

cusolverDnCCgels	cusolverDnCCgels_bufferSize	cusolverDnCCgesv
cusolverDnCCgesv_bufferSize	cusolverDnCEgels	cusolverDnCEgels_bufferSize
cusolverDnCEgesv	cusolverDnCEgesv_bufferSize	cusolverDnCgebrd
cusolverDnCgebrd_bufferSize	cusolverDnCgeqrf	cusolverDnCgeqrf_bufferSize
cusolverDnCgesvd	cusolverDnCgesvdaStridedBatched	cusolverDnCgesvdaStridedBatched_bufferSize
cusolverDnCgesvd_bufferSize	cusolverDnCgesvdj	cusolverDnCgesvdjBatched
cusolverDnCgesvdjBatched_bufferSize	cusolverDnCgesvdj_bufferSize	cusolverDnCgetrf
cusolverDnCgetrf_bufferSize	cusolverDnCgetrs	cusolverDnCheevd
cusolverDnCheevd_bufferSize	cusolverDnCheevdx	cusolverDnCheevdx_bufferSize
cusolverDnCheevj	cusolverDnCheevjBatched	cusolverDnCheevjBatched_bufferSize
cusolverDnCheevj_bufferSize	cusolverDnChegvd	cusolverDnChegvd_bufferSize
cusolverDnChegvdx	cusolverDnChegvdx_bufferSize	cusolverDnChegvj
cusolverDnChegvj_bufferSize	cusolverDnChetrd	cusolverDnChetrd_bufferSize
cusolverDnCKgels	cusolverDnCKgels_bufferSize	cusolverDnCKgesv
cusolverDnCKgesv_bufferSize	cusolverDnClaswp	cusolverDnCpotrf
cusolverDnCpotrfBatched	cusolverDnCpotrf_bufferSize	cusolverDnCpotri
cusolverDnCpotri_bufferSize	cusolverDnCpotrs	cusolverDnCpotrsBatched
cusolverDnCreate	cusolverDnCreateGesvdjInfo	cusolverDnCreateParams
cusolverDnCreateSyevjInfo	cusolverDnCsytrf	cusolverDnCsytrf_bufferSize
cusolverDnCungbr	cusolverDnCungbr_bufferSize	cusolverDnCungqr
cusolverDnCungqr_bufferSize	cusolverDnCungtr	cusolverDnCungtr_bufferSize
cusolverDnCunmqr	cusolverDnCunmqr_bufferSize	cusolverDnCunmtr

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cusolverDnCunmtr_bufferSize	cusolverDnCYgels	cusolverDnCYgels_bufferSize
cusolverDnCYgesv	cusolverDnCYgesv_bufferSize	cusolverDnDBgels
cusolverDnDBgels_bufferSize	cusolverDnDBgesv	cusolverDnDBgesv_bufferSize
cusolverDnDDgels	cusolverDnDDgels_bufferSize	cusolverDnDDgesv
cusolverDnDDgesv_bufferSize	cusolverDnDestroy	cusolverDnDestroyGesvdjInfo
cusolverDnDestroyParams	cusolverDnDestroySyevjInfo	cusolverDnDgebrd
cusolverDnDgebrd_bufferSize	cusolverDnDgeqrf	cusolverDnDgeqrf_bufferSize
cusolverDnDgesvd	cusolverDnDgesvdaStridedBatched	cusolverDnDgesvdaStridedBatched_bufferSize
cusolverDnDgesvd_bufferSize	cusolverDnDgesvdj	cusolverDnDgesvdjBatched
cusolverDnDgesvdjBatched_bufferSize	cusolverDnDgesvdj_bufferSize	cusolverDnDgetrf
cusolverDnDgetrf_bufferSize	cusolverDnDgetrs	cusolverDnDHgels
cusolverDnDHgels_bufferSize	cusolverDnDHgesv	cusolverDnDHgesv_bufferSize
cusolverDnDlaswp	cusolverDnDorgbr	cusolverDnDorgbr_bufferSize
cusolverDnDorgqr	cusolverDnDorgqr_bufferSize	cusolverDnDorgtr
cusolverDnDorgtr_bufferSize	cusolverDnDormqr	cusolverDnDormqr_bufferSize
cusolverDnDormtr	cusolverDnDormtr_bufferSize	cusolverDnDpotrf
cusolverDnDpotrfBatched	cusolverDnDpotrf_bufferSize	cusolverDnDpotri
cusolverDnDpotri_bufferSize	cusolverDnDpotrs	cusolverDnDpotrsBatched
cusolverDnDSgels	cusolverDnDSgels_bufferSize	cusolverDnDSgesv
cusolverDnDSgesv_bufferSize	cusolverDnDsyevd	cusolverDnDsyevd_bufferSize
cusolverDnDsyevdx	cusolverDnDsyevdx_bufferSize	cusolverDnDsyevj
cusolverDnDsyevjBatched	cusolverDnDsyevjBatched_bufferSize	cusolverDnDsyevj_bufferSize
cusolverDnDsygvd	cusolverDnDsygvd_bufferSize	cusolverDnDsygvdx
cusolverDnDsygvdx_bufferSize	cusolverDnDsygvj	cusolverDnDsygvj_bufferSize

cusolverDnDsytrd	cusolverDnDsytrd_bufferSize	cusolverDnDsytrf
cusolverDnDsytrf_bufferSize	cusolverDnDXgels	cusolverDnDXgels_bufferSize
cusolverDnDXgesv	cusolverDnDXgesv_bufferSize	cusolverDnGetDeterministicMode
cusolverDnGetStream	cusolverDnIRSInfosCreate	cusolverDnIRSInfosDestroy
cusolverDnIRSInfosGetMaxIters	cusolverDnIRSInfosGetNiters	cusolverDnIRSInfosGetOuterNiters
cusolverDnIRSInfosGetResidualHistory	cusolverDnIRSInfosRequestResidual	cusolverDnIRSParamsCreate
cusolverDnIRSParamsDestroy	cusolverDnIRSParamsDisableFallback	cusolverDnIRSParamsEnableFallback
cusolverDnIRSParamsGetMaxIters	cusolverDnIRSParamsSetMaxIters	cusolverDnIRSParamsSetMaxItersInner
cusolverDnIRSParamsSetRefinementSolver	cusolverDnIRSParamsSetSolverLowestPrecision	cusolverDnIRSParamsSetSolverMainPrecision
cusolverDnIRSParamsSetSolverPrecisions	cusolverDnIRSParamsSetTol	cusolverDnIRSParamsSetTolInner
cusolverDnIRSXgels	cusolverDnIRSXgels_bufferSize	cusolverDnIRSXgesv
cusolverDnIRSXgesv_bufferSize	cusolverDnSBgels	cusolverDnSBgels_bufferSize
cusolverDnSBgesv	cusolverDnSBgesv_bufferSize	cusolverDnSetAdvOptions
cusolverDnSetDeterministicMode	cusolverDnSetStream	cusolverDnSgebrd
cusolverDnSgebrd_bufferSize	cusolverDnSgeqrf	cusolverDnSgeqrf_bufferSize
cusolverDnSgesvd	cusolverDnSgesvdaStridedBatched	cusolverDnSgesvdaStridedBatched_bufferSize
cusolverDnSgesvd_bufferSize	cusolverDnSgesvdj	cusolverDnSgesvdjBatched

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cusolverDnSgesvdjBatched_bufferSize	cusolverDnSgesvdj_bufferSize	cusolverDnSgetrf
cusolverDnSgetrf_bufferSize	cusolverDnSgetrs	cusolverDnSHgels
cusolverDnSHgels_bufferSize	cusolverDnSHgesv	cusolverDnSHgesv_bufferSize
cusolverDnSlaswp	cusolverDnSorgbr	cusolverDnSorgbr_bufferSize
cusolverDnSorgqr	cusolverDnSorgqr_bufferSize	cusolverDnSorgtr
cusolverDnSorgtr_bufferSize	cusolverDnSormqr	cusolverDnSormqr_bufferSize
cusolverDnSormtr	cusolverDnSormtr_bufferSize	cusolverDnSpotrf
cusolverDnSpotrfBatched	cusolverDnSpotrf_bufferSize	cusolverDnSpotri
cusolverDnSpotri_bufferSize	cusolverDnSpotrs	cusolverDnSpotrsBatched
cusolverDnSSgels	cusolverDnSSgels_bufferSize	cusolverDnSSgesv
cusolverDnSSgesv_bufferSize	cusolverDnSsyevd	cusolverDnSsyevd_bufferSize
cusolverDnSsyevdx	cusolverDnSsyevdx_bufferSize	cusolverDnSsyevj
cusolverDnSsyevjBatched	cusolverDnSsyevjBatched_bufferSize	cusolverDnSsyevj_bufferSize
cusolverDnSsygvd	cusolverDnSsygvd_bufferSize	cusolverDnSsygvdx
cusolverDnSsygvdx_bufferSize	cusolverDnSsygvj	cusolverDnSsygvj_bufferSize
cusolverDnSsytrd	cusolverDnSsytrd_bufferSize	cusolverDnSsytrf
cusolverDnSsytrf_bufferSize	cusolverDnSXgels	cusolverDnSXgels_bufferSize
cusolverDnSXgesv	cusolverDnSXgesv_bufferSize	cusolverDnXgeev
cusolverDnXgeev_bufferSize	cusolverDnXgeqrf	cusolverDnXgeqrf_bufferSize
cusolverDnXgesvd	cusolverDnXgesvd_bufferSize	cusolverDnXgesvdjGetResidual
cusolverDnXgesvdjGetSweeps	cusolverDnXgesvdjSetMaxSweeps	cusolverDnXgesvdjSetSortEig
cusolverDnXgesvdjSetTolerance	cusolverDnXgesvdp	cusolverDnXgesvdp_bufferSize
cusolverDnXgesvdr	cusolverDnXgesvdr_bufferSize	cusolverDnXgetrf
cusolverDnXgetrf_bufferSize	cusolverDnXgetrs	cusolverDnXlarft
cusolverDnXlarft_bufferSize	cusolverDnXpotrf	cusolverDnXpotrf_bufferSize

cusolverDnXpotrs	cusolverDnXsyevBatched	cusolverDnXsyevBatched_bufferSize
cusolverDnXsyevd	cusolverDnXsyevd_bufferSize	cusolverDnXsyevdx
cusolverDnXsyevdx_bufferSize	cusolverDnXsyevjGetResidual	cusolverDnXsyevjGetSweeps
cusolverDnXsyevjSetMaxSweeps	cusolverDnXsyevjSetSortEig	cusolverDnXsyevjSetTolerance
cusolverDnXsytrs	cusolverDnXsytrs_bufferSize	cusolverDnXtrtri
cusolverDnXtrtri_bufferSize	cusolverDnZCgels	cusolverDnZCgels_bufferSize
cusolverDnZCgesv	cusolverDnZCgesv_bufferSize	cusolverDnZEgels
cusolverDnZEgels_bufferSize	cusolverDnZEgesv	cusolverDnZEgesv_bufferSize
cusolverDnZgebrd	cusolverDnZgebrd_bufferSize	cusolverDnZgeqrf
cusolverDnZgeqrf_bufferSize	cusolverDnZgesvd	cusolverDnZgesvdaStridedBatched
cusolverDnZgesvdaStridedBatched_bufferSize	cusolverDnZgesvd_bufferSize	cusolverDnZgesvdj
cusolverDnZgesvdjBatched	cusolverDnZgesvdjBatched_bufferSize	cusolverDnZgesvdj_bufferSize
cusolverDnZgetrf	cusolverDnZgetrf_bufferSize	cusolverDnZgetrs
cusolverDnZheevd	cusolverDnZheevd_bufferSize	cusolverDnZheevdx
cusolverDnZheevdx_bufferSize	cusolverDnZheevj	cusolverDnZheevjBatched
cusolverDnZheevjBatched_bufferSize	cusolverDnZheevj_bufferSize	cusolverDnZhegvd
cusolverDnZhegvd_bufferSize	cusolverDnZhegvdx	cusolverDnZhegvdx_bufferSize
cusolverDnZhegvj	cusolverDnZhegvj_bufferSize	cusolverDnZhetrd
cusolverDnZhetrd_bufferSize	cusolverDnZKgels	cusolverDnZKgels_bufferSize
cusolverDnZKgesv	cusolverDnZKgesv_bufferSize	cusolverDnZlaswp
cusolverDnZpotrf	cusolverDnZpotrfBatched	cusolverDnZpotrf_bufferSize
cusolverDnZpotri	cusolverDnZpotri_bufferSize	cusolverDnZpotrs

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cusolverDnZpotrsBatched	cusolverDnZsytrf	cusolverDnZsytrf_bufferSize
cusolverDnZungbr	cusolverDnZungbr_bufferSize	cusolverDnZungqr
cusolverDnZungqr_bufferSize	cusolverDnZungtr	cusolverDnZungtr_bufferSize
cusolverDnZunmqr	cusolverDnZunmqr_bufferSize	cusolverDnZunmtr
cusolverDnZunmtr_bufferSize	cusolverDnZYgels	cusolverDnZYgels_bufferSize
cusolverDnZYgesv	cusolverDnZYgesv_bufferSize	cusolverDnZZgels
cusolverDnZZgels_bufferSize	cusolverDnZZgesv	cusolverDnZZgesv_bufferSize
cusolverGetProperty	cusolverGetVersion	cusolverMgCreate
cusolverMgDestroy	cusolverMgDeviceSelect	cusolverMgCreateDeviceGrid
cusolverMgDestroyGrid	cusolverMgCreateMatrixDesc	cusolverMgDestroyMatrixDesc
cusolverMgSyevd_bufferSize	cusolverMgSyevd	cusolverMgGetrf_bufferSize
cusolverMgGetrf	cusolverMgGetrs_bufferSize	cusolverMgGetrs
cusolverMgPotrf_bufferSize	cusolverMgPotrf	cusolverMgPotrs_bufferSize
cusolverMgPotrs	cusolverMgPotri_bufferSize	cusolverMgPotri
cusolverRfGetNumericProperties	cusolverRfGetResetValuesFastMode	cusolverRfRefactor
cusolverRfResetValues	cusolverRfSetAlgs	cusolverRfSetMatrixFormat
cusolverRfSetNumericProperties	cusolverRfSetResetValuesFastMode	cusolverRfSetupDevice
cusolverRfSetupHost	cusolverRfSolve	cusolverSpCreate
cusolverSpDestroy	cusolverSpSetStream	cusolverSpGetStream
cusolverSpXcsrissymHost	cusolverSpScsrlsvluHost	cusolverSpDcsrlsvluHost
cusolverSpCcsrlsvluHost	cusolverSpZcsrlsvluHost	cusolverSpScsrlsvqr
cusolverSpDcsrlsvqr	cusolverSpCcsrlsvqr	cusolverSpZcsrlsvqr
cusolverSpScsrlsvqrHost	cusolverSpDcsrlsvqrHost	cusolverSpCcsrlsvqrHost
cusolverSpZcsrlsvqrHost	cusolverSpScsrlsvcholHost	cusolverSpDcsrlsvcholHost
cusolverSpCcsrlsvcholHost	cusolverSpZcsrlsvcholHost	cusolverSpScsrlsvchol
cusolverSpDcsrlsvchol	cusolverSpCcsrlsvchol	cusolverSpZcsrlsvchol

cusolverSpScsrlsqvqrHost	cusolverSpDcsrlsqvqrHost	cusolverSpCcsrlsqvqrHost
cusolverSpZcsrlsqvqrHost	cusolverSpScsreigvsiHost	cusolverSpDcsreigvsiHost
cusolverSpCcsreigvsiHost	cusolverSpZcsreigvsiHost	cusolverSpScsreigvsi
cusolverSpDcsreigvsi	cusolverSpCcsreigvsi	cusolverSpZcsreigvsi
cusolverSpScsreigsHost	cusolverSpDcsreigsHost	cusolverSpCcsreigsHost
cusolverSpZcsreigsHost	cusolverSpXcsrsymrcmHost	cusolverSpXcsrsymmdqHost
cusolverSpXcsrsymamdHost	cusolverSpXcsrmetisndHost	cusolverSpScsrzfdHost
cusolverSpDcsrzfdHost	cusolverSpCcsrzfdHost	cusolverSpZcsrzfdHost
cusolverSpXcsrperm_bufferSizeHost	cusolverSpXcsrpermHost	cusolverSpCreateCsrqrInfo
cusolverSpDestroyCsrqrInfo	cusolverSpXcsrqrAnalysisBatched	cusolverSpScsrqrBufferInfoBatched
cusolverSpDcsrqrBufferInfoBatched	cusolverSpCcsrqrBufferInfoBatched	cusolverSpZcsrqrBufferInfoBatched
cusolverSpScsrqrsvBatched	cusolverSpDcsrqrsvBatched	cusolverSpCcsrqrsvBatched
cusolverSpZcsrqrsvBatched		

5.10.2 未支持的函数

cusolverDnClauum	cusolverDnClauum_bufferSize	cusolverDnCsytri
cusolverDnCsytri_bufferSize	cusolverDnDlauum	cusolverDnDlauum_bufferSize
cusolverDnDsytri	cusolverDnDsytri_bufferSize	cusolverDnGeqrf
cusolverDnGeqrf_bufferSize	cusolverDnGesvd	cusolverDnGesvd_bufferSize
cusolverDnGetrf	cusolverDnGetrf_bufferSize	cusolverDnGetrs
cusolverDnLoggerForceDisable	cusolverDnLoggerOpenFile	cusolverDnLoggerSetCallback
cusolverDnLoggerSetFile	cusolverDnLoggerSetLevel	cusolverDnLoggerSetMask
cusolverDnPotrf	cusolverDnPotrf_bufferSize	cusolverDnPotrs
cusolverDnSlauum	cusolverDnSlauum_bufferSize	cusolverDnSsytri
cusolverDnSsytri_bufferSize	cusolverDnSyevd	cusolverDnSyevd_bufferSize
cusolverDnSyevdx	cusolverDnSyevdx_bufferSize	cusolverDnZlauum

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cusolverDnZlauum_bufferSize	cusolverDnZsytri	cusolverDnZsytri_bufferSize

5.11 cuDNN 函数

5.11.1 支持的函数

cudnnActivationBackward	cudnnActivationForward	cudnnAddTensor
cudnnBatchNormalizationBackward	cudnnBatchNormalizationForwardInference	cudnnBatchNormalizationForwardTraining
cudnnCTCLoss	cudnnConvolutionBackwardBias	cudnnConvolutionBackwardData
cudnnConvolutionBackwardFilter	cudnnConvolutionBiasActivationForward	cudnnConvolutionForward
cudnnCreate	cudnnCreateActivationDescriptor	cudnnCreateCTCLossDescriptor
cudnnCreateConvolutionDescriptor	cudnnCreateDropoutDescriptor	cudnnCreateFilterDescriptor
cudnnCreateLRNDescriptor	cudnnCreateOpTensorDescriptor	cudnnCreatePersistentRNNPlan
cudnnCreatePoolingDescriptor	cudnnCreateRNNDescriptor	cudnnCreateReduceTensorDescriptor
cudnnCreateTensorDescriptor	cudnnDeriveBNTensorDescriptor	cudnnDestroy
cudnnDestroyActivationDescriptor	cudnnDestroyCTCLossDescriptor	cudnnDestroyConvolutionDescriptor
cudnnDestroyDropoutDescriptor	cudnnDestroyFilterDescriptor	cudnnDestroyLRNDescriptor
cudnnDestroyOpTensorDescriptor	cudnnDestroyPersistentRNNPlan	cudnnDestroyPoolingDescriptor
cudnnDestroyRNNDescriptor	cudnnDestroyReduceTensorDescriptor	cudnnDestroyTensorDescriptor

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cudnnDropoutBackward	cudnnDropoutForward	cudnnDropoutGetReserveSpaceSize
cudnnDropoutGetStatesSize	cudnnFindConvolutionBackwardDataAlgorithm	cudnnFindConvolutionBackwardDataAlgorithmEx
cudnnFindConvolutionBackwardFilterAlgorithm	cudnnFindConvolutionBackwardFilterAlgorithmEx	cudnnFindConvolutionForwardAlgorithm
cudnnFindConvolutionForwardAlgorithmEx	cudnnGetActivationDescriptor	cudnnGetCTCLossDescriptor
cudnnGetCTCLossWorkspaceSize	cudnnGetConvolution2dDescriptor	cudnnGetConvolution2dForwardOutputDim
cudnnGetConvolutionBackwardDataWorkspaceSize	cudnnGetConvolutionBackwardFilterWorkspaceSize	cudnnGetConvolutionForwardWorkspaceSize
cudnnGetConvolutionMathType	cudnnGetConvolutionNdDescriptor	cudnnGetConvolutionNdForwardOutputDim
cudnnGetCudartVersion	cudnnGetDropoutDescriptor	cudnnGetErrorString
cudnnGetFilter4dDescriptor	cudnnGetFilterNdDescriptor	cudnnGetFilterSizeInBytes
cudnnGetLRNDescriptor	cudnnGetMaxDeviceVersion	cudnnGetOpTensorDescriptor
cudnnGetPooling2dDescriptor	cudnnGetPooling2dForwardOutputDim	cudnnGetPoolingNdDescriptor
cudnnGetPoolingNdForwardOutputDim	cudnnGetRNNLinLayerBiasParams	cudnnGetRNNLinLayerMatrixParams
cudnnGetRNNParamsSize	cudnnGetRNNTrainingReserveSize	cudnnGetRNNWorkspaceSize
cudnnGetReduceTensorDescriptor	cudnnGetReductionIndicesSize	cudnnGetReductionWorkspaceSize
cudnnGetStream	cudnnGetTensor4dDescriptor	cudnnGetTensorNdDescriptor
cudnnGetTensorSizeInBytes	cudnnGetVersion	cudnnLRNCrossChannelBackward

cudnnLRNCrossChannelForward	cudnnOpTensor	cudnnPoolingBackward
cudnnPoolingForward	cudnnRNNBackwardData	cudnnRNNBackwardWeights
cudnnRNNForwardInference	cudnnRNNForwardTraining	cudnnReduceTensor
cudnnRestoreDropoutDescriptor	cudnnScaleTensor	cudnnSetActivationDescriptor
cudnnSetCTCLossDescriptor	cudnnSetConvolution2dDescriptor	cudnnSetConvolutionGroupCount
cudnnSetConvolutionMathType	cudnnSetConvolutionNdDescriptor	cudnnSetDropoutDescriptor
cudnnSetFilter4dDescriptor	cudnnSetFilterNdDescriptor	cudnnSetLRNDescriptor
cudnnSetOpTensorDescriptor	cudnnSetPersistentRNNPlan	cudnnSetPooling2dDescriptor
cudnnSetPoolingNdDescriptor	cudnnSetRNNDescriptor_v6	cudnnSetReduceTensorDescriptor
cudnnSetStream	cudnnSetTensor	cudnnSetTensor4dDescriptor
cudnnSetTensor4dDescriptorEx	cudnnSetTensorNdDescriptor	cudnnSoftmaxBackward
cudnnSoftmaxForward	cudnnTransformTensor	cudnnBuildRNNDynamic
cudnnBatchNormalizationBackwardEx	cudnnBatchNormalizationForwardTrainingEx	cudnnGetBatchNormalizationBackwardExWorkspaceSize
cudnnCTCLoss_v8	cudnnGetActivationDescriptorSwishBeta	cudnnGetCTCLossDescriptorEx
cudnnGetBatchNormalizationForwardTrainingExWorkspaceSize	cudnnGetBatchNormalizationTrainingExReserveSpaceSize	cudnnGetConvolutionBackwardDataAlgorithm_v7
cudnnGetCTCLossDescriptor_v8	cudnnGetCTCLossWorkspaceSize_v8	cudnnGetConvolutionGroupCount
cudnnGetConvolutionBackwardFilterAlgorithm_v7	cudnnGetConvolutionForwardAlgorithm_v7	cudnnGetRNNDataDescriptor
cudnnGetConvolutionReorderType	cudnnGetRNNBiasMode	cudnnGetRNNProjectionLayers

cudnnGetRNNMatrixMathType	cudnnGetRNNPaddingMode	cudnnRNNBackwardDataEx
cudnnGetRNNTempSpaceSizes	cudnnGetRNNWeightParams	cudnnRNNBackwardWeights_v8
cudnnRNNBackwardData_v8	cudnnRNNBackwardWeightsEx	cudnnReorderFilterAndBias
cudnnRNNForwardInferenceEx	cudnnRNNForwardTrainingEx	cudnnSetCTCLossDescriptor_v8
cudnnSetActivationDescriptorSwishBeta	cudnnSetCTCLossDescriptorEx	cudnnSetRNNDescriptor_v8
cudnnSetConvolutionReorderType	cudnnSetRNNBiasMode	cudnnSetRNNProjectionLayers
cudnnSetRNNMatrixMathType	cudnnSetRNNPaddingMode	cudnnSetTensorNdDescriptorEx

5.11.2 未支持的函数

cudnnAdvInferVersionCheck	cudnnAdvTrainVersionCheck	cudnnBackendCreateDescriptor
cudnnBackendDestroyDescriptor	cudnnBackendExecute	cudnnBackendFinalize
cudnnBackendGetAttribute	cudnnBackendInitialize	cudnnBackendSetAttribute
	cudnnCnnInferVersionCheck	cudnnCnnTrainVersionCheck
cudnnCopyAlgorithmDescriptor	cudnnCreateAlgorithmDescriptor	cudnnCreateAlgorithmPerformance
cudnnCreateAttnDescriptor	cudnnCreateFusedOpsConstParamPack	cudnnCreateFusedOpsPlan
cudnnCreateFusedOpsVariantParamPack	cudnnCreateTensorTransformDescriptor	cudnnCreateSeqDataDescriptor
cudnnCreateSpatialTransformerDescriptor	cudnnDestroyAlgorithmPerformance	cudnnDeriveNormTensorDescriptor
cudnnDestroyAlgorithmDescriptor	cudnnDestroyFusedOpsPlan	cudnnDestroyAttnDescriptor
cudnnDestroyFusedOpsConstPa	cudnnDestroySeqDataDescripto	cudnnDestroyFusedOpsVariantP

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ramPack	r	aramPack
cudnnDestroyRNNDataDescriptor	cudnnDivisiveNormalizationBackward	cudnnDestroySpatialTransformerDescriptor
cudnnDestroyTensorTransformDescriptor	cudnnFindRNNBackwardWeightsAlgorithmEx	cudnnDivisiveNormalizationForward
cudnnFindRNNBackwardDataAlgorithmEx	cudnnFusedOpsExecute	cudnnFindRNNForwardInferenceAlgorithmEx
cudnnFindRNNForwardTrainingAlgorithmEx	cudnnGetAlgorithmPerformance	cudnnGetAlgorithmSpaceSize
cudnnGetAlgorithmDescriptor	cudnnGetCallback	cudnnGetConvolutionBackwardDataAlgorithmMaxCount
cudnnGetAttnDescriptor	cudnnGetConvolutionBackwardFilterAlgorithmMaxCount	cudnnGetFusedOpsConstParamPackAttribute
cudnnGetConvolutionForwardAlgorithmMaxCount	cudnnGetFoldedConvBackwardDataDescriptors	cudnnGetMultiHeadAttnWeights
cudnnMultiHeadAttnBackwardData	cudnnGetMultiHeadAttnBuffers	cudnnGetNormalizationTrainingReserveSpaceSize
cudnnGetFusedOpsVariantParamPackAttribute	cudnnGetNormalizationForwardTrainingWorkspaceSize	cudnnGetRNNBackwardWeightsAlgorithmMaxCount
cudnnGetNormalizationBackwardWorkspaceSize	cudnnGetRNNBackwardDataAlgorithmMaxCount	cudnnGetRNNDescriptor_v6
cudnnGetProperty	cudnnGetTensorTransformDescriptor	cudnnGetRNNForwardTrainingAlgorithmMaxCount
cudnnGetRNNDescriptor_v8	cudnnGetRNNForwardInferenceAlgorithmMaxCount	cudnnIm2Col
cudnnGetSeqDataDescriptor	cudnnMakeFusedOpsPlan	cudnnNormalizationBackward
cudnnInitTransformDest	cudnnMultiHeadAttnForward	cudnnOpsInferVersionCheck
cudnnMultiHeadAttnBackwardWeights	cudnnNormalizationForwardTraining	cudnnOpsTrainVersionCheck

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cudnnNormalizationForwardInference	cudnnQueryRuntimeError	cudnnRNNSetClip
cudnnRNNGetClip	cudnnRNNGetClip_v8	cudnnRestoreAlgorithm
cudnnRNNSetClip_v8	cudnnSetRNNAlgorithmDescriptor	cudnnSetAlgorithmDescriptor
cudnnSaveAlgorithm	cudnnSetAttnDescriptor	cudnnSpatialTfGridGeneratorForward
cudnnSetAlgorithmPerformance	cudnnSetCallback	cudnnTransformFilter
cudnnSetFusedOpsConstParamPackAttribute	cudnnSetRNNDataDescriptor	cudnnSetFusedOpsVariantParamPackAttribute
cudnnSpatialTfSamplerBackward	cudnnSetSpatialTransformerNdDescriptor	cudnnTransformTensorEx
cudnnSetSeqDataDescriptor	cudnnSpatialTfGridGeneratorBackward	cudnnSpatialTfSamplerForward
cudnnSetTensorTransformDescriptor		

5.12 NCCL 函数

5.12.1 支持的函数

ncclAllGather	ncclAllReduce	ncclBroadcast
ncclCommAbort	ncclCommCount	ncclCommCuDevice
ncclCommDestroy	ncclCommGetAsyncError	ncclCommInitAll
ncclCommInitRank	ncclCommUserRank	ncclGetErrorString
ncclGetUniqueId	ncclGetVersion	ncclGroupEnd
ncclGroupStart	ncclRecv	ncclReduce
ncclReduceScatter	ncclSend	

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5.13 NVRTC 函数

5.13.1 支持的函数

nvrtcAddNameExpression	nvrtcCompileProgram	nvrtcCreateProgram
nvrtcDestroyProgram	nvrtcGetErrorString	nvrtcGetLoweredName
nvrtcGetPTX	nvrtcGetPTXSize	nvrtcGetProgramLog
nvrtcGetProgramLogSize	nvrtcVersion	nvrtcGetCUBINSize
nvrtcGetCUBIN	nvrtcGetNumSupportedArchs	nvrtcGetSupportedArchs

5.13.2 未支持的函数

nvrtcSetFlowCallback	nvrtcGetNVVMSize	nvrtcGetNVVM
nvrtcGetLTOIRSize	nvrtcGetLTOIR	nvrtcGetOptiXIRSize
nvrtcGetOptiXIR	nvrtcGetPCHHeapSize	nvrtcSetPCHHeapSize
nvrtcGetPCHCreateStatus	nvrtcGetPCHHeapSizeRequired	

5.14 NVML 函数

5.14.1 支持的函数

nvmlDeviceGetArchitecture	nvmlInitWithFlags	nvmlInit_v2
nvmlDeviceGetCount_v2	nvmlDeviceGetHandleByIndex_v2	nvmlShutdown
nvmlErrorString	nvmlSystemGetDriverVersion	nvmlSystemGetNVMLVersion
nvmlSystemGetCudaDriverVersion	nvmlSystemGetCudaDriverVersion_v2	nvmlDeviceGetVbiosVersion
nvmlDeviceGetName	nvmlDeviceGetBrand	nvmlDeviceGetIndex
nvmlDeviceGetUUID	nvmlDeviceGetSerial	nvmlDeviceGetPciInfo_v3
nvmlDeviceGetHandleByPciBusId_v2	nvmlDeviceGetHandleBySerial	nvmlDeviceGetHandleByUUID
nvmlDeviceGetIndexByUUID	nvmlDeviceGetMemoryInfo	nvmlDeviceGetMemoryInfo_v2

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nvmlDeviceGetSupportedMemoryClocks	nvmlDeviceResetMemoryLockedClocks	nvmlDeviceSetMemoryLockedClocks
nvmlDeviceGetRetiredPages	nvmlDeviceGetRetiredPages_v2	nvmlDeviceGetRetiredPagesPendingStatus
nvmlDeviceGetComputeRunningProcesses_v3	nvmlDeviceGetAccountingPids	nvmlDeviceGetAccountingStats
nvmlDeviceGetProcessUtilization	nvmlDeviceGetUtilizationRates	nvmlSystemGetProcessName
nvmlDeviceGetAccountingMode	nvmlDeviceSetAccountingMode	nvmlDeviceGetEnforcedPowerLimit
nvmlDeviceGetPowerManagementDefaultLimit	nvmlDeviceGetPowerManagementLimit	nvmlDeviceGetPowerManagementLimitConstraints
nvmlDeviceGetPowerState	nvmlDeviceGetPowerUsage	nvmlDeviceSetPowerManagementLimit
nvmlDeviceGetTotalEnergyConsumption	nvmlDeviceGetFanSpeed	nvmlDeviceGetFanSpeed_v2
nvmlDeviceGetTargetFanSpeed	nvmlDeviceSetFanSpeed_v2	nvmlDeviceSetDefaultFanSpeed_v2
nvmlDeviceGetMinMaxFanSpeed	nvmlDeviceGetTemperature	nvmlDeviceGetPcieReplayCounter
nvmlDeviceGetPcieThroughput	nvmlDeviceGetCurrPcieLinkWidth	nvmlDeviceGetMaxPcieLinkWidth
nvmlDeviceGetCurrPcieLinkGeneration	nvmlDeviceGetMaxPcieLinkGeneration	nvmlDeviceGetClockInfo
nvmlDeviceGetClock	nvmlDeviceResetGpuLockedClocks	nvmlDeviceSetGpuLockedClocks
nvmlDeviceGetApplicationsClock	nvmlDeviceGetDefaultApplicationsClock	nvmlDeviceResetApplicationsClocks
nvmlDeviceGetMaxClockInfo	nvmlDeviceGetMaxCustomerBo	nvmlDeviceGetPerformanceSta

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	ostClock	te
nvmlDeviceGetDynamicPstatesInfo	nvmlDeviceGetMinorNumber	nvmlDeviceGetComputeMode
nvmlDeviceSetComputeMode	nvmlDeviceGetGpuOperationMode	nvmlDeviceSetGpuOperationMode

5.14.2 未支持的函数

nvmlSystemGetDriverBranch	nvmlDeviceGetInforomImageVersion	nvmlDeviceGetInforomVersion
nvmlDeviceGetGspFirmwareVersion	nvmlDeviceGetMemoryErrorCounter	nvmlDeviceGetBAR1MemoryInfo
nvmlDeviceGetMemoryBusWidth	nvmlDeviceGetAccountingBufferSize	nvmlDeviceClearAccountingPids
nvmlDeviceGetGraphicsRunningProcesses_v3	nvmlDeviceGetMPSComputeRunningProcesses_v3	nvmlDeviceGetSupportedPerformanceStates
nvmlDeviceGetPowerSource	nvmlDeviceGetNumFans	nvmlDeviceGetTemperatureThreshold
nvmlDeviceSetTemperatureThreshold	nvmlDeviceGetPcieLinkMaxSpeed	nvmlDeviceGetPcieSpeed
nvmlDeviceGetAutoBoostedClocksEnabled	nvmlDeviceGetCurrentClocksThrottleReasons	nvmlDeviceGetSupportedClocksThrottleReasons
nvmlDeviceGetMinMaxClockOfPState	nvmlDeviceGetSupportedGraphicsClocks	nvmlDeviceSetAutoBoostedClocksEnabled
nvmlDeviceSetDefaultAutoBoostedClocksEnabled	nvmlDeviceSetApplicationsClocks	nvmlDeviceGetAdaptiveClockInfoStatus
nvmlDeviceGetClkMonStatus	nvmlDeviceGetThermalSettings	nvmlDeviceGetEccMode
nvmlDeviceSetEccMode	nvmlDeviceGetDefaultEccMode	nvmlDeviceGetTotalEccErrors
nvmlDeviceGetDetailedEccErrors	nvmlDeviceClearEccErrorCounts	nvmlEventSetCreate

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nvmlDeviceRegisterEvents	nvmlDeviceGetSupportedEventTypes	nvmlEventSetWait_v2
nvmlEventSetFree	nvmlDeviceGetP2PStatus	nvmlDeviceGetAttributes_v2
nvmlDeviceGetRemappedRows	nvmlDeviceGetRowRemapperHistogram	nvmlDeviceGetTopologyCommonAncestor
nvmlDeviceGetTopologyNearestGpus	nvmlSystemGetTopologyGpuSet	nvmlDeviceRemoveGpu_v2
nvmlDeviceDiscoverGpus	nvmlDeviceGetFieldValues	nvmlDeviceGetBoardId
nvmlDeviceGetBoardPartNumber	nvmlDeviceGetBridgeChipInfo	nvmlDeviceGetFBCStats
nvmlDeviceGetFBCSessions	nvmlDeviceGetDisplayActive	nvmlDeviceGetDisplayMode
nvmlDeviceSetDriverModel	nvmlDeviceGetInforomConfigurationChecksum	nvmlDeviceGetMultiGpuBoard
nvmlDeviceGetSamples	nvmlDeviceGetViolationStatus	nvmlDeviceGetIrqNum
nvmlDeviceGetNumGpuCores	nvmlDeviceGetBusType	nvmlDeviceOnSameBoard
nvmlDeviceValidateInforom	nvmlDeviceModifyDrainState	nvmlDeviceQueryDrainState
nvmlDeviceGetPowerManagementMode	nvmlDeviceGetAPIRestriction	nvmlDeviceSetAPIRestriction
nvmlDeviceGetPersistenceMode	nvmlDeviceSetPersistenceMode	nvmlDeviceGetCudaComputeCapability
nvmlDeviceGetGspFirmwareMode	nvmlGetExcludedDeviceCount	nvmlGetExcludedDeviceInfoByIndex
nvmlDeviceGetGpcClkVfOffset	nvmlDeviceSetGpcClkVfOffset	nvmlDeviceGetMemClkVfOffset
nvmlDeviceSetMemClkVfOffset	nvmlDeviceGetGpcClkMinMaxVfOffset	nvmlDeviceGetMemClkMinMaxVfOffset
nvmlGpmMetricsGet	nvmlGpmSampleFree	nvmlGpmSampleAlloc
nvmlGpmSampleGet	nvmlGpmQueryDeviceSupport	nvmlDeviceGetMemoryAffinit

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nvmlDeviceGetCpuAffinityWithinScope	nvmlDeviceGetCpuAffinity	nvmlDeviceSetCpuAffinity
nvmlDeviceClearCpuAffinity	nvmlDeviceGetNumaNodeId	nvmlDeviceGetEncoderUtilization
nvmlDeviceGetEncoderCapacity	nvmlDeviceGetEncoderStats	nvmlDeviceGetEncoderSessions
nvmlDeviceGetDecoderUtilization	nvmlUnitGetCount	nvmlUnitGetHandleByIndex
nvmlUnitGetUnitInfo	nvmlUnitGetLedState	nvmlUnitGetPsuInfo
nvmlUnitGetTemperature	nvmlUnitGetFanSpeedInfo	nvmlUnitGetDevices
nvmlSystemGetHicVersion	nvmlUnitSetLedState	nvmlDeviceGetNvLinkState
nvmlDeviceGetNvLinkVersion	nvmlDeviceGetNvLinkCapability	nvmlDeviceGetNvLinkRemotePciInfo_v2
nvmlDeviceGetNvLinkErrorCounter	nvmlDeviceResetNvLinkErrorCounters	nvmlDeviceSetNvLinkUtilizationControl
nvmlDeviceGetNvLinkUtilizationControl	nvmlDeviceGetNvLinkUtilizationCounter	nvmlDeviceFreezeNvLinkUtilizationCounter
nvmlDeviceResetNvLinkUtilizationCounter	nvmlDeviceGetNvLinkRemoteDeviceType	nvmlDeviceGetVirtualizationMode
nvmlDeviceGetHostVgpuMode	nvmlDeviceSetVirtualizationMode	nvmlDeviceGetGridLicensableFeatures_v4
nvmlDeviceGetSupportedVgpus	nvmlDeviceGetCreatableVgpus	nvmlVgpuTypeGetClass
nvmlVgpuTypeGetName	nvmlVgpuTypeGetGpuInstanceProfileId	nvmlVgpuTypeGetDeviceID
nvmlVgpuTypeGetFramebufferSize	nvmlVgpuTypeGetNumDisplayHeads	nvmlVgpuTypeGetResolution

nvmlVgpuTypeGetLicense	nvmlVgpuTypeGetFrameRateLimit	nvmlVgpuTypeGetMaxInstances
nvmlVgpuTypeGetMaxInstancesPerVm	nvmlDeviceGetActiveVgpus	nvmlVgpuInstanceGetVmID
nvmlVgpuInstanceGetUUID	nvmlVgpuInstanceGetMdevUUID	nvmlVgpuInstanceGetVmDriverVersion
nvmlVgpuInstanceGetFbUsage	nvmlVgpuInstanceGetLicenseStatus	nvmlVgpuInstanceGetType
nvmlVgpuInstanceGetFrameRateLimit	nvmlVgpuInstanceGetEccMode	nvmlVgpuInstanceGetEncoderCapacity
nvmlVgpuInstanceSetEncoderCapacity	nvmlVgpuInstanceGetEncoderStats	nvmlVgpuInstanceGetEncoderSessions
nvmlVgpuInstanceGetFBCStats	nvmlVgpuInstanceGetFBCSessions	nvmlVgpuInstanceGetGpuInstanceId
nvmlVgpuInstanceGetGpuPciId	nvmlVgpuTypeGetCapabilities	nvmlVgpuInstanceGetMetadata
nvmlDeviceGetVgpuMetadata	nvmlGetVgpuCompatibility	nvmlDeviceGetPgpuMetadataString
nvmlGetVgpuVersion	nvmlSetVgpuVersion	nvmlDeviceGetVgpuUtilization
nvmlDeviceGetVgpuProcessUtilization	nvmlVgpuInstanceGetAccountingMode	nvmlVgpuInstanceGetAccountingPids
nvmlVgpuInstanceGetAccountingStats	nvmlVgpuInstanceClearAccountingPids	nvmlVgpuInstanceGetLicenseInfo_v2
nvmlDeviceSetMigMode	nvmlDeviceGetMigMode	nvmlDeviceGetGpuInstanceProfileInfo
nvmlDeviceGetGpuInstanceProfileInfoV	nvmlDeviceGetGpuInstancePossiblePlacements_v2	nvmlDeviceGetGpuInstanceRemainingCapacity
nvmlDeviceCreateGpuInstance	nvmlDeviceCreateGpuInstanceWithPlacement	nvmlGpuInstanceDestroy
nvmlDeviceGetGpuInstances	nvmlDeviceGetGpuInstanceByI	nvmlGpuInstanceGetInfo

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nvmlGpuInstanceGetComputeInstanceProfileInfo	nvmlGpuInstanceGetComputeInstanceProfileInfoV	nvmlGpuInstanceGetComputeInstanceRemainingCapacity
nvmlGpuInstanceCreateComputeInstance	nvmlComputeInstanceDestroy	nvmlGpuInstanceGetComputeInstances
nvmlGpuInstanceGetComputeInstanceById	nvmlComputeInstanceGetInfo_v2	nvmlDeviceIsMigDeviceHandle
nvmlDeviceGetGpuInstanceId	nvmlDeviceGetComputeInstanceId	nvmlDeviceGetMaxMigDeviceCount
nvmlDeviceGetMigDeviceHandleByIndex	nvmlDeviceGetDeviceHandleFromMigDeviceHandle	nvmlDeviceGetModuleId
nvmlDeviceGetC2cModeInfoV	nvmlDeviceGetLastBBXFlushTime	nvmlDeviceGetPciInfoExt
nvmlDeviceGetGpuMaxPcieLinkGeneration	nvmlDeviceGetFanControlPolicy_v2	nvmlDeviceGetCurrentClocksEventReasons
nvmlDeviceGetSupportedClocksEventReasons	nvmlDeviceGetClockOffsets	nvmlDeviceSetClockOffsets
nvmlDeviceGetJpgUtilization	nvmlDeviceGetOfaUtilization	nvmlDeviceGetRunningProcessDetailList
nvmlDeviceGetGpuFabricInfo	nvmlDeviceGetGpuFabricInfoV	nvmlDeviceSetPowerManagementLimit_v2
nvmlDeviceGetSramEccErrorStatus	nvmlSystemGetConfComputeCapabilities	nvmlDeviceGetConfComputeMemSizeInfo
nvmlSystemGetConfComputeGpusReadyState	nvmlDeviceGetConfComputeProtectedMemoryUsage	nvmlDeviceGetConfComputeGpuCertificate
nvmlDeviceGetConfComputeGpuAttestationReport	nvmlSystemGetConfComputeKeyRotationThresholdInfo	nvmlDeviceSetConfComputeUnprotectedMemSize
nvmlSystemSetConfComputeGpusReadyState	nvmlSystemSetConfComputeKeyRotationThresholdInfo	nvmlSystemGetConfComputeSettings

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nvmlDeviceGetProcessesUtilizationInfo	nvmlDeviceSetFanControlPolicy	nvmlDeviceSetNvLinkDeviceLowPowerThreshold
nvmlSystemSetNvlinkBwMode	nvmlSystemGetNvlinkBwMode	nvmlDeviceClearFieldValues
nvmlDeviceGetVgpuHeterogeneousMode	nvmlDeviceSetVgpuHeterogeneousMode	nvmlVgpuInstanceGetPlacementId
nvmlDeviceGetVgpuTypeSupportedPlacements	nvmlDeviceGetVgpuTypeCreatablePlacements	nvmlVgpuTypeGetGspHeapSize
nvmlVgpuTypeGetFbReservation	nvmlDeviceSetVgpuCapabilities	nvmlGetVgpuDriverCapabilities
nvmlDeviceGetVgpuCapabilities	nvmlVgpuTypeGetBAR1Info	nvmlDeviceGetVgpuSchedulerLog
nvmlDeviceGetVgpuSchedulerState	nvmlDeviceGetVgpuSchedulerCapabilities	nvmlDeviceGetVgpuInstancesUtilizationInfo
nvmlDeviceGetVgpuProcessesUtilizationInfo	nvmlGpuInstanceGetComputeInstancePossiblePlacements	nvmlGpuInstanceCreateComputeInstanceWithPlacement
nvmlGpmMigSampleGet	nvmlGpmQueryIfStreamingEnabled	nvmlGpmSetStreamingEnabled
nvmlDeviceGetCapabilities	nvmlDeviceGetCoolerInfo	nvmlDeviceGetCurrentClockFreqs
nvmlDeviceGetDramEncryptionMode	nvmlDeviceGetDriverModel_v2	nvmlDeviceGetFanSpeedRPM
nvmlDeviceGetHandleByUUIDV	nvmlDeviceGetMarginTemperature	nvmlDeviceGetNvlinkBwMode
nvmlDeviceGetNvlinkSupportedBwModes	nvmlDeviceGetPerformanceModes	nvmlDeviceGetTemperatureV
nvmlDevicePowerSmoothingActivatePresetProfile	nvmlDevicePowerSmoothingSetState	nvmlDevicePowerSmoothingUpdatePresetProfileParam
nvmlDeviceSetDramEncryptionMode	nvmlDeviceSetNvlinkBwMode	nvmlDeviceSetVgpuSchedulerState

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nvmlDeviceWorkloadPowerProfileClearRequestedProfiles	nvmlDeviceWorkloadPowerProfileGetCurrentProfiles	nvmlDeviceWorkloadPowerProfileGetProfilesInfo
nvmlDeviceWorkloadPowerProfileSetRequestedProfiles	nvmlGpuInstanceGetActiveVgpus	nvmlGpuInstanceGetCreatableVgpus
nvmlGpuInstanceGetVgpuHeterogeneousMode	nvmlGpuInstanceGetVgpuSchedulerLog	nvmlGpuInstanceGetVgpuSchedulerState
nvmlGpuInstanceGetVgpuTypeCreatablePlacements	nvmlGpuInstanceSetVgpuHeterogeneousMode	nvmlGpuInstanceSetVgpuSchedulerState
nvmlSystemEventSetCreate	nvmlSystemEventSetFree	nvmlSystemEventSetWait
nvmlSystemGetConfComputeState	nvmlSystemRegisterEvents	nvmlVgpuInstanceGetRuntimeStateSize
nvmlVgpuTypeGetMaxInstancesPerGpuInstance	nvmlDeviceGetPlatformInfo	

5.15 NVVM 函数

5.15.1 支持的函数

nvvmGetErrorString	nvvmVersion	nvvmIRVersion
nvvmCreateProgram	nvvmDestroyProgram	nvvmAddModuleToProgram
nvvmLazyAddModuleToProgram	nvvmCompileProgram	nvvmGetCompiledResultSize
nvvmGetCompiledResult	nvvmSetArch	

5.15.2 未支持的函数

nvvmGetProgramLogSize	nvvmGetProgramLog	nvvmVerifyProgram

5.16 NVCOMP 函数

5.16.1 支持的函数

nvcompBatchedANSCompressGetTempSize	nvcompBatchedANSCompressGetMaxOutputChunkSize	nvcompBatchedANSCompressAsync

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nvcompBatchedANSDecompressGetTempSize	nvcompBatchedANSGetDecompressSizeAsync	nvcompBatchedANSDecompressAsync
nvcompBitcompCompressConfigure	nvcompBitcompCompressAsync	nvcompBitcompDecompressConfigure
nvcompBitcompDecompressAsync	nvcompIsBitcompData	nvcompBatchedBitcompCompressGetMaxOutputChunkSize
nvcompBatchedBitcompCompressAsync	nvcompBatchedBitcompDecompressAsync	nvcompBatchedBitcompGetDecompressSizeAsync
nvcompBatchedBitcompCompressGetTempSize	nvcompBatchedBitcompDecompressGetTempSize	nvcompBatchedCascadedCompressGetTempSize
nvcompBatchedCascadedCompressGetMaxOutputChunkSize	nvcompBatchedCascadedCompressAsync	nvcompBatchedCascadedDecompressGetTempSize
nvcompBatchedCascadedDecompressAsync	nvcompBatchedCascadedGetDecompressSizeAsync	nvcompBatchedGdeflateCompressGetTempSize
nvcompBatchedGdeflateCompressGetMaxOutputChunkSize	nvcompBatchedGdeflateCompressAsync	nvcompBatchedGdeflateDecompressGetTempSize
nvcompBatchedGdeflateDecompressAsync	nvcompBatchedGdeflateGetDecompressSizeAsync	nvcompBatchedLZ4CompressGetTempSize
nvcompBatchedLZ4CompressGetMaxOutputChunkSize	nvcompBatchedLZ4CompressAsync	nvcompBatchedLZ4DecompressGetTempSize
nvcompBatchedLZ4DecompressAsync	nvcompBatchedLZ4GetDecompressSizeAsync	nvcompBatchedSnappyDecompressGetTempSize
nvcompBatchedSnappyGetDecompressSizeAsync	nvcompBatchedSnappyDecompressAsync	nvcompBatchedSnappyCompressGetTempSize
nvcompBatchedSnappyCompressGetMaxOutputChunkSize	nvcompBatchedSnappyCompressAsync	

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5.17 NVJPEG 函数

5.17.1 支持的函数

nvjpegCreate	nvjpegDestroy	nvjpegCreateSimple
nvjpegJpegStateCreate	nvjpegDecodeBatchedPreAllocate	nvjpegJpegStreamDestroy
nvjpegJpegStateDestroy	nvjpegDecodeParamsSetOutputFormat	nvjpegDecodeParamsDestroy
nvjpegDecoderDestroy	nvjpegGetImageInfo	nvjpegDecode
nvjpegDecodeParamsSetROI	nvjpegDecodeBatchedSupported	nvjpegDecodeBatchedSupportedEx
nvjpegDecodeBatchedInitialize	nvjpegDecodeBatched	nvjpegDecodeBatchedEx
nvjpegGetProperty	nvjpegGetHardwareDecoderInfo	nvjpegJpegStreamGetSamplePrecision
nvjpegJpegStreamParseHeader	nvjpegCreateEx	nvjpegCreateExV2
nvjpegDecoderJpegSupported	nvjpegEncodeImage	nvjpegEncodeRetrieveBitstream
nvjpegEncoderParamsCreate	nvjpegEncoderParamsDestroy	nvjpegEncoderParamsSetEncoding
nvjpegEncoderParamsSetQuality	nvjpegEncoderParamsSetSamplingFactors	nvjpegEncoderStateCreate
nvjpegEncoderStateDestroy	nvjpegEncodeYUV	nvjpegJpegStreamGetJpegEncoding

5.17.2 未支持的函数

nvjpegSetDeviceMemoryPadding	nvjpegSetPinnedMemoryPadding	nvjpegJpegStreamCreate
nvjpegBufferPinnedCreate	nvjpegBufferPinnedResize	nvjpegBufferPinnedDestroy
nvjpegBufferDeviceResize	nvjpegBufferDeviceDestroy	nvjpegDecodeParamsCreate
nvjpegDecodeParamsSetAllow	nvjpegDecoderCreate	nvjpegDecoderStateCreate

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nvjpegStateAttachPinnedBuffer	nvjpegStateAttachDeviceBuffer	nvjpegDecodeJpegTransferToDevice
nvjpegDecodeJpegHost	nvjpegDecodeJpegDevice	nvjpegJpegStreamParse
nvjpegBufferDeviceCreate	nvjpegBufferDeviceCreateV2	nvjpegBufferPinnedCreateV2
nvjpegDecodeParamsSetExifOrientation	nvjpegEncoderParamsSetOptimizedHuffman	nvjpegJpegStreamGetFrameDimensions
nvjpegJpegStreamGetComponentsNum	nvjpegJpegStreamGetChromaSubsampling	nvjpegJpegStreamGetComponentDimensions
nvjpegJpegStreamGetExifOrientation		

5.18 nvImageCodec 函数

5.18.1 支持的函数

nvimgcodecCodeStreamCreateFromFile	nvimgcodecCodeStreamCreateFromHostMem	nvimgcodecCodeStreamCreateToFile
nvimgcodecCodeStreamCreateToHostMem	nvimgcodecCodeStreamDestroy	nvimgcodecCodeStreamGetImageInfo
nvimgcodecDebugMessengerCreate	nvimgcodecDebugMessengerDestroy	nvimgcodecDecoderCanDecode
nvimgcodecDecoderCreate	nvimgcodecDecoderDecode	nvimgcodecDecoderDestroy
nvimgcodecEncoderCanEncode	nvimgcodecEncoderCreate	nvimgcodecEncoderDestroy
nvimgcodecEncoderEncode	nvimgcodecExtensionCreate	nvimgcodecExtensionDestroy
nvimgcodecFutureDestroy	nvimgcodecFutureGetProcessingStatus	nvimgcodecFutureWaitForAll
nvimgcodecGetProperties	nvimgcodecImageCreate	nvimgcodecImageDestroy
nvimgcodecImageGetImageInf	nvimgcodecInstanceCreate	nvimgcodecInstanceDestroy

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5.19 CUVID 函数

5.19.1 支持的函数

cuvidCreateDecoder	cuvidCreateVideoParser	cuvidCreateVideoSource
cuvidCreateVideoSourceW	cuvidCtxLock	cuvidCtxLockCreate
cuvidCtxLockDestroy	cuvidCtxUnlock	cuvidDecodePicture
cuvidDestroyDecoder	cuvidDestroyVideoParser	cuvidDestroyVideoSource
cuvidGetDecoderCaps	cuvidGetDecodeStatus	cuvidGetSourceAudioFormat
cuvidGetSourceVideoFormat	cuvidGetVideoSourceState	cuvidMapVideoFrame
cuvidMapVideoFrame64	cuvidParseVideoData	cuvidReconfigureDecoder
cuvidSetVideoSourceState	cuvidUnmapVideoFrame	cuvidUnmapVideoFrame64

5.20 MMA 函数

5.20.1 支持的函数

load_matrix_sync	store_matrix_sync	mma_sync

5.21 NPP 函数

5.21.1 支持的函数

nppiAdd_8u_C1RSfs_Ctx	nppiGammaFwd_8u_P3R	nppiRShiftC_32s_C1R
nppiAdd_8u_C1RSfs	nppiGammaFwd_8u_IP3R_Ctx	nppiRShiftC_8u_C4R
nppiAdd_8u_C1IRSfs_Ctx	nppiGammaFwd_8u_IP3R	nppiRShiftC_8s_C4R
nppiAdd_8u_C1IRSfs	nppiColorTwist32f_8u_C3R	nppiRShiftC_16u_C4R
nppiAdd_8u_C3RSfs_Ctx	nppiColorTwist32f_8u_C3R_Ctx	nppiRShiftC_16s_C4R
nppiAdd_8u_C3RSfs	nppiColorTwist32f_8u_C3IR_Ctx	nppiRShiftC_32s_C4R

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nppiAdd_32f_C3RSfs	nppiColorTwist32f_8u_C3IR	nppiMean_StdDev_8u_C1MR
nppiAdd_8u_C3IRSfs_Ctx	nppiColorTwist32f_8u_P3R_Ctx	nppiMean_StdDev_32f_C1MR
nppiAdd_8u_C3IRSfs	nppiColorTwist32f_8u_P3R	nppiMean_StdDev_32f_C1R
nppiAddSquare_8u32f_C1IR	nppiColorTwist32f_8u_IP3R_Ctx	nppiMeanStdDevGetBufferHostSize_8u_C1R
nppiAddSquare_8u32f_C1IR_Ctx	nppiColorTwist32f_8u_IP3R	nppiMeanStdDevGetBufferHostSize_8u_C1MR
nppiAddWeighted_8u32f_C1IR	nppiLUT_8u_C3R	nppiMeanStdDevGetBufferHostSize_32f_C1R
nppiAddWeighted_8u32f_C1IR_Ctx	nppiLUT_8u_C1R_Ctx	nppiMeanStdDevGetBufferHostSize_32f_C1MR
nppiMul_8u_C3RSfs	nppiLUT_8u_C1R	nppiTranspose_8u_C1R
nppiMul_8u_C1RSfs_Ctx	nppiLUT_8u_C1IR_Ctx	nppiMagnitudeSqr_32fc32f_C1R
nppiMul_8u_C1RSfs	nppiLUT_8u_C1IR	nppiMagnitude_32fc32f_C1R
nppiMul_8u_C1IRSfs_Ctx	nppiLUT_8u_C3R_Ctx	nppGetStream
nppiMul_8u_C1IRSfs	nppiLUT_8u_C3IR_Ctx	nppSetStream
nppiMul_8u_C3RSfs_Ctx	nppiLUT_8u_C3IR	nppiFilterMin_8u_C1R
nppiMul_8u_C3IRSfs_Ctx	nppiCopy_8u_C3R	nppiFilterMin_8u_C4R
nppiMul_8u_C3IRSfs	nppiCopy_8u_C1R_Ctx	nppiFilterMax_8u_C1R
nppiSub_8u_C3RSfs	nppiCopy_8u_C1R	nppiFilterMax_8u_C4R
nppiSub_8u_C1RSfs_Ctx	nppiCopy_8u_C3R_Ctx	nppiErode_8u_C4R
nppiSub_8u_C1RSfs	nppiSwapChannels_8u_C3R	nppiErode_32f_C1R
nppiSub_8u_C1IRSfs_Ctx	nppiSwapChannels_8u_C3R_Ctx	nppiErode_32f_C4R
nppiSub_8u_C1IRSfs	nppiSwapChannels_8u_C3R	nppiDilate_8u_C4R
nppiSub_8u_C3RSfs_Ctx	nppiSwapChannels_8u_C3IR_Ct	nppiDilate_32f_C4R

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nppiSub_8u_C3IRSfs_Ctx	nppiSwapChannels_8u_C3IR	nppiDilate_32f_C1R
nppiSub_8u_C3IRSfs	nppiRGBToGray_8u_C3C1R_Ctx	nppiFilterBox_8u_C4R
nppiAbsDiff_8u_C3R	nppiRGBToGray_8u_C3C1R	nppiFilterBox_32f_C1R
nppiAbsDiff_8u_C1R_Ctx	nppiRGBToGray_32f_C3C1R_Ctx	nppiWarpAffine_32f_C1R
nppiAbsDiff_8u_C1R	nppiRGBToGray_32f_C3C1R	nppiWarpAffine_32f_C3R
nppiAbsDiff_8u_C3R_Ctx	nppiFilterBox_8u_C3R	nppiWarpAffine_8u_C4R
nppiAnd_8u_C3R	nppiFilterBox_8u_C1R_Ctx	nppiWarpAffine_32f_C4R
nppiAnd_8u_C1R_Ctx	nppiFilterBox_8u_C1R	nppiWarpAffine_16u_C1R
nppiAnd_8u_C1R	nppiFilterBox_8u_C3R_Ctx	nppiWarpAffine_16u_C3R
nppiAnd_8u_C1IR_Ctx	nppiWarpAffine_8u_C3R	nppiWarpAffine_16u_C4R
nppiAnd_8u_C1IR	nppiWarpAffine_8u_C1R_Ctx	nppiWarpAffine_32s_C1R
nppiAnd_8u_C3R_Ctx	nppiWarpAffine_8u_C1R	nppiWarpAffine_32s_C3R
nppiAnd_8u_C3IR_Ctx	nppiWarpAffine_8u_C3R	nppiWarpAffine_32s_C4R
nppiAnd_8u_C3IR	nppiWarpAffine_8u_C3R_Ctx	nppiWarpAffineBack_8u_C1R
nppiNot_8u_C3R	nppiWarpAffine_8u_P3R_Ctx	nppiWarpAffineBack_8u_C3R
nppiNot_8u_C1R_Ctx	nppiWarpAffine_8u_P3R	nppiWarpAffineBack_8u_C4R
nppiNot_8u_C1R	nppiResize_8u_C3R	nppiWarpAffineBack_16u_C1R
nppiNot_8u_C1IR_Ctx	nppiResize_8u_C1R_Ctx	nppiWarpAffineBack_16u_C3R
nppiNot_8u_C1IR	nppiResize_8u_C1R	nppiWarpAffineBack_16u_C4R
nppiNot_8u_C3R_Ctx	nppiResize_8u_C3R_Ctx	nppiWarpAffineBack_32f_C1R
nppiNot_8u_C3IR_Ctx	nppiResize_8u_P3R_Ctx	nppiWarpAffineBack_32f_C3R
nppiNot_8u_C3IR	nppiResize_8u_P3R	nppiWarpAffineBack_32f_C4R

nppiXor_8u_C3R	nppiErode3x3_8u_C3R	nppiWarpAffineBack_32s_C1R
nppiXor_8u_C1R_Ctx	nppiErode3x3_8u_C1R_Ctx	nppiWarpAffineBack_32s_C3R
nppiXor_8u_C1R	nppiErode3x3_8u_C1R	nppiWarpAffineBack_32s_C4R
nppiXor_8u_C1IR_Ctx	nppiErode3x3_8u_C3R_Ctx	nppiWarpPerspective_8u_C4R
nppiXor_8u_C1IR	nppiDilate3x3_8u_C3R	nppiWarpPerspective_16u_C1R
nppiXor_8u_C3R_Ctx	nppiDilate3x3_8u_C1R_Ctx	nppiWarpPerspective_16u_C3R
nppiXor_8u_C3IR_Ctx	nppiDilate3x3_8u_C1R	nppiWarpPerspective_16u_C4R
nppiXor_8u_C3IR	nppiDilate3x3_8u_C3R_Ctx	nppiWarpPerspective_32f_C1R
nppiOr_8u_C3R	nppiSum_8u_C3R	nppiWarpPerspective_32f_C3R
nppiOr_8u_C1R_Ctx	nppiSum_8u_C1R_Ctx	nppiWarpPerspective_32f_C4R
nppiOr_8u_C1R	nppiSum_8u_C1R	nppiWarpPerspective_32s_C1R
nppiOr_8u_C1IR_Ctx	nppiSum_8u_C3R_Ctx	nppiWarpPerspective_32s_C3R
nppiOr_8u_C1IR	nppiMin_8u_C3R	nppiWarpPerspective_32s_C4R
nppiOr_8u_C3R_Ctx	nppiMin_8u_C1R_Ctx	nppiWarpPerspectiveBack_8u_C1R
nppiOr_8u_C3R	nppiMin_8u_C1R	nppiWarpPerspectiveBack_8u_C3R
nppiOr_8u_C3IR_Ctx	nppiMin_8u_C3R_Ctx	nppiWarpPerspectiveBack_8u_C4R
nppiOr_8u_C3IR	nppiMax_8u_C3R	nppiWarpPerspectiveBack_16u_C1R
nppiDiv_8u_C3RSfs	nppiMax_8u_C1R_Ctx	nppiWarpPerspectiveBack_16u

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nppiAdd_8u_AC4RSfs	nppiMax_8u_C1R	nppiWarpPerspectiveBack_16u_C4R
nppiMalloc_8u_C1	nppiMax_8u_C3R_Ctx	nppiWarpPerspectiveBack_32f_C1R
nppiMalloc_16s_C1	nppiRemap_8u_C1R_Ctx	nppiWarpPerspectiveBack_32f_C3R
nppiFree	nppiRemap_8u_C3R_Ctx	nppiWarpPerspectiveBack_32f_C4R
nppiLUT_Linear_8u_C1R	nppiRemap_16u_C1R_Ctx	nppiWarpPerspectiveBack_32s_C1R
nppiCopyConstBorder_8u_C1R	nppiRemap_16u_C3R_Ctx	nppiWarpPerspectiveBack_32s_C3R
nppiFilterCannyBorder_8u_C1R	nppiRemap_16s_C1R_Ctx	nppiWarpPerspectiveBack_32s_C4R
nppiFilterCannyBorderGetBufferSize	nppiRemap_16s_C3R_Ctx	nppiRotate_8u_C1R
nppi_prewitt_filter_u8_pln1_batchPD_gpu	nppiRemap_32f_C1R_Ctx	nppiRotate_8u_C3R
nppiGradientVectorPrewittBorder_8u16s_C1R	nppiRemap_32f_C3R_Ctx	nppiRotate_8u_C4R
nppiLabelMarkersUFGetBufferSize_32u_C1R	nppiRemap_64f_C1R_Ctx	nppiRotate_16u_C1R
nppiCompressMarkerLabelsGetBufferSize_32u_C1R	nppiRemap_64f_C3R_Ctx	nppiRotate_16u_C3R
nppiLabelMarkersUF_8u32u_C1R_Ctx	nppiCFAToRGB_8u_C1C3R_Ctx	nppiRotate_16u_C4R
nppiLabelMarkersUFBatch_8u32u_C1R_Advanced_Ctx	nppiCFAToRGB_16u_C1C3R_Ctx	nppiRotate_32f_C1R

nppiCompressMarkerLabelsUFBatch_32u_C1IR_Advanced_Ctx	nppiColorTwist32f_8u_C1R	nppiRotate_32f_C3R
nppGetLibVersion	nppiMirror_8u_C4R	nppiRotate_32f_C4R
nppiCompressMarkerLabelsUF_32u_C1IR	nppiMirror_8u_C4IR	nppiHistogramEven_8u_C4R
nppiFilterBoxBorder_8u_C1R	nppiMirror_32s_C4R	nppiHistogramEven_16u_C1R
nppi_box_filter_u8_pln1_batchPD_gpu	nppiMirror_32f_C4R	nppiHistogramEven_16u_C4R
nppiFilterSobelHoriz_8u_C3R	nppiMirror_16u_C4R	nppiHistogramEven_16s_C1R
nppiFilterSobelHoriz_8u_C1R_Ctx	nppiMirror_32f_C4IR	nppiHistogramEven_16s_C4R
nppiFilterSobelHoriz_8u_C1R	nppiMirror_32s_C4IR	nppiHistogramEvenGetBufferSize_8u_C1R
nppiFilterSobelHoriz_8u_C3R_Ctx	nppiMirror_16u_C4IR	nppiHistogramEvenGetBufferSize_8u_C4R
nppiFilterSobelVert_8u_C3R	nppiMirror_32s_C3R	nppiHistogramEvenGetBufferSize_16u_C1R
nppiFilterSobelVert_8u_C1R_Ctx	nppiMirror_32f_C3R	nppiHistogramEvenGetBufferSize_16u_C4R
nppiFilterSobelVert_8u_C1R	nppiMirror_32s_C3IR	nppiHistogramEvenGetBufferSize_16s_C1R
nppiFilterSobelVert_8u_C3R_Ctx	nppiMirror_32f_C3IR	nppiHistogramEvenGetBufferSize_16s_C4R
nppiFilterMedian_8u_C3R	nppiMirror_16u_C3IR	nppiHistogramRange_8u_C1R
nppiFilterMedian_8u_C1R_Ctx	nppiMirror_32s_C1R	nppiHistogramRange_8u_C4R
nppiFilterMedian_8u_C1R	nppiMirror_32f_C1R	nppiHistogramRange_16u_C1R

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nppiFilterMedian_8u_C3R_Ctx	nppiMirror_16u_C1R	nppiHistogramRange_16u_C4R
nppiFilterGauss_8u_C3R	nppiMirror_32s_C1IR	nppiHistogramRange_16s_C1R
nppiFilterGauss_8u_C1R_Ctx	nppiMirror_32f_C1IR	nppiHistogramRange_16s_C4R
nppiFilterGauss_8u_C1R	nppiMirror_16u_C1IR	nppiHistogramRange_32f_C1R
nppiFilterGauss_8u_C3R_Ctx	nppiAndC_8u_C3R	nppiHistogramRange_32f_C4R
nppiWarpPerspective_8u_C3R	nppiAndC_16u_C3R	nppiHistogramRangeGetBufferSize_8u_C1R
nppiWarpPerspective_8u_C1R_Ctx	nppiAndC_32s_C3R	nppiHistogramRangeGetBufferSize_8u_C4R
nppiWarpPerspective_8u_C1R	nppiAndC_16u_C4R	nppiHistogramRangeGetBufferSize_16u_C1R
nppiWarpPerspective_8u_C3R_Ctx	nppiAndC_32s_C4R	nppiHistogramRangeGetBufferSize_16u_C4R
nppiWarpPerspective_8u_P3R_Ctx	nppiXorC_8u_C3R	nppiHistogramRangeGetBufferSize_16s_C1R
nppiWarpPerspective_8u_P3R	nppiXorC_16u_C3R	nppiHistogramRangeGetBufferSize_16s_C4R
nppiMirror_8u_C3R	nppiXorC_32s_C3R	nppiHistogramRangeGetBufferSize_32f_C1R
nppiMirror_8u_C1R_Ctx	nppiXorC_16u_C4R	nppiHistogramRangeGetBufferSize_32f_C4R
nppiMirror_8u_C1R	nppiXorC_32s_C4R	nppiGammaFwd_8u_AC4IR
nppiMirror_8u_C1IR_Ctx	nppiOrC_8u_C3R	nppiGammaFwd_8u_AC4R
nppiMirror_8u_C1IR	nppiOrC_16u_C3R	nppiGammaInv_8u_C3R
nppiMirror_8u_C3R_Ctx	nppiOrC_32s_C3R	nppiGammaInv_8u_C3IR
nppiMirror_8u_C3IR_Ctx	nppiOrC_16u_C4R	nppiGammaInv_8u_AC4R
nppiMirror_8u_C3IR	nppiOrC_32s_C4R	nppiGammaInv_8u_AC4IR

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nppiErode_8u_C3R	nppiLShiftC_8u_C3R	nppiAlphaComp_8u_AC4R
nppiErode_8u_C1R_Ctx	nppiLShiftC_8u_C1R	nppiAlphaComp_16u_AC4R
nppiErode_8u_C1R	nppiLShiftC_8u_C4R	nppiAlphaComp_32s_AC4R
nppiErode_8u_C3R_Ctx	nppiLShiftC_16u_C3R	nppiAlphaComp_32f_AC4R
nppiDilate_8u_C3R	nppiLShiftC_16u_C1R	nppiAlphaPremul_8u_AC4R
nppiDilate_8u_C1R_Ctx	nppiLShiftC_16u_C4R	nppiAlphaPremul_16u_AC4R
nppiDilate_8u_C1R	nppiLShiftC_32s_C3R	nppiSwapChannels_8u_C4IR
nppiDilate_8u_C3R_Ctx	nppiLShiftC_32s_C1R	nppiRectStdDev_32s32f_C1R
nppiHistogramEven_8u_C1R	nppiLShiftC_32s_C4R	nppiSumWindowRow_8u32f_C1R
nppiCompareC_16s_C1R	nppiRShiftC_8u_C3R	nppiSumWindowColumn_8u32f_C1R
nppiHistogramEvenGetBufferSize_8u_C1R	nppiRShiftC_8s_C3R	nppiMirror_16u_C3R
nppiEvenLevelsHost_32s	nppiRShiftC_16u_C3R	nppiThreshold_32f_C1R
nppiMean_StdDev_8u_C1R_Ctx	nppiRShiftC_16s_C3R	nppiNV12ToBGR_709HDTV_8u_P2C3R_Ctx
nppiMean_StdDev_8u_C1R	nppiRShiftC_32s_C3R	nppiNV12ToBGR_709CSC_8u_P2C3R_Ctx
nppiGammaFwd_8u_C3R	nppiRShiftC_8u_C1R	nppiNV12ToBGR_709HDTV_8u_P2C3R
nppiGammaFwd_8u_C3R_Ctx	nppiRShiftC_8s_C1R	nppiNV12ToBGR_8u_P2C3R
nppiGammaFwd_8u_C3IR_Ctx	nppiRShiftC_16s_C1R	nppiNV12ToBGR_8u_P2C3R_Ctx
nppiGammaFwd_8u_C3IR	nppiRShiftC_16u_C1R	nppGetStreamContext
nppiGammaFwd_8u_P3R_Ctx		

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5.22 CUDA 12 新增函数

Runtime 新增函数

cudaDeviceRegisterAsyncNotification	cudaDeviceUnregisterAsyncNotification	cudaGetDeviceProperties_v2
cudaInitDevice	cudaFuncGetName	cudaFuncGetParamInfo
cudaGetDriverEntryPointByVersion	cudaGetDriverEntryPointByVersion_ptsz	cudaGetKernel
cudaGraphAddDependencies_v2	cudaGraphAddNode	cudaGraphAddNode_v2
cudaGraphConditionalHandleCreate	cudaGraphExecGetFlags	cudaGraphExecNodeSetParams
cudaGraphGetEdges_v2	cudaGraphInstantiateWithParams	cudaGraphInstantiateWithParams_ptsz
cudaGraphNodeGetDependencies_v2	cudaGraphNodeGetDependentNodes_v2	cudaGraphNodeSetParams
cudaGraphRemoveDependencies_v2	cudaMemAdvise_v2	cudaMemPrefetchAsync_v2
cudaMemPrefetchAsync_v2_ptsz	cudaStreamBeginCaptureToGraph	cudaStreamBeginCaptureToGraph_ptsz
cudaStreamGetCaptureInfo_v3	cudaStreamGetCaptureInfo_v3_ptsz	cudaStreamGetId
cudaStreamGetId_ptsz	cudaStreamUpdateCaptureDependencies_v2	cudaStreamUpdateCaptureDependencies_v2_ptsz
cudaEventElapsedTime_v2	cudaLibraryEnumerateKernels	cudaLibraryGetGlobal
cudaLibraryGetKernel	cudaLibraryGetManaged	cudaLibraryLoadData
cudaLibraryUnload	cudaMemcpy3DBatchAsync	cudaMemcpyBatchAsync
cudaStreamGetDevice		

Runtime 废弃函数

cudaUnbindTexture	cudaBindTextureToMipmappe	cudaGetTextureObjectTextureDe

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	dArray	sc_v2
cudaGetTextureReference	cudaGetSurfaceReference	cudaBindTextureToArray
cudaGetTextureAlignmentOffset	cudaBindTexture	cudaBindSurfaceToArray
__cudaRegisterSurface	cudaBindTexture2D	cudaProfilerInitialize
cudaCreateTextureObject_v2	__cudaRegisterTexture	cudaGraphicsUnregisterResource
cudaGLGetDevices	cudaGraphicsGLRegisterBuffer	cudaGraphicsGLRegisterImage
cudaGraphicsMapResources	cudaGraphicsResourceGetMappedPointer	cudaGraphicsSubResourceGetMappedArray

Driver 新增函数

cuCtxCreate_v4	cuCtxGetId	cuCtxRecordEvent
cuCtxSetFlags	cuCtxWaitEvent	cuDeviceRegisterAsyncNotification
cuDeviceUnregisterAsyncNotification	cuFuncGetName	cuFuncGetParamInfo
cuFuncIsLoaded	cuFuncLoad	cuGetProcAddress_v2
cuGraphAddDependencies_v2	cuGraphAddKernelNode_v2	cuGraphAddNode
cuGraphAddNode_v2	cuGraphConditionalHandleCreate	cuGraphExecGetFlags
cuGraphExecKernelNodeSetParams_v2	cuGraphExecNodeSetParams	cuGraphExecUpdate_v2
cuGraphGetEdges_v2	cuGraphInstantiateWithParams	cuGraphInstantiateWithParams_ptsz
cuGraphKernelNodeGetParams_v2	cuGraphKernelNodeSetParams_v2	cuGraphNodeGetDependencies_v2
cuGraphNodeGetDependentNodes_v2	cuGraphNodeSetParams	cuGraphRemoveDependencies_v2
cuGreenCtxCreate	cuGreenCtxDestroy	cuGreenCtxGetDevResource
cuGreenCtxRecordEvent	cuGreenCtxStreamCreate	cuGreenCtxWaitEvent

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cuDevResourceGenerateDesc	cuDevSmResourceSplitByCount	cuDeviceGetDevResource
cuCtxFromGreenCtx	cuStreamGetGreenCtx	cuCtxGetDevResource
cuKernelGetAttribute	cuKernelGetFunction	cuKernelGetLibrary
cuKernelGetName	cuKernelGetParamInfo	cuKernelSetAttribute
cuLibraryEnumerateKernels	cuLibraryGetGlobal	cuLibraryGetKernel
cuLibraryGetKernelCount	cuLibraryGetManaged	cuLibraryGetModule
cuLibraryLoadData	cuLibraryLoadFromFile	cuLibraryUnload
cuMemAdvise_v2	cuMemPrefetchAsync_v2	cuMemPrefetchAsync_v2_ptsz
cuModuleEnumerateFunctions	cuModuleGetFunctionCount	cuStreamBeginCaptureToGraph
cuStreamBeginCaptureToGraph_ptsz	cuStreamGetCaptureInfo_v3	cuStreamGetCaptureInfo_v3_ptsz
cuStreamGetCtx_v2	cuStreamGetCtx_v2_ptsz	cuStreamGetId
cuStreamGetId_ptsz	cuStreamUpdateCaptureDependencies_v2	cuStreamUpdateCaptureDependencies_v2_ptsz
cuEventElapsedTime_v2	cuMemcpy3DBatchAsync	cuMemcpyBatchAsync
cuStreamGetDevice		

Driver 废弃函数

cuGLGetDevices	cuGLGetDevices_v2	cuGraphicsGLRegisterBuffer
cuGraphicsGLRegisterImage	cuGraphicsResourceGetMappedPointer	cuGraphicsResourceGetMappedPointer_v2
cuGraphicsSubResourceGetMappedArray	cuGraphicsUnregisterResource	

Cublas 新增函数

cublasAxpyEx_64	cublasCaxpy_v2_64	cublasCcopy_v2_64	cublasCdgmm_64
cublasCdotc_v2_64	cublasCdotu_v2_64	cublasCgbmv_v2_64	cublasCgeam_64
cublasCgemm_v2_64	cublasCgemm3m_64	cublasCgemm3mStridedBatched_64	cublasCgemmBatched_64

cublasCgemmEx_64	cublasCgemmStridedBatched_64	cublasCgemv_v2_64	cublasCgemvBatched_64
cublasCgemvStridedBatched_64	cublasCgerc_v2_64	cublasCgeru_v2_64	cublasChbmv_v2_64
cublasChemm_v2_64	cublasChemv_v2_64	cublasCher_v2_64	cublasCher2_v2_64
cublasCher2k_v2_64	cublasCherk_v2_64	cublasCherk3mEx_64	cublasCherkEx_64
cublasCherkx_64	cublasChpmv_v2_64	cublasChpr_v2_64	cublasChpr2_v2_64
cublasCrot_v2_64	cublasCscal_v2_64	cublasCsrot_v2_64	cublasCsscal_v2_64
cublasCswap_v2_64	cublasCsymm_v2_64	cublasCsymv_v2_64	cublasCsyr_v2_64
cublasCsyr2_v2_64	cublasCsyr2k_v2_64	cublasCsyrk_v2_64	cublasCsyrk3mEx_64
cublasCsyrkEx_64	cublasCsyrkx_64	cublasCtbmv_v2_64	cublasCtbsv_v2_64
cublasCtpmv_v2_64	cublasCtpsv_v2_64	cublasCtrmm_v2_64	cublasCtrmv_v2_64
cublasCtrsm_v2_64	cublasCtrsmBatched_64	cublasCtrsv_v2_64	cublasDasum_v2_64
cublasDaxpy_v2_64	cublasDcopy_v2_64	cublasDdgmm_64	cublasDdot_v2_64
cublasDgbmv_v2_64	cublasDgeam_64	cublasDgemm_v2_64	cublasDgemmBatched_64
cublasDgemmGroupedBatched	cublasDgemmGroupedBatched_64	cublasDgemmStridedBatched_64	cublasDgemv_v2_64
cublasDgemvBatched_64	cublasDgemvStridedBatched_64	cublasDger_v2_64	cublasDnrm2_v2_64
cublasDotcEx_64	cublasDotEx_64	cublasDrot_v2_64	cublasDrotm_v2_64
cublasDsbmv_v2_64	cublasDscal_v2_64	cublasDspmv_v2_64	cublasDspr_v2_64
cublasDspr2_v2_64	cublasDswap_v2_64	cublasDsymm_v2_64	cublasDsymv_v2_64
cublasDsyr_v2_64	cublasDsyr2_v2_64	cublasDsyr2k_v2_64	cublasDsyrk_v2_64
cublasDsyrkx_64	cublasDtbmv_v2_64	cublasDtbsv_v2_64	cublasDtpmv_v2_64
cublasDtpsv_v2_64	cublasDtrmm_v2_64	cublasDtrmv_v2_64	cublasDtrsm_v2_64
cublasDtrsmBatched_6	cublasDtrsv_v2_64	cublasDzasum_v2_64	cublasDznrm2_v2_64

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cublasGemmBatchedEx_64	cublasGemmEx_64	cublasGemmGroupedBatchedEx_64	cublasGemmStridedBatchedEx_64
cublasGetMatrix_64	cublasGetMatrixAsync_64	cublasGetVector_64	cublasGetVectorAsync_64
cublasHgemm_64	cublasHgemmBatched_64	cublasHgemmStridedBatched_64	cublasHSHgemvBatched_64
cublasHSHgemvStridedBatched_64	cublasHSSgemvBatched_64	cublasHSSgemvStridedBatched_64	cublasIcamax_v2_64
cublasIcamin_v2_64	cublasIdamax_v2_64	cublasIdamin_v2_64	cublasIsamax_v2_64
cublasIsamin_v2_64	cublasIzamax_v2_64	cublasIzamin_v2_64	cublasNrm2Ex_64
cublasRotEx_64	cublasSasum_v2_64	cublasSaxpy_v2_64	cublasScalEx_64
cublasScasum_v2_64	cublasScnrm2_v2_64	cublasScopy_v2_64	cublasSdgmm_64
cublasSdot_v2_64	cublasSetMatrix_64	cublasSetMatrixAsync_64	cublasSetVector_64
cublasSetVectorAsync_64	cublasSgbmv_v2_64	cublasSgeam_64	cublasSgemm_v2_64
cublasSgemmBatched_64	cublasSgemmEx_64	cublasSgemmGroupedBatched	cublasSgemmGroupedBatched_64
cublasSgemmStridedBatched_64	cublasSgemv_v2_64	cublasSgemvBatched_64	cublasSgemvStridedBatched_64
cublasSger_v2_64	cublasSnrm2_v2_64	cublasSrot_v2_64	cublasSrotm_v2_64
cublasSsbmv_v2_64	cublasSscal_v2_64	cublasSspmv_v2_64	cublasSspr_v2_64
cublasSspr2_v2_64	cublasSswap_v2_64	cublasSsymm_v2_64	cublasSsymv_v2_64
cublasSsyr_v2_64	cublasSsyr2_v2_64	cublasSsyr2k_v2_64	cublasSsyrk_v2_64
cublasSsyrkx_64	cublasStbmv_v2_64	cublasStbsv_v2_64	cublasStpmv_v2_64
cublasStpsv_v2_64	cublasStrmm_v2_64	cublasStrmv_v2_64	cublasStrsm_v2_64
cublasStrsmBatched_6	cublasStrsv_v2_64	cublasTSSgemvBatche	cublasTSSgemvStrided

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4		d_64	Batched_64
cublasTSTgemvBatched_64	cublasTSTgemvStridedBatched_64	cublasZaxpy_v2_64	cublasZcopy_v2_64
cublasZdgmm_64	cublasZdotc_v2_64	cublasZdotu_v2_64	cublasZdrot_v2_64
cublasZdscal_v2_64	cublasZgbmv_v2_64	cublasZgeam_64	cublasZgemm_v2_64
cublasZgemm3m_64	cublasZgemmBatched_64	cublasZgemmStridedBatched_64	cublasZgemv_v2_64
cublasZgemvBatched_64	cublasZgemvStridedBatched_64	cublasZgerc_v2_64	cublasZgeru_v2_64
cublasZhbmv_v2_64	cublasZhemm_v2_64	cublasZhemv_v2_64	cublasZher_v2_64
cublasZher2_v2_64	cublasZher2k_v2_64	cublasZherk_v2_64	cublasZherkx_64
cublasZhpmv_v2_64	cublasZhpr_v2_64	cublasZhpr2_v2_64	cublasZrot_v2_64
cublasZscal_v2_64	cublasZswap_v2_64	cublasZsymm_v2_64	cublasZsymv_v2_64
cublasZsyr_v2_64	cublasZsyr2_v2_64	cublasZsyr2k_v2_64	cublasZsyrk_v2_64
cublasZsyrkx_64	cublasZtbmv_v2_64	cublasZtbsv_v2_64	cublasZtpmv_v2_64
cublasZtpsv_v2_64	cublasZtrmm_v2_64	cublasZtrmv_v2_64	cublasZtrsm_v2_64
cublasZtrsmBatched_64	cublasZtrsv_v2_64		

Cufft 新增函数

cufftGetPlanPropertyInt64	cufftResetPlanProperty	cufftSetPlanPropertyInt64	cufftXtSetJITCallback_12_7

Cusparse 新增函数

cusparseBsrSetStridedBatch	cusparseConstBlockedEllGet	cusparseConstCooGet	cusparseConstCscGet
cusparseConstCsrGet	cusparseConstDnMatGet	cusparseConstDnMatGetValues	cusparseConstDnVecGet
cusparseConstDnVecGetValues	cusparseConstSpMatGetValues	cusparseConstSpVecGet	cusparseConstSpVecGetValues

cusparseCreateBsr	cusparseCreateConstBlockedEll	cusparseCreateConstBsr	cusparseCreateConstCoo
cusparseCreateConstCsc	cusparseCreateConstCsr	cusparseCreateConstDnMat	cusparseCreateConstDnVec
cusparseCreateConstSlicedEll	cusparseCreateConstSpVec	cusparseCreateSlicedEll	cusparseSpGEMM_estimateMemory
cusparseSpGEMM_getNumProducts	cusparseSpMV_preprocess	cusparseSpSM_updateMatrix	cusparseSpSV_updateMatrix

CusolverDn 新增函数

cusolverDnGetDeterministicMode	cusolverDnSetDeterministicMode	cusolverDnXgeev	cusolverDnXgeev_bufferSize
cusolverDnXlarft	cusolverDnXlarft_bufferSize	cusolverDnXsyevBatched	cusolverDnXsyevBatched_bufferSize

Device 新增函数

__vimax_s32_relu	__vimax_s16x2_relu	__vimin_s32_relu	__vimin_s16x2_relu	__vimax3_s32
__vimax3_s16x2	__vimax3_u32	__vimax3_u16x2	__vimin3_s32	__vimin3_s16x2
__vimin3_u32	__vimin3_u16x2	__vimax3_s32_relu	__vimax3_s16x2_relu	__vimin3_s32_relu
__vimin3_s16x2_relu	__viaddmax_s32	__viaddmax_s16x2	__viaddmax_u32	__viaddmax_u16x2
__viaddmin_s32	__viaddmin_s16x2	__viaddmin_u32	__viaddmin_u16x2	__viaddmax_s32_relu
__viaddmax_s16x2_relu	__viaddmin_s32_relu	__viaddmin_s16x2_relu	__vibmax_s32	__vibmax_u32
__vibmin_s32	__vibmin_u32	__vibmax_s16x2	__vibmax_u16x2	__vibmin_s16x2
__vibmin_u16x2	__half2char_rz	__half2uchar_rz	__heq2_mask	__hne2_mask

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__hle2_mask	__hge2_mask	__hlt2_mask	__hgt2_mask	__hequ2_mask
__hneu2_mask	__hleu2_mask	__hgeu2_mask	__hltu2_mask	__hgtu2_mask
__bfloat162char_rz	__bfloat162uchar_rz	__heq2_mask	__hne2_mask	__hle2_mask
__hge2_mask	__hlt2_mask	__hgt2_mask	__hequ2_mask	__hneu2_mask
__hleu2_mask	__hgeu2_mask	__hltu2_mask	__hgtu2_mask	atomicAdd
atomicAdd_block	atomicAdd_system	atomicAdd	atomicAdd_block	atomicAdd_system

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